Why FA and MA Tin Iodides Tilt Oppositely: Cage Shape, Cooling Path and the Operating Band Gap

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Abstract

The band gap of a tin halide perovskite absorber is set by the film in the device, which bonds to its substrate at the anneal and is cooled while clamped. How the film responds depends on its octahedral tilts, which differ between compounds: caesium and formamidinium (FA) tin iodide tilt in-phase, methylammonium (MA) tin iodide antiphase. We simulate $CsSnI_3$, $MASnI_3$ and $FASnI_3$ with our machine-learned force field UrsaMLFF, fine-tuned on meta-GGA density functional theory labels and about a hundred times faster than MACE models. On cooling, all three compounds form their experimental tilt systems, with tetragonal tilt angles within 0.3° of experiment. Free-energy calculations explain the opposite senses: antiphase tilting crosses the long axes of the upper and lower faces of the cation cage, which frustrates FA, with an amine group at each end, but suits MA, which binds to one face; $FASnI_3$ favours in-phase tilting by 1.1 mHa and $MASnI_3$ antiphase tilting by 0.6 mHa per formula unit. Under clamping, residual tension opens the gap of tilted $CsSnI_3$ by about 20 meV, whereas

films operated above their tilt transition respond only elastically. The distance from the tilt transition thus sets how strongly processing shifts the operating band gap.

Keywords

perovskite, band gap, tin halide, lead-free, residual stress, substrate clamping, phase transition, machine-learning force field, density functional theory

Introduction

Hybrid and all-inorganic halide perovskites are leading candidates for next-generation photovoltaic absorbers, with certified single-junction power-conversion efficiencies above 26 %.[1] The best absorbers still contain lead, whose toxicity is an obstacle to commercial deployment.[2] Tin halide perovskites, the all-inorganic $CsSnI_3$ and the hybrid methylammonium (MA) and formamidinium (FA) compounds $MASnI_3$ and $FASnI_3$, are the main lead-free alternatives, combining band gaps of 1.2–1.4 eV with high carrier mobilities.[3,4] They are also chemically fragile and strongly polymorphic, and their electronic structure is tightly coupled to their crystal structure.[5,6]

The structural motif that connects their phases is the cooperative tilting of the SnI_6 octahedra. Below the cubic phase all three compounds tilt their octahedra about one pseudocubic axis by 6 to 9° (Table 1), but they differ in how the tilt is stacked along that axis. In $CsSnI_3$ and $FASnI_3$ successive layers rotate in the same sense (in-phase, $a^0a^0c^+$, $P4/mbm$); in $MASnI_3$ the sense alternates from layer to layer (antiphase, $a^0a^0c^-$, $I4cm$ or $I4/mcm$), which doubles the cell along the tilt axis.[4,7,8] The lead iodides and their mixtures follow the same rule,[9–12] so the stacking is set by the A-site cation rather than by the metal. The inorganic framework itself barely distinguishes the two senses: in DFT, the in-phase and antiphase tilts of $CsPbI_3$ and $CsSnI_3$ differ by about 1 meV per formula unit.[13] Several explanations have been proposed, a general preference of halide perovskites for in-phase tilts,[14] hydrogen

bonds of MA that drive the antiphase mode,[15] an FA arrangement that templates the in-phase tilt[16] and the near-degeneracy of the two modes,[11] but none accounts for both cations. It is also not known whether the sense is fixed by the static energetics of the cation in its cage or emerges only from the thermal ensemble and from the path along which the crystal is cooled.

Table 1: Tilt systems of the tin iodide perovskites below the cubic phase. Tilt angles (degrees): amplitude about the tilt axis, computed from the deposited structures with the analysis used for the simulations; for $P4bm$ the tilts of the two alternating layers. Transitions: tilt order (upper) and the next transition on cooling (lower), heating / cooling where both are reported.

compound	phase (structure)	tilt system	tilt angle	upper (K)	lower (K)	ref.
$CsSnI_3$	β $P4/mbm$ (380 K)	$a^0a^0c^+$	6.0	440 / 426	362 / 351	[7,17]
	γ $Pnma$ (300 K)	$a^-a^-c^+$	10.2 (c), 4.8 (a, b)			
$FASnI_3$	$P4/mbm$ (200 K)	$a^0a^0c^+$	6.3	225–255	125–155	[8,18]
	$P4bm$ (100 K)	c^+, alternating layers	11.6 / 3.7			
$MASnI_3$	$I4cm$ (200 K)	$a^0a^0c^-$	8.6	∼275	114 / 108	[4,19–21]
	$I4/mcm$ (140 K)	$a^0a^0c^-$	9.3			

Finite-temperature theory has not resolved these questions for the tin compounds. Molecular dynamics (MD) with machine-learned force fields now reproduces the phase sequences of halide perovskites,[22–25] and anharmonic lattice dynamics describes the stabilisation of their high-temperature phases.[26,27] Predicted transition temperatures exist almost only for $CsSnI_3$, where they span more than 100 K;[25–27] to our knowledge there is no finite-temperature simulation of the phase sequence of $MASnI_3$ and little for $FASnI_3$. Studies of orientational effects have used ordered or frozen cation arrangements rather than thermal ensembles.[10,28]

We address the tilt sense with MD driven by a machine-learned force field trained on meta-GGA (r^{2}SCAN) density functional theory (DFT) labels, which we first validate against the measured phase sequences and tilt angles. We then separate the static from the thermal contributions: the energy difference between the two tilt senses is computed for many cation arrangements in fixed tilted frameworks and related to the geometry of the A-site cage, whose upper and lower faces share the same long axis under in-phase tilting but are crossed under antiphase tilting, and the cation orientations are followed in the MD. We test the

hypothesis that this cage shape, together with the number and placement of the hydrogen-bonding groups of the cation, accounts for the opposite stacking in FA and MA compounds, and we examine how the cooling path decides the tilt sense where the preference is weak.

The same structural sensitivity separates the band gap of a device absorber from the idealised gap of a free, equilibrated crystal. A film is deposited and annealed at an elevated temperature, where it crystallises on and bonds to its substrate, and is then cooled to its operating temperature while the substrate restricts its in-plane dimensions. The mismatch of thermal expansion leaves residual tensile stresses of tens of MPa, up to about 150 MPa for an anneal at 160 °C, and in-plane strains of about 0.5 %.[29–32] The direct elastic effect of such a strain on the gap is small, but a film that crosses a structural transition while clamped cannot develop the spontaneous strain of order 1 % that accompanies it.[7] Clamping then competes with the transition: it can shift the transition temperature, select or frustrate tilt variants and bias the cation orientations, and the band gap follows the structure. Through the anneal temperature and the cooling under the constraint of the substrate, the manufacturing process can therefore shift the operating band gap away from its idealised value.

We use the same simulation framework, with a force field trained on PBEsol labels, to determine what happens to the band gap when a film is cooled during manufacturing. The three compounds place the anneal and the operating temperature at different positions relative to their transitions. $CsSnI_3$, annealed near 373 K in its tetragonal β phase, crosses the $\gamma \leftrightarrow \beta$ transition at 362 K on cooling, only a few kelvin above the 358 K damp-heat qualification test.[7] $MASnI_3$ is annealed and operated in its cubic phase, a few tens of kelvin above its tilt transition near 275 K.[4,19–21] $FASnI_3$, cubic above 225 to 250 K,[18] crosses no transition between anneal and operation and isolates the effect of clamping alone. For each compound we simulate the film free-standing, under an in-plane tension of 50 MPa and rigidly clamped in-plane at the anneal, with the simulation temperatures anchored to the model's own transition temperatures so that anneal and operation lie in the same phase and at the same distance from T_c as in a device, and we compute band gaps with DFT on

MD frames chosen by stratified sampling. Every headline quantity is a difference between matched states, the shift of a transition temperature ΔT_c or of a band gap ΔE_g, which cancels systematic errors of absolute values such as the missing spin–orbit coupling. We also test whether the orientational response to clamping orders $MASnI_3 > FASnI_3 > CsSnI_3$ $(= 0)$.

Both questions require a force field that is accurate across phases, strains and cation orientations, yet fast enough for nanoseconds of MD in cells of thousands of atoms. We select the DFT training structures from a large MD pool by their coverage of distinct atomic environments in a space of Gaussian multipole descriptors,[33,34] which captures the rare configurations near transitions and in strained cells at a fraction of the cost of random selection, and we use equivariant message-passing models (MACE[35]) to validate a faster in-house potential that carries the production simulations.

The Computational Methods describe the DFT settings, the selection of the training structures, the fine-tuning of the force fields, the MD protocols and the identification of the phases. The Results and Discussion justify the choice of the exchange–correlation functional, compare the simulated phase sequences and tilt systems with experiment, analyse why the cation sets the tilt sense, and apply the model to films cooled under the constraint of a substrate and to their band gaps.

Computational Methods

Overview

The workflow has four parts (Figure 1). First, a broad pool of configurations of the three compounds is generated by MD with an existing force field and down-sampled to a training set, which is labelled with DFT. Second, machine-learned potentials are fine-tuned on these labels and validated against held-out DFT data and against DFT calculations on configurations from its own MD. Third, UrsaMLFF drives large-scale MD: heating and cooling ramps

locate the transition temperatures of the free-standing crystals, and anneal-and-cooling runs under two substrate constraints and without one describe a clamped film. Fourth, frames of these ensembles are selected by stratified sampling and their band gaps are computed with DFT. Every quantity reported as a result is a difference between two matched states.

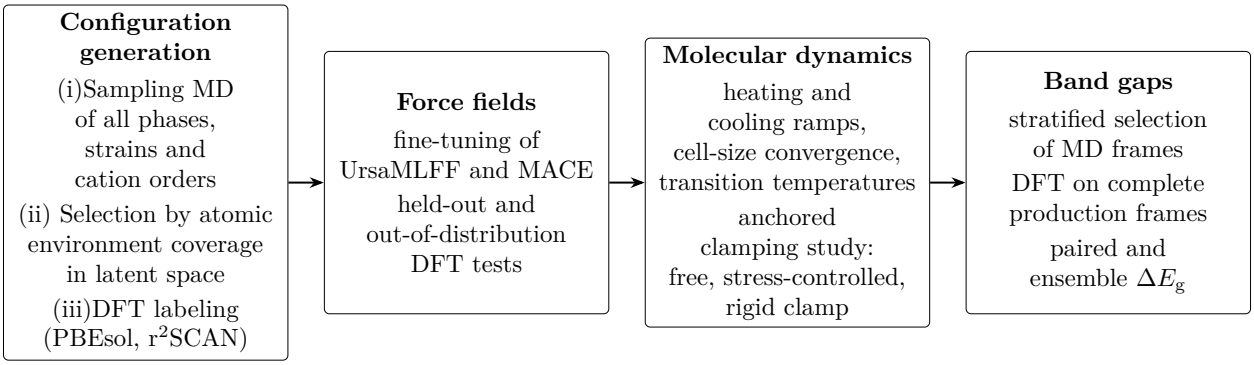


Figure 1: Workflow: configuration generation and DFT labelling, fine-tuning and validation of the force fields, MD of free-standing and clamped cells, and stratified DFT band gaps of selected MD frames.

Electronic structure

The energies, forces and stresses used to train the force fields, and the band gaps of selected MD frames, are obtained from Kohn–Sham density functional theory (DFT) with our in-house code UrsaDFT. The Kohn–Sham equations may be discretised in any complete basis, such as finite elements or plane waves; here we employ plane waves together with Blöchl's projector augmented-wave (PAW) method[36] and the JTH PAW datasets (version 2.0),[37,38] which are generated with the PBE functional. Exchange and correlation are treated at two levels of approximation for both the labels and the band gaps: the generalised-gradient approximation (GGA) with PBEsol[39] and the meta-GGA with r^{2}SCAN.[40] The r^{2}SCAN meta-GGA proved more robust and accounts for most of the experimental observations (*Choice of the exchange–correlation functional*). The main text therefore reports r^{2}SCAN results, including the MD of the clamping study (*Effect of process conditions on band-gap deviations*), whose band gaps are still those of the earlier PBEsol study; the remaining PBEsol results

are given in the Supporting Information. Spin–orbit coupling is neglected throughout. The plane-wave cutoffs were chosen such that the discretisation errors remain below 2×10^{-4} Ha per atom in the energy, 2×10^{-4} Ha Bohr^{-1} in the forces and 5×10^{-6} Ha Bohr^{-3} in the stress; further details are given in the Supporting Information.

Identifying structures for DFT labelling

The structures labelled with DFT are drawn from MD trajectories that span, with margin, every state the production simulations visit: the α, β and γ phases and the transitions between them, the strains of thermal expansion, of the transitions and of substrate clamping, and disordered orientations of the molecular cations. With the UrsaMLFF foundation model we ran constant-volume and constant-pressure simulations of the phases of $CsSnI_3$, $MASnI_3$ and $FASnI_3$. The starting structures were the three phases of each compound built from their Glazer tilt patterns,[41] structures from the Materials Project,[42] and frameworks of documented Pb-based perovskites in which Pb was replaced by Sn (Table S2). The foundation model only proposes structures, since every selected structure is recomputed with DFT. The trajectories form a pool of 55 928 structures (Supporting Information).

Our workflow then identifies the distinct atomic environments of the pool and selects the structures that cover all of them at the lowest DFT cost. Every atom is described by Gaussian multipole descriptors;[33,34] environments closer than a cutoff in this descriptor space are treated as equivalent, which leaves 3816 distinct environments, and a greedy set cover weighted by the cell volume selects the cheapest set of structures that contains all of them. Together with one structure from each sampling stratum that the cover missed and a second round of small structures, 718 structures of 36 to 432 atoms were labelled: 320 of $MASnI_3$, 247 of $FASnI_3$, 136 of $CsSnI_3$ and 15 of reference compounds.

Large-scale MD with the first potentials revealed configurations that these labels do not cover and in which the force field failed: protons detached from their cation, broken C–N bonds and two cations in one cage in the hybrids, and Cs ions passing into a neighbouring

cage in $CsSnI_3$. Such configurations were labelled in an active-learning loop: cells cut around the defect events of the MD, images in which an N–H or C–H bond is stretched or a Cs–Sn, C–I, N–I or H–I contact is shortened, and frames of the hottest MD. Table 2 gives the tally of the DFT labels; the details are given in the Supporting Information.

Table 2: DFT labels of this work. Initial: structures selected from the sampling pool; active learning: structures added after failures of the force field in large-scale MD; used: training, validation and held-out structures after the exclusions given in the Supporting Information.

	PBEsol	r^{2}SCAN
initial selection	718	721
active learning: molecular defects	212	205
active learning: Cs–Sn, C–I, N–I and H–I contacts, hot MD	–	120
total computed	930	1046

Fine-tuning the machine-learned potentials

We fine-tune two classes of machine-learned interatomic potentials on the labels of each functional. The first is the equivariant message-passing MACE architecture,[35] initialised from the foundation models MACE-MH-1[43] and MACE-OMAT-0-medium.[44,45] The second is our in-house force field UrsaMLFF, whose radial and angular descriptors and per-element networks follow refs 46 and 47, extended by one equivariant message-passing step. UrsaMLFF carries all production MD, as it is fast enough for nanosecond trajectories of cells with thousands of atoms; the more accurate but slower MACE models serve to validate it. The potentials are fitted to cohesive energies, forces and the full stress tensor. The labelled structures are divided by MD trajectory, and stratified by compound, into training, validation and held-out sets; the final UrsaMLFF is trained on the complete r^{2}SCAN training set of 866 structures, including the active-learning labels. Architectures and training settings are given in the Supporting Information.

Table 3 and Figure 2 report the accuracy of the r^{2}SCAN potentials on the held-out structures. All three reproduce the DFT forces with $R^2 \geq 0.99$ and the energies within a

compound with $R^2 \geq 0.98$. The MACE models predict forces about twice as accurately as UrsaMLFF, whereas UrsaMLFF predicts stresses 2.3 to 4.7 times more accurately, the stress error of the MACE models being dominated by a constant pressure offset. On a single A100 GPU, UrsaMLFF performs 960 to 2540 MD steps per second for cells of up to 2300 atoms, 70 to 135 times more than the faster MACE model. Dispersion is included at run time through Grimme's D3 correction with Becke–Johnson damping[48,49] and the parameters of the training functional; it is not part of the fit. The PBEsol potentials and the throughput benchmark are reported in the Supporting Information.

Table 3: Errors of the potentials trained on the r^{2}SCAN labels on the 99 held-out structures: root-mean-square errors of the energy per atom, the force components and the stress components, and coefficients of determination R^2 (for the energy after subtracting the mean DFT energy of each compound). UrsaMLFF: final potential, trained on 866 structures including the active-learning labels; MACE: fine-tuned on the 546 initial structures.

model	E (Ha/atom)	F (Ha Bohr^{-1})	σ (Ha Bohr^{-3})	R^2_E	R^2_F	R^2_σ
UrsaMLFF	9.23×10^{-5}	5.20×10^{-4}	8.51×10^{-7}	0.9865	0.9986	0.9974
MACE-MH-1	4.37×10^{-5}	2.48×10^{-4}	4.01×10^{-6}	0.9970	0.9997	0.9429
MACE-OMAT	9.78×10^{-5}	2.67×10^{-4}	1.99×10^{-6}	0.9848	0.9996	0.9860

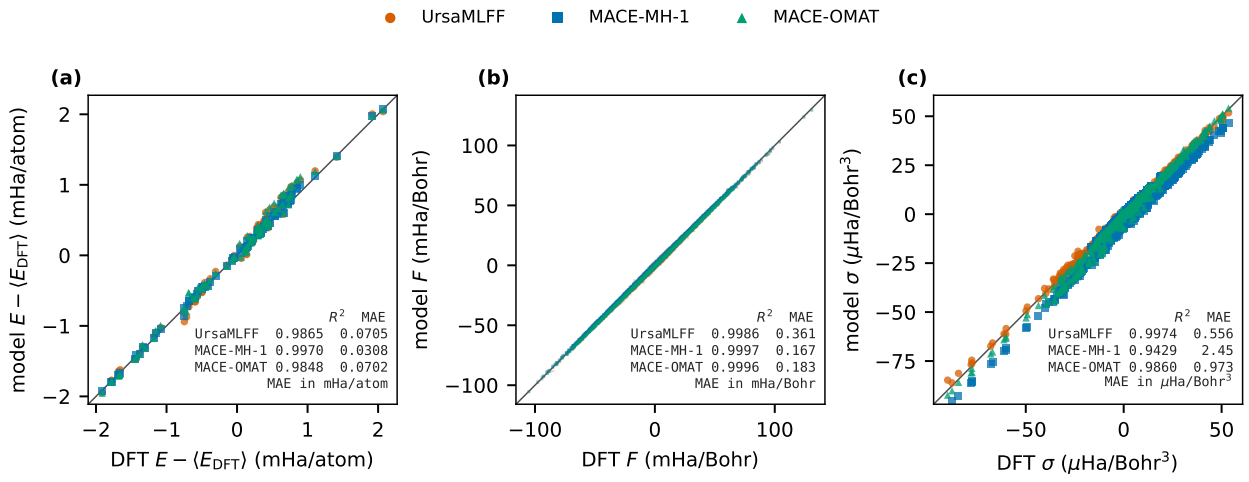


Figure 2: Predicted against DFT values on the 99 held-out structures for the potentials trained on the r^{2}SCAN labels: (a) energy per atom relative to the mean DFT energy of each compound, (b) force components, (c) stress components. Insets: coefficient of determination R^2 and mean absolute error (MAE) of each potential.

Molecular dynamics

MD is run with the GPU engine of UrsaMLFF in double precision. Temperature is controlled by stochastic velocity rescaling[50] and pressure by stochastic cell rescaling,[51] with the three orthogonal cell axes coupled independently and an external pressure that offsets the Pulay stress of the labels. All cells are supercells of the orthogonal $\sqrt{2} \times \sqrt{2} \times 2$ setting of the cubic perovskite, in which the cubic, tetragonal ($P4/mbm$, $I4/mcm$) and orthorhombic ($Pnma$) phases are all orthogonal, so every transition can proceed by rescaling the three axes, without shear. The production cells contain 2000 atoms for $CsSnI_3$ and 2304 atoms for $MASnI_3$ and $FASnI_3$. Transitions are located with linear temperature ramps at constant pressure at 0.25 K/ps. Each compound is cooled from the cubic phase, from 450 to 500 K ($CsSnI_3$) or 400 K (hybrids) down to 200 K or 50 K, and the cooled crystal is heated back over the same range at the same rate. $MASnI_3$ is also heated from an in-phase tilted crystal to test the role of the thermal path. Every protocol is run three times with different initial velocities. Ramps of 0.1 to 2 K/ps and cells of up to 23 040 atoms, given in the Supporting Information, show how the transition temperatures depend on rate and cell size. a_{pc} denotes the cube root of the volume per formula unit. Ensembles, starting structures and run parameters are given in the Supporting Information.

Phase identification and transition temperatures

We analyse the octahedral tilts in the MD trajectories with PDynA.[52] For every frame, PDynA computes the tilt angles of every SnI_6 octahedron about the three pseudo-cubic axes and the correlation between the tilts of nearest-neighbour octahedra along each axis. The sign of this correlation distinguishes in-phase (+) from antiphase (−) tilting and so identifies the Glazer tilt pattern.[41] The correlations are averaged over all octahedra and over 10 K windows of the heating and cooling ramps. The phases and transition temperatures are then assigned from the loss of tilt order, with the criteria of Table 4. Because the order parameters are the absolute correlations sorted by size, the assignment does not depend on which cell

axis carries which tilt, which a cooling run chooses at random. The low-temperature phase of $FASnI_3$ ($P4bm$) adds no tilted axis but alternates the tilt amplitude between successive layers; we detect it with the layer alternation S, half the difference between the mean rms tilts of even and odd layers about the tilt axis in 14 ps time averages, computed every 10 K.

Table 4: Assignment of phases and transitions from the nearest-neighbour tilt correlations C_a, C_b and C_c about the three pseudo-cubic axes (degrees), averaged over all octahedra in windows of 10 K spaced by 5 K. $|C|_{(1)} \geq |C|_{(2)} \geq |C|_{(3)}$ are the absolute correlations sorted by size; α: cubic, β: tetragonal, γ: orthorhombic. The thresholds are at least twice the values in the phase above the transition. A transition temperature is the hottest window of an ordered stretch of at least 20 K (three consecutive windows above the threshold); we report the mean and standard error over the runs, on cooling and on heating.

	order parameter	criterion	compounds
tilt sense	C about the tilted axis	+ in-phase, − antiphase	all
$\beta \leftrightarrow \alpha$ transition	$\|C\|_{(1)}$, any tilt	falls below 3°	all
$\gamma \leftrightarrow \beta$ transition	$\|C\|_{(2)}$, second tilted axis	falls below 2°	$CsSnI_3$, $MASnI_3$
layered tilt ($P4bm$)	S, layer alternation	falls below 1°	$FASnI_3$

Free energy of the tilt sense

The free-energy difference between the antiphase and the in-phase tilt is computed by thermodynamic integration[53] in fixed tilted frameworks, in which only the cations move. The cells contain 128 formula units (1536 atoms, $4\times4\times2$ repeats of the $\sqrt{2}\times\sqrt{2}\times2$ setting) with the volume of the force field at 200 K, a cubic metric of 6.292 Å for $FASnI_3$ and 6.237 Å for $MASnI_3$. The octahedra are rotated about c by the experimental tetragonal tilt (Table 1), 6.3° for $FASnI_3$ and 8.6° for $MASnI_3$, in phase ($a^0a^0c^+$) or in antiphase ($a^0a^0c^-$). The two frameworks share the Sn atoms, the axial I atoms and the equatorial I atoms of every second layer, and differ only in the equatorial I atoms of the other layers, whose rotation is reversed. These atoms are moved along the straight path $\mathbf{x}(\lambda) = \mathbf{x}_+ + \lambda(\mathbf{x}_- - \mathbf{x}_+)$ from the in-phase ($\lambda = 0$) to the antiphase framework ($\lambda = 1$), and

$$\Delta F = F_- - F_+ = \int_0^1 \left\langle \frac{\partial U}{\partial \lambda} \right\rangle_\lambda \mathrm{d}\lambda, \qquad \frac{\partial U}{\partial \lambda} = -\sum_i \mathbf{F}_i \cdot (\mathbf{x}_{-,i} - \mathbf{x}_{+,i}), \tag{1}$$

where the sum runs over the moving I atoms and $\mathbf{F}_i$ are the forces on them. Because the framework is held fixed, ΔF isolates the contribution of the cations, through their orientations and their contacts with the iodide cage; a positive ΔF favours the in-phase tilt.

The cations are propagated at constant volume with stochastic velocity rescaling (time step 1 fs, time constant 100 fs), without the dispersion correction, which changes the static energy difference between the two senses by less than 0.07 mHa per formula unit (Supporting Information). Each window starts from a random orientation of every cation, the same in both frameworks, is held for 20 ps at 350 K, and is then cooled in steps of 25 K from 325 to 100 K, with 20 ps of equilibration and 100 ps of production at each temperature. The integrand is sampled at the nine Gauss–Lobatto nodes of λ, with five further windows between $\lambda = 0.75$ and 0.97 for $FASnI_3$, where it changes rapidly, and integrated by Gauss–Lobatto quadrature; as an independent estimate, the potential energy of each configuration is evaluated every 1 ps at every λ of the path and ΔF is obtained with MBAR.[54] The energy difference follows from the end windows, $\Delta U = \langle U \rangle_- - \langle U \rangle_+$, and the entropic part as $-T\Delta S = \Delta F - \Delta U$. Every calculation is repeated for three independent cation arrangements, and we report the mean and standard error. For $FASnI_3$, the windows above $\lambda = 0.84$ are repeated from the configuration equilibrated at $\lambda = 0.84$, to test for hysteresis along the path, and the two end states at $\lambda = 1$ are joined in one cell (half of each) to test which cation arrangement is stable in the antiphase framework.

Results and Discussion

Choice of the exchange–correlation functional

As described in the Computational Methods, we trained UrsaMLFF on DFT labels from both the GGA functional PBEsol and the meta-GGA functional r^{2}SCAN. Table 5 compares the two potentials, cooled and reheated at 0.25 K/ps, with the experimental tilt systems and transition temperatures of Table 1. The r^{2}SCAN potential reproduces the pseudo-cubic

lattice constants to within 0.4 to 1.0 % of experiment, whereas PBEsol underestimates them by 2.0 to 2.6 %. On cooling, the r^{2}SCAN potential orders $MASnI_3$ in the antiphase tilt of the experimental tetragonal phase, with an amplitude of 8.5° at 200 K against 8.6° measured. With PBEsol the antiphase tilt of $MASnI_3$ remains weak, 1.4 to 4.5° at 200 K, and in most runs it stays below the tilt-order threshold of Table 4. Both potentials reproduce the in-phase tilt of $FASnI_3$, 5.9° (r^{2}SCAN) and 6.4° (PBEsol) at 200 K against 6.3°, and the sequence $a^0a^0c^+$, $a^-a^-c^+$ of $CsSnI_3$.

Table 5: UrsaMLFF trained on PBEsol and r^{2}SCAN labels against experiment (Table 1), 0.25 K/ps, three runs per direction; hybrids' tilt systems at 200 K. Transitions: heating / cooling, criteria of Table 4. [a]: not every run orders; [b]: loss of the layer alternation of the *P4bm* structure. a_{pc}: deviation from experiment[7,18,55] in parentheses.

compound	quantity	experiment	PBEsol	r^{2}SCAN
$CsSnI_3$	tilt system	$a^0a^0c^+$, $a^-a^-c^+$	$a^0a^0c^+$, $a^-a^-c^+$	$a^0a^0c^+$, $a^-a^-c^+$
	upper transition: tilt lost (K)	440 / 426	465 ± 8 / 435 ± 13	385 ± 3 / 370 ± 18
	lower transition: second tilt lost (K)	362 / 351	387 ± 21 / 380 ± 26	323 ± 2 / 317 ± 2
	a_{pc} at 500 K (Å)	6.206	6.084 (−2.0 %)	6.268 (+1.0 %)
$MASnI_3$	tilt system	$a^0a^0c^-$	$a^0a^0c^-$ [a]	$a^0a^0c^-$
	upper transition: tilt lost (K)	∼275	240[a] / 230[a]	290 ± 6 / 277 ± 7
	lower transition: second tilt lost (K)	114 / 108	not formed	not formed
	a_{pc} at 298 K (Å)	6.240	6.103 (−2.2 %)	6.265 (+0.4 %)
$FASnI_3$	tilt system	$a^0a^0c^+$	$a^0a^0c^+$	$a^0a^0c^+$
	upper transition: tilt lost (K)	225–255	272 ± 2 / 265 ± 6	255 ± 3 / 247 ± 2
	lower transition: second tilt lost (K)	125–155	137 ± 6 / 80 ± 5[a]	133 ± 3 / 127 ± 13[b]
	a_{pc} at 298 K (Å)	6.316	6.155 (−2.6 %)	6.340 (+0.4 %)

The transition temperatures do not favour one functional throughout. For the hybrids, the r^{2}SCAN potential places the loss of tilt order at (277 ± 7) K ($MASnI_3$) and (247 ± 2) K ($FASnI_3$) on cooling, against about 275 K and 225 to 255 K measured, whereas PBEsol places that of $FASnI_3$ 10 to 17 K above the measured range. For $CsSnI_3$, the r^{2}SCAN potential places both transitions 34 to 56 K below experiment, while PBEsol places them 9 to 29 K above it. Neither potential forms the low-temperature phase of $MASnI_3$ near 110 K on cooling. With the r^{2}SCAN potential, a crystal prepared in the ordered low-temperature structure keeps its second tilt on heating up to 190 K, and between 100 and 160 K the ordered and the tetragonal structures both persist for 1.5 ns (Supporting Information), so this transition is

kinetically blocked on the simulated time scales. The second in-phase tilt that PBEsol develops in $FASnI_3$ below about 140 K has no counterpart in the experimental low-temperature phase, which instead modulates the tilt from layer to layer, a modulation the r^{2}SCAN potential reproduces below (133 ± 3) K. Because the r^{2}SCAN potential reproduces the lattice constants, the tilt systems of all three compounds and the tilt-ordering temperatures of the hybrids, the results on the free crystals below use the r^{2}SCAN potential. The PBEsol results, and an earlier comparison of the two label sets at 2 K/ps, are given in the Supporting Information.

Octahedral tilt patterns

Figure 3 shows the deposited structures of Table 1, analysed with the code used below for the simulations. In-phase stacking repeats the tilt pattern from layer to layer; antiphase stacking inverts it. On further cooling $CsSnI_3$ adds antiphase tilts about the other two axes ($a^-a^-c^+$, *Pnma*), and $FASnI_3$ doubles c with layers tilted alternately by 11.6° and 3.7° ($P4bm$).

Each compound is cooled from the cubic phase at 0.25 K/ps and heated back at the same rate, three times in cells of 2000 ($CsSnI_3$) and 2304 atoms (hybrids) (Computational Methods). Phases and transitions are assigned with the criteria of Table 4. Figure 4 shows the tilt correlation along the ramps and tilt maps of the ordered phases, and Table 6 compares the tilt systems and angles with the experimental structures at the same distance below the transition.

The potential forms the experimental tilt systems. $CsSnI_3$ orders as $a^0a^0c^+$ and then as $a^-a^-c^+$, $FASnI_3$ as $a^0a^0c^+$, and every $MASnI_3$ run as $a^0a^0c^-$, with successive layers inverting the pattern as in the experimental structures. The single-tilt angles agree with experiment to within 0.3°: −8.5° and −9.6° against −8.6° and −9.3° for the two tetragonal structures of $MASnI_3$, 6.4° against 6.3° for tetragonal $FASnI_3$, and 6.2° against 6.0° for β-$CsSnI_3$. In γ-$CsSnI_3$ the apical Sn–I–Sn bending, 6.7°, matches the measured 6.8°, while the equatorial bending, 9.9°, is smaller than the measured 11.3°. Below about 130 K $FASnI_3$ develops the

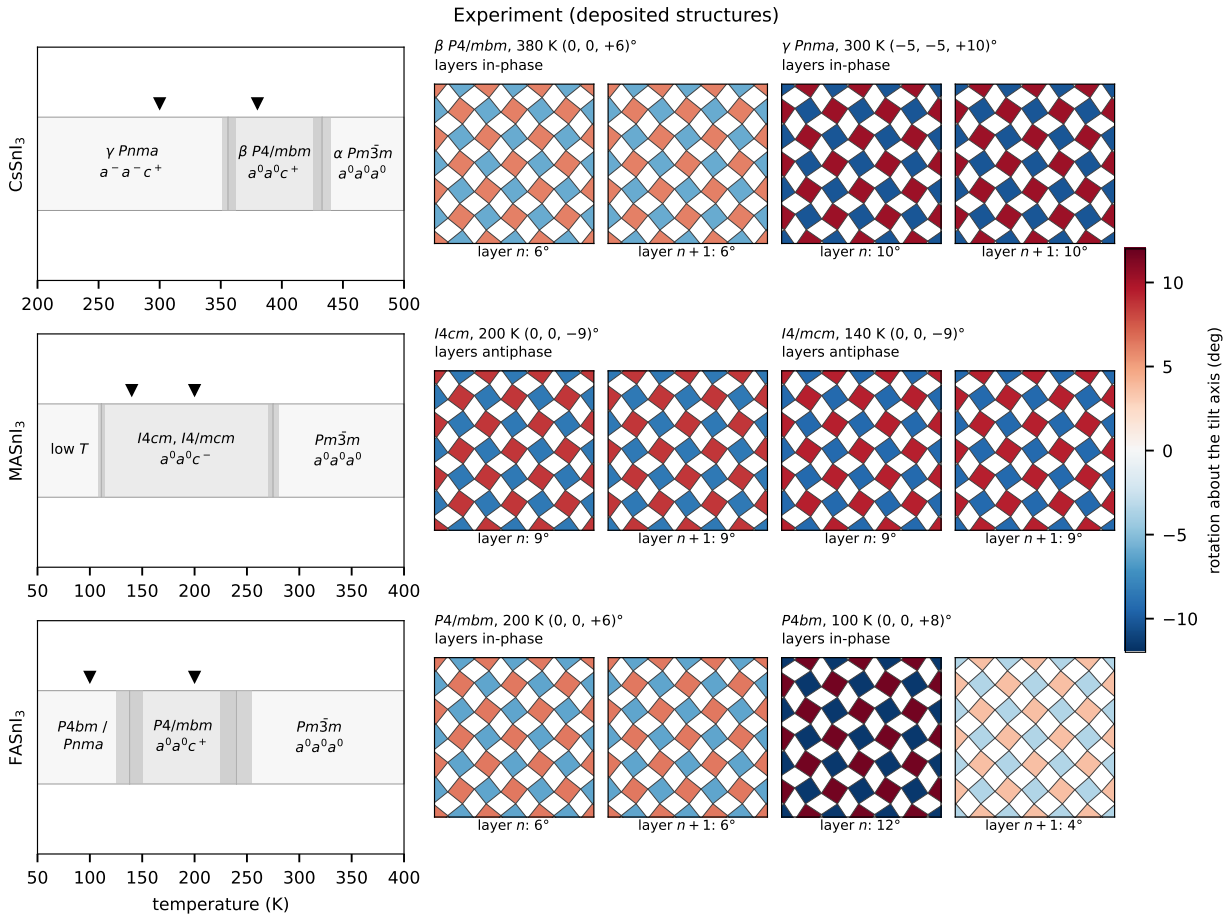


Figure 3: Tilt patterns of the deposited structures of $CsSnI_3$,[7] $MASnI_3$[4,19] and $FASnI_3$[8] (COD 4117958, 4117959, 4335636, 7015449, 4517852, 4517853). Left: phase sequences; grey bands: reported transition temperatures; triangles: temperatures of the structures shown. Right: two consecutive layers perpendicular to the tilt axis. Each square is an SnI_6 octahedron drawn through its four equatorial I atoms and coloured by its rotation about the tilt axis; in-phase tilting repeats the pattern from layer to layer, antiphase tilting inverts it. Titles: tilt amplitudes about the three pseudo-cubic axes (+ in-phase, − antiphase, 0 below 2°); below each map: rms tilt of the layer.

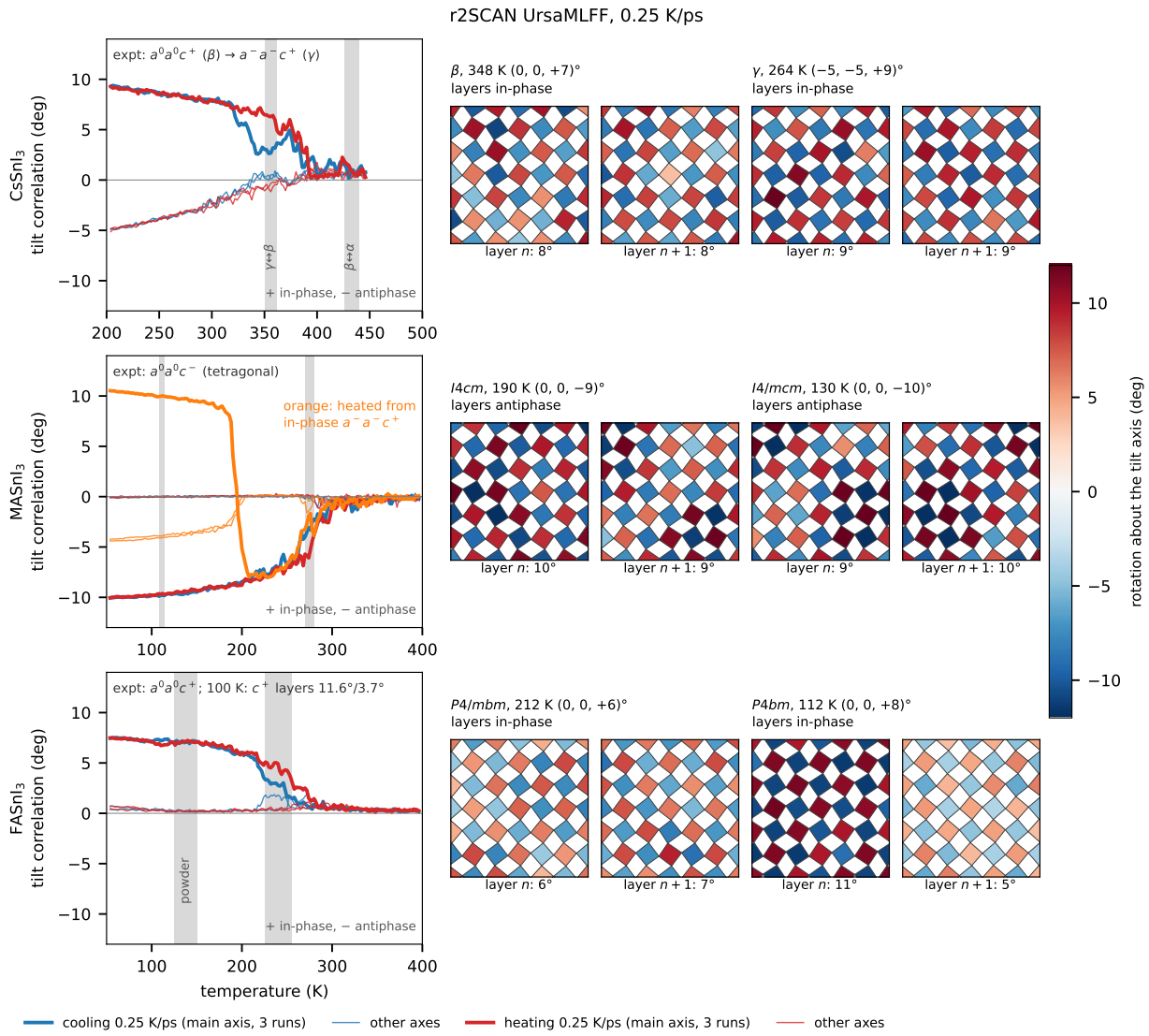


Figure 4: Tilt patterns with the r^{2}SCAN potential at 0.25 K/ps ($CsSnI_3$ 2000 atoms, hybrids 2304). Left: nearest-neighbour tilt correlation about the main tilt axis (thick) and the other two axes (thin), mean of three runs, on cooling from the cubic phase (blue) and on heating the cooled crystal (red); orange: $MASnI_3$ heated from an in-phase ($a^-a^-c^+$) crystal at 50 K; + in-phase, − antiphase; grey bands: experimental transitions. Right: tilt maps of one cooling run at the model temperatures of Table 6, positions averaged over 14 ps, drawn as in Figure 3; titles: experimental structure compared, model temperature and tilt amplitudes about the three axes.

layered tilt of the $P4bm$ structure: on heating, successive layers tilt by 10.9° and 4.9° at 110 K, against 11.6° and 3.7° measured at 100 K, and the alternation is lost at (133 ± 3) K ((127 ± 13) K on cooling), within the measured range of 125 to 155 K.

Table 5 gives the transition temperatures. On heating, the tilt is lost 6 to 15 K above the temperature at which it forms on cooling. The tilt-ordering temperatures of the hybrids lie within the measured ranges, whereas both transitions of $CsSnI_3$ remain 34 to 56 K below experiment; at 0.5 K/ps they lie 36 to 66 K low (Table S12), so the deficit is not a rate effect. Cells of about 20 000 atoms give the same tilt systems and transition temperatures within 15 K (Supporting Information).

The antiphase tilt of $MASnI_3$ does not depend on the thermal path. A crystal prepared with the in-phase $a^-a^-c^+$ tilt of $CsSnI_3$ at 50 K and heated at 0.25 K/ps keeps this pattern up to (190 ± 3) K, converts to $a^0a^0c^-$ and stays antiphase until the tilt is lost at (275 ± 8) K. Cooled from the cubic phase or heated from the opposite stacking, $MASnI_3$ ends in the antiphase phase. Why the cation selects the stacking is the subject of the next section.

Table 6: Tilt systems and tilt angles (degrees) with the r^{2}SCAN potential on cooling at 0.25 K/ps (2000 atoms for $CsSnI_3$, 2304 for the hybrids, three runs each) against the deposited structures (Table 1). Each structure is compared at the same distance below its transition as in experiment (model temperature T). Angles of 14 ps time averages, about each axis from the rotation correlation of neighbouring octahedra, which tilt domains do not cancel. Single tilt: amplitude about the tilt axis (+ in-phase, − antiphase); $Pnma$: equatorial / apical Sn–I–Sn bending $(180° - \theta)/2$; $P4bm$: tilts of the alternating layers.

		experiment		r^{2}SCAN, 0.25 K/ps		
compound	phase, T	tilt system	angle	T (K)	tilt system	angle
$CsSnI_3$	β ($P4/mbm$), 380 K[7]	$a^0a^0c^+$	+6.0	343	$a^0a^0c^+$	+6.2
	γ ($Pnma$), 300 K[7]	$a^-a^-c^+$	11.3 / 6.8	266	$a^-a^-c^+$	9.9 / 6.7
$MASnI_3$	tetragonal ($I4cm$), 200 K[4]	$a^0a^0c^-$	−8.6	202	$a^0a^0c^-$	−8.5
	tetragonal ($I4/mcm$), 140 K[19]	$a^0a^0c^-$	−9.3	142	$a^0a^0c^-$	−9.6
$FASnI_3$	tetragonal ($P4/mbm$), 200 K[8]	$a^0a^0c^+$	+6.3	209	$a^0a^0c^+$	+6.4
	low T ($P4bm$), 100 K[8]	c^+, layers	11.6 / 3.7	109	c^+, layers	9.7 / 6.2

Origin of the tilt sense

The inorganic framework hardly distinguishes the two tilt senses: in DFT the in-phase and antiphase tilts of $CsPbI_3$ and $CsSnI_3$ differ by about 1 meV per formula unit,[13] and in our DFT calculations those of $CsSnI_3$ at 8 and 12° differ by at most 0.04 mHa per formula unit. The sense is therefore chosen by the A-site cation. To separate this choice from the kinetics of the MD, we computed the free-energy difference $\Delta F = F_{\mathrm{anti}} - F_{\mathrm{in}}$ between frameworks fixed at the experimental tetragonal tilts, in which only the cations move (Computational Methods).

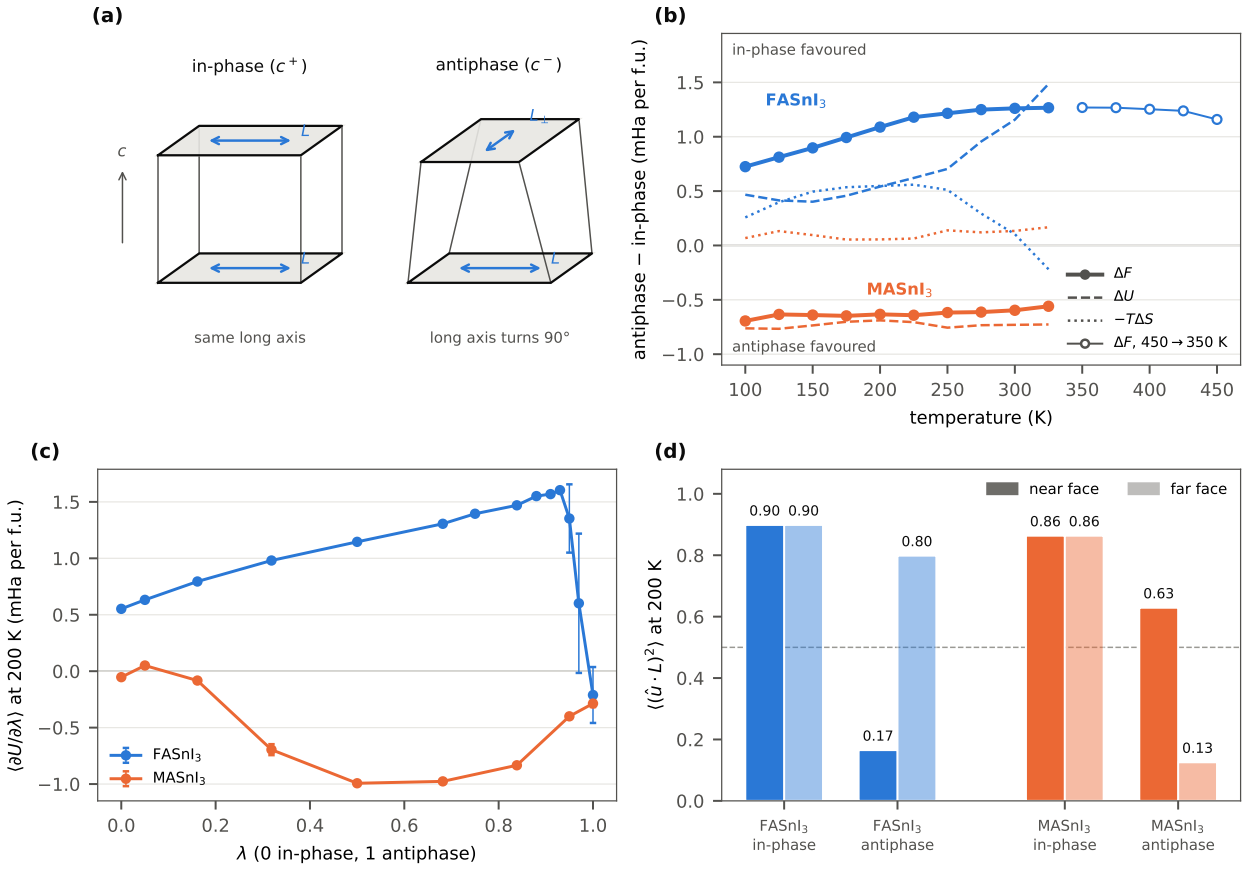


Figure 5: Free energy of the tilt sense with the r^{2}SCAN potential, frameworks fixed at the experimental tetragonal tilts ($FASnI_3$ 6.3°, $MASnI_3$ 8.6°). (a) A tilt about c makes each face of the A-site cage a rectangle with long axis L; in-phase stacking keeps L at the lower and upper faces, antiphase stacking turns it by 90°. (b) $\Delta F = F_{\mathrm{anti}} - F_{\mathrm{in}}$ per formula unit (mean ± standard error over three cation arrangements) with its energy and entropy parts; open symbols: $FASnI_3$ on a 450 to 350 K ladder. (c) Integrand $\langle \partial U/\partial\lambda \rangle$ along the path at 200 K. (d) Alignment of the cation axis (FA N–N, MA C–N) with the long axis of the near and far face of its cage at 200 K; 0.5: no preference within the plane.

The free energy reproduces the experimental senses (Figure 5b). At 200 K, $FASnI_3$ is lower in-phase by (1.09 ± 0.03) mHa per formula unit and $MASnI_3$ lower antiphase by (0.63 ± 0.01) mHa. The preferences hold over the whole range: $FASnI_3$ favours in-phase by 0.72 to 1.27 mHa between 100 and 450 K, and $MASnI_3$ favours antiphase by 0.56 to 0.69 mHa between 100 and 325 K. The three cation arrangements agree to within 0.1 mHa, and MBAR reproduces the integrated values to within 0.03 mHa for $FASnI_3$ and 0.07 mHa for $MASnI_3$ from 275 K on; at lower temperatures the nine $MASnI_3$ windows overlap too little for MBAR. The preferences are comparable to $k_\mathrm{B}T$ (0.63 mHa at 200 K) per formula unit, but they act on every formula unit of a tilt domain.

The two preferences have different origins. In $MASnI_3$ the energy difference, $\Delta U = -0.69$ mHa at 200 K, accounts for the whole preference, and the entropy opposes it slightly ($-T\Delta S = 0.06$ mHa). In $FASnI_3$ energy and entropy contribute equally at 200 K ($\Delta U = +0.54$, $-T\Delta S = 0.55$ mHa); the entropic part vanishes towards 300 K, where the energy carries the preference.

The cause is the shape of the cage (Figure 5a). A tilt about c turns each face of the A-site cage into a rectangle, long along one in-plane axis L. In-phase tilting gives the lower and upper faces the same long axis; antiphase tilting crosses them. In the in-phase cage both cations lie along the common long axis (Figure 5d; $\langle(\hat{u} \cdot L)^2\rangle = 0.90$ for the N–N axis of FA and 0.86 for the C–N axis of MA, against 0.5 without preference). In the crossed cage they behave differently. FA, which carries an NH_2 group at each end and spans the cage, can follow only one face: it aligns with the long axis of the far face (0.80) and lies across that of the near face (0.17), so one end is always squeezed, which costs energy and orientational freedom. MA, with a single NH_3 group, sits close to one face and aligns with it (0.63, against 0.13 for the far face), and turns its C–N axis further out of the plane; this arrangement is lower in energy than in the in-phase cage. The integrand along the path shows the same contrast (Figure 5c): it is positive over almost the whole path for $FASnI_3$ and negative for $MASnI_3$.

Three checks support these results. For $FASnI_3$, the windows above $\lambda = 0.84$ restarted from the configuration equilibrated at $\lambda = 0.84$ change ΔF by less than 0.05 mHa, and the two end states at $\lambda = 1$, joined in one cell, relax to the same cation order, so the path has no trapped hysteresis. The free energies agree with the MD: $MASnI_3$ heated from an in-phase crystal converts to the antiphase tilt at 190 K, and $FASnI_3$ orders in-phase at every rate. The static energies of relaxed cation arrangements give the same signs (Supporting Information).

Effect of process conditions on band-gap deviations

Film processing and residual strain. Perovskite absorbers are deposited from solution onto a substrate that carries the transport layers, crystallised by a thermal anneal and cooled to room temperature.[32,56] The film bonds to the substrate during the anneal. On cooling it would contract with its own thermal expansion, about $5\times10^{-5}\,K^{-1}$ for lead iodide perovskites against 4×10^{-6}–$9\times10^{-6}\,K^{-1}$ for glass and indium tin oxide, but the substrate restricts the in-plane contraction and the film ends under in-plane tension.[57]

Measurements confirm the residual tension. It reaches about 50 MPa in lead halide films[29,30] and grows with the anneal temperature: $CsPbI_2Br$ annealed at 160 °C carries about 150 MPa.[31] Black $CsPbI_3$ cooled on its substrate from 300 to 100 °C acquires an in-plane biaxial strain of about 1 %.[57] The tension matters for stability: the tensile $CsPbI_2Br$ film converts to its non-perovskite phase and segregates its halides at 85 °C, whereas a film held in compression does not.[31] Its effect on the band gap is less clear, because the reported shifts differ between compounds and the unstrained reference state is hard to define.[32] For the tin iodides we are not aware of a strain-resolved band-gap measurement.

Two models of the clamped film. We describe the substrate in two ways and compare both with a free-standing crystal that goes through the same thermal history; every band-gap change below is a difference from this free crystal. The first setup takes the residual stress

from experiment: the film carries an in-plane tension of 50 MPa while its normal relaxes freely. It does not depend on how the stress arose and stands for a film whose bond to the substrate has partly relaxed. The second setup simulates the process: the in-plane cell is fixed at its value at the end of the anneal, as for a film bonded to a rigid substrate, and only the film normal relaxes on cooling. The residual stress then follows from the thermal contraction that the substrate blocks. Because the contraction of the substrate itself is neglected, the second setup gives the largest mismatch strain the process can produce.

Protocol. Every film follows the same history with the r^{2}SCAN potential and D3, in cells of 2000 ($CsSnI_3$) and 2304 atoms (hybrids) with the a and b axes in the film plane, three runs per setup:

1. Anneal: 5 ps at constant volume and 50 ps at the external pressure P_{ext} on all axes, at the anneal temperature (373 K in the device).

2. Substrate: from the end of the anneal, the free crystal keeps P_{ext} on all axes; setup 1 applies $P_{xx} = P_{yy} = P_{\text{ext}} - 0.05$ GPa; setup 2 fixes a and b at their values at the end of the anneal. The normal stays at P_{ext} in all three.

3. Cooling: at 0.25 K/ps to the damp-heat test temperature (358 K, held 50 ps) and on to the operating temperature (300 K, held 150 ps), from which the band-gap frames are drawn by stratified sampling.

Each device temperature is shifted by the difference between the model's and the experimental transition on heating (Table 5; $CsSnI_3$ −39 K, $MASnI_3$ and $FASnI_3$ +15 K), so that the films anneal and operate at the same distance from their transitions as in a device. $CsSnI_3$ then anneals in its β phase and crosses into γ under the substrate constraint, whereas $MASnI_3$ and $FASnI_3$ stay untilted from anneal to operation.

Mechanical state. Both setups leave the films stretched in the plane by 0.2 to 0.4 % and thinner along the normal (Table 7). The rigid clamp builds up an in-plane tension of 63

to 77 MPa, close to the 50 MPa of the first setup and to the stresses measured in films, so the two setups reach similar states by different routes. In $CsSnI_3$ the constraint weakens the in-phase tilt about the normal from 8.5° to 7.8° but keeps the $a^-a^-c^+$ pattern of the free crystal in every clamped run; under the imposed tension one of the three films moved its in-phase axis into the plane. $MASnI_3$ and $FASnI_3$ stay untilted in every run. At these strains the FA cations turn slightly into the film plane, while the MA orientations do not change; the response of the tilt and of the cation orientation over a wider range of strain is given in the Supporting Information.

Band gaps. Table 8 gives the band-gap changes at the operating temperature. In $CsSnI_3$ the imposed tension opens the gap by 20 ± 13 meV and the rigid clamp by 6 ± 12 meV. Straining free frames to the mean cells of the two setups isolates the elastic part, 18 ± 6 and 19 ± 7 meV; it follows the volume, with a deformation potential $a_V = \mathrm{d}E_\mathrm{g}/\mathrm{d}\ln V$ of 3.5 eV, as expected for a valence-band maximum of antibonding Sn 5s–I 5p character.[6] The gap of $MASnI_3$ opens by 29 ± 13 and 40 ± 12 meV, and that of $FASnI_3$ stays within 11 meV of the free crystal (-6 ± 10 and -11 ± 10 meV). The ensemble changes carry the scatter of the gap from frame to frame, 25 to 38 meV, so that differences below about 20 meV are not resolved with the present number of frames.

Table 7: State at the operating temperature of the films under the two substrate setups, relative to the free crystal with the same history (r^{2}SCAN potential with D3, 0.25 K/ps, three runs each). $\Delta\varepsilon_\parallel$, $\Delta\varepsilon_\perp$: in-plane and normal strain (%); $\Delta\sigma_\parallel$: in-plane stress (MPa, tensile positive). Tilt: amplitude about the film normal from the rotation correlation of neighbouring octahedra (deg; + in-phase). Biaxial moduli $Y/(1-\nu)$ from strain ladders at the operating temperature: $CsSnI_3$ 23.1, $MASnI_3$ 20.3, $FASnI_3$ 21.5 GPa.

compound	film	$\Delta\varepsilon_\parallel$	$\Delta\varepsilon_\perp$	$\Delta\sigma_\parallel$	tilt
$CsSnI_3$	free	0	0	0	+8.5
	setup 1 (50 MPa)	+0.40[a]	−0.47[a]	+50	+8.0[a]
	setup 2 (clamp)	+0.29	−0.23	+63	+7.8
$MASnI_3$	free	0	0	0	<1
	setup 1 (50 MPa)	+0.21	−0.14	+53	<1
	setup 2 (clamp)	+0.32	−0.24	+77	<1
$FASnI_3$	free	0	0	0	<1
	setup 1 (50 MPa)	+0.27	−0.17	+47	<1
	setup 2 (clamp)	+0.36	−0.26	+66	<1

[a] One of the three films moved its in-phase axis into the plane; strains: mean of all three films, tilt: mean of the other two.

Table 8: Band-gap change against the free crystal at the operating temperature (meV, mean ± standard error). Ensemble: MD frames of each setup. Elastic: free frames strained affinely to the mean cell of the setup. Values of the earlier PBEsol study (PBEsol force field and gaps; $MASnI_3$ frames at 235 K).

	setup 1 (50 MPa)		setup 2 (clamp)	
compound	ensemble	elastic	ensemble	elastic
$CsSnI_3$	+20 ± 13	+18 ± 6	+6 ± 12	+19 ± 7
$MASnI_3$	+29 ± 13	–	+40 ± 12	–
$FASnI_3$	−6 ± 10	−3 ± 4	−11 ± 10	+1 ± 5

Conclusions

We fine-tuned two classes of machine-learned potentials on PBEsol and r^{2}SCAN labels of $CsSnI_3$, $MASnI_3$ and $FASnI_3$. On held-out structures our in-house UrsaMLFF reaches errors of 9.2×10^{-5} Ha per atom in the energy, 5.2×10^{-4} Ha Bohr^{-1} in the forces and 8.5×10^{-7} Ha Bohr^{-3} in the stress. The fine-tuned MACE models predict forces about twice as accurately but stresses 2.3 to 4.7 times less accurately, and they run 70 to 135 times slower: on one A100 GPU UrsaMLFF performs 960 to 2540 MD steps per second for cells of up to 2300 atoms, so that the production MD of this work takes about 10 h of GPU time instead of 40 to 59 days. With active-learning labels for the molecular defects and close contacts that large-scale MD first reached, UrsaMLFF carries nanosecond MD of thousands of atoms at meta-GGA accuracy, and the MACE models serve to validate it.

The r^{2}SCAN potential reproduces the octahedral tilt patterns of the three compounds. Cooled from the cubic phase, $CsSnI_3$ orders as $a^0a^0c^+$ and then $a^-a^-c^+$, $FASnI_3$ in-phase ($a^0a^0c^+$) and $MASnI_3$ antiphase ($a^0a^0c^-$), with single-tilt angles within 0.3° of the deposited structures and tilt-ordering temperatures of the hybrids within the measured ranges; it also forms the layered low-temperature phase of $FASnI_3$, at 133 K against 125 to 155 K measured. It describes the hybrids better than the PBEsol potential, which underestimates the lattice constants by 2.0 to 2.6 % (against 0.4 to 1.0 %), leaves the antiphase tilt of $MASnI_3$ weak and adds to $FASnI_3$ a second in-phase tilt that its single-crystal structure does not show; PBEsol places the transitions of $CsSnI_3$ closer to experiment. Free-energy calculations at the experimental tilts explain the opposite senses: at 200 K $FASnI_3$ is lower in-phase by (1.09 ± 0.03) mHa and $MASnI_3$ lower antiphase by (0.63 ± 0.01) mHa per formula unit. Antiphase tilting crosses the long axes of the lower and upper faces of the A-site cage. FA, with an amine group at each end, spans the cage and can follow only one face, which costs energy and orientational entropy; MA, with a single ammonium group, sits at one face and is lower in energy in the crossed cage. Consistently, an in-phase $MASnI_3$ crystal converts to the antiphase tilt on heating, at 190 K.

Through the anneal and the cooling on the substrate, the manufacturing process leaves the films under an in-plane tension of 50 to 80 MPa, a strain of 0.2 to 0.4 %, close to the levels measured in films, whether the residual stress is taken from experiment or follows from cooling with the in-plane cell fixed at the anneal. The constraint weakens the tilt of $CsSnI_3$ without changing its pattern. The band gap opens with the expansion, by about 20 meV in $CsSnI_3$ and 30 to 40 meV in $MASnI_3$, whereas that of $FASnI_3$ stays within 11 meV of the free crystal. These shifts are comparable to the scatter of the gap from frame to frame, so resolving them requires the thermal ensemble at the operating point.

Three extensions would bring the model closer to a device. Polycrystalline films contain grains and grain boundaries, which relieve strain, nucleate tilt domains and may pin the tilt variant that forms on cooling; cells of 20 000 atoms already form such domains. An explicit substrate would replace the idealised boundary conditions by the interface and the mismatch of thermal expansion along the actual cooling path. Spin–orbit coupling, strong for iodine and tin, lowers the absolute gaps; the differences reported here cancel it to first order, but its effect on the deformation potentials and on the tilt dependence of the gap remains to be tested. For manufacturers and designers of tin perovskite solar cells, the approach offers a way to estimate, before a film is made, how the choice of cation, the anneal temperature and the substrate shift the operating band gap, and to select compositions and process windows that keep the operating point far from a tilt transition, where the gap is least sensitive to processing.

Data Availability Statement

The DFT labels, the fine-tuned UrsaMLFF and MACE models, the MD input files, the frames sent to DFT with their results, and the analysis scripts are available from the corresponding author upon reasonable request. UrsaDFT-PW and UrsaMLFF are in-house codes of Ursa Simulation.

Acknowledgement

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Supporting Information

Configuration generation and sampling

The training data of the machine-learned force fields are built in three steps: configuration generation, which is described here, down-sampling in descriptor space, and DFT labelling (main text, *Identifying structures for DFT labelling*). Configuration generation produces a deliberately oversampled pool of 55 928 structures of $CsSnI_3$, $MASnI_3$ and $FASnI_3$. The pool has to span, with margin, every state that the production simulations visit: the α, β and γ perovskite phases and the transitions between them, temperatures from the orthorhombic phase to above the cubic transition, the lattice strains produced by thermal expansion, by the spontaneous strain of each transition and by substrate clamping, and disordered orientations of the molecular cations.

Sampling model and molecular-dynamics engine

Sampling model. The structures are sampled with our in-house foundation model. For sampling, the neighbour-list capacities in the header of the model file were enlarged from (76, 40) to (120, 60) neighbours. These numbers size buffers only: energies, forces and stresses are bitwise identical to those of the original file, but compressed low-temperature hybrid cells need more than the original radial capacity. The model only proposes plausible structures. Every structure selected for labelling is recomputed with DFT, so the errors of the sampling model affect where configuration space is sampled but never the labels.

Rejected alternative. We first tested a public general-purpose foundation model[58] and rejected it. In 50 ps constant-volume runs at the experimental volume it collapsed the $CsSnI_3$ framework at 300 K and 500 K and the $MASnI_3$ and $FASnI_3$ frameworks already at 200 K, producing Sn–Sn distances of 2.7 Å and mean Sn–I coordination numbers of 2.9 to 5.4.

Engine. All sampling MD uses the batched GPU engine of UrsaMLFF (commit `4f4f697e1`). It integrates the equations of motion in double precision and accumulates forces in 64-bit fixed point, so that a run is bitwise reproducible for a given input and seed, and it evaluates many independent systems concurrently on one GPU. Two engine defects found during this work are fixed in that commit and are documented and tested in the engine repository. First, trajectory frames written during constant-pressure runs carried the cell of the start of the run segment. Second, in cells thinner than $2.5(r_\mathrm{c} + r_\mathrm{skin})$ the engine uses explicit periodic-image atoms, and a system that outgrew the image region sized at start-up was integrated with missing periodic images until the next output frame; atoms are now wrapped into the cell at every regeneration of the images, and an overflowing system is stopped immediately. All data reported here were generated after both fixes.

Qualification. Before sampling we checked the model in three ways. First, along an isotropic energy–volume scan the pressure obtained from the stress equals $-\mathrm{d}E/\mathrm{d}V$; the zero-pressure pseudo-cubic lattice constants are 6.29 Å for $CsSnI_3$ and about 6.40 Å and 6.49 Å for $MASnI_3$ and $FASnI_3$, 1 to 3 % larger than experiment, as expected for a PBE-based model. Second, in constant-volume MD at the experimental volume the Sn–I framework stays intact from 200–500 K for all three compounds, with a mean Sn–I coordination of at least 5.9 and no Sn–Sn distance below 5 Å; these runs were repeated with the corrected engine. Third, in flexible constant-pressure MD at 0 GPa, $CsSnI_3$ at 500 K and both hybrids at 300 K expand without bound (pseudo-cubic lattice constants above 7.2 Å for the hybrids) and their frameworks decompose (Sn–Sn distances of about 2.7 Å), because the model lacks restoring forces on expansion of these compounds. At 0.5 GPa $MASnI_3$ still decomposes at 400 K. We therefore apply a constant pressure correction of $P^* = +1.0$ GPa to every constant-pressure run. At this pressure the model reproduces the experimental volume with every framework intact: the pseudo-cubic lattice constant is 6.200 Å for $CsSnI_3$ at 500 K (experiment 6.206 Å), and 6.310 Å and 6.381 Å for $MASnI_3$ and $FASnI_3$ at 400 K (experiment 6.240 Å and 6.316 Å

at room temperature). Stage S2 also samples $P^* \pm 0.2\,\mathrm{GPa}$, so that the pool is not tied to a single pressure.

Seed structures

Cell family. The barostat of the engine acts on axis-aligned orthorhombic cells, so all constant-pressure sampling uses a single cell family, the $\sqrt{2}a \times \sqrt{2}a \times 2a$ pseudo-cubic cell of four formula units with the doubled axis along z. This cell hosts the $a^0a^0a^0$ ($Pm\bar{3}m$), $a^0a^0c^+$ ($P4/mbm$), $a^0a^0c^-$ ($I4/mcm$) and $a^-a^-c^+$ ($Pnma$) tilt systems exactly orthogonally, so a heating or cooling run can cross $\alpha \leftrightarrow \beta \leftrightarrow \gamma$ inside one cell. A pseudo-cubic $2 \times 2 \times 2$ cell would not do, because it is monoclinic by 0.33° for $Pnma$. The sampling cells are $2 \times 2 \times 2$ repetitions for $CsSnI_3$ (160 atoms) and $2 \times 2 \times 1$ repetitions for $MASnI_3$ and $FASnI_3$ (192 atoms), both below the limit of 200 atoms set for DFT labelling.

Glazer-tilt frameworks. The three phases of each compound are built by rigid octahedral rotations with the Glazer pattern of the phase[41] and then rescaled axis by axis to the experimental lattice parameters of $CsSnI_3$,[7] $MASnI_3$[4,55] and $FASnI_3$[18] (Table S1). No verified lattice of the low-temperature orthorhombic phase of $MASnI_3$ is available, so for that phase the β lattice contracted by 0.3 to 0.6 % is used. The tilt angles of the seeds are only starting values and are re-equilibrated at temperature before any frame is kept.

Molecular cations. MA^+ and FA^+ are placed as rigid ideal cations with independent, uniformly random orientations on every A site. The orientations are drawn after the supercell is built, so that no period-two pattern is inherited, and placements with H···I distances below 2.35 Å or intermolecular H···H distances below 1.9 Å are rejected. Each hybrid phase is seeded with two independent orientation draws, and one polar-aligned $MASnI_3$ seed with all C→N vectors along $+z$ is added at low weight.

Materials Project seeds. The Materials Project[42] contributes structures that tilting an ideal perovskite cannot produce: the edge-sharing yellow δ phase of $CsSnI_3$, the DFT-relaxed γ phase of $CsSnI_3$ with its A-site displacements and Sn off-centring, cation-ordered $MASnI_3$ (antipolar $Pnma$ and polar) and $FASnI_3$ ($Amm2$) structures, and, at low weight, the decomposition products Cs_2SnI_6, SnI_2 and CsI. Their Materials Project identifiers are listed in the main text.

Substitution-derived seeds. Nine further frameworks are derived from ABX_3 entries of neighbouring chemistries, Pb analogues, bromides and chlorides, by transmutation to Sn and I. Deduplication with a symmetry-tolerant structure matcher reduced 37 candidates to 14 distinct frameworks, of which nine are used. Each is rescaled isotropically to the pseudo-cubic lattice constant of the target compound, with the molecular cations moving as rigid units. Transmutation transfers the topology (connectivity, tilt pattern and cation ordering) but not the geometry, and the tilt amplitudes are reset by equilibration. These seeds add three further antipolar MA orderings, a partially aligned MA ordering (Cc) and the $a^0a^0c^-$ tilt pattern for $CsSnI_3$.

Non-diagonal supercells. A diagonal supercell forces every tilt and cation-orientation pattern to repeat every two pseudo-cubic units. We therefore also build the smallest supercells beyond that period-two repeat, the non-diagonal supercells of a pseudo-cubic $2 \times 2 \times 2$ cell,[59] which at a comparable atom count hold more independent cation orientations and tilt patterns. The framework inside is the same α, β or γ perovskite, so these cells are a choice of periodicity, not a different phase. The supercell matrix $Q = [[0, 1, 1], [1, 0, 1], [1, 1, 0]]$ gives 16 formula units (80 atoms for $CsSnI_3$, 192 for the hybrids) with a minimum width of about $2.3a_{\mathrm{pc}}$, against $2.0a_{\mathrm{pc}}$ for the diagonal 16-formula-unit hybrid cell; $Q = [[\bar{1}, 1, 1], [1, \bar{1}, 1], [1, 1, \bar{1}]]$ gives 32 formula units (160 atoms, $CsSnI_3$ only) with a minimum width of about $2.8a_{\mathrm{pc}}$. Because these cells are not orthogonal, they are sampled at constant volume.

Table S1: Glazer-tilt seed frameworks. Lattice parameters are those of the orthorhombic $\sqrt{2} \times \sqrt{2} \times 2$ setting (Å), with the temperature at which they were measured. Tilt angles are starting values only; they relax during equilibration. The Materials Project IDs of the direct and substitution-derived seeds are listed in Table S2.

	Phase	Glazer	Lattice (Å)	T (K)	Ref.
$CsSnI_3$	α, $Pm\bar{3}m$	$a^0a^0a^0$	$a = 6.206$	≈ 500	[7]
	β, $P4/mbm$	$a^0a^0c^+$	8.712, 6.191	≈ 380	[7]
	γ, $Pnma$	$a^-a^-c^+$	8.688, 8.643, 12.378	300	[7]
$MASnI_3$	α	$a^0a^0a^0$	$a = 6.240$	298	[55]
	β, $I4cm$	$a^0a^0c^-$	8.758, 12.429	200	[4]
	γ	$a^-a^-c^+$	$\beta \times (0.997, 0.994, 0.997)$	–	–
$FASnI_3$	α, $Pm\bar{3}m$	$a^0a^0a^0$	$a = 6.316$	298	[18]
	β, $P4/mbm$	$a^0a^0c^+$	8.862, 6.249	175	[18]
	γ, $Pnma$	$a^-a^-c^+$	8.817, 8.858, 12.416	100	[18]

Sampling stages

Seven complementary sampling stages (S1 to S7), one stage of larger cells (S5′) and one dipole-targeted stage (S8) target different parts of configuration space; Table S4 summarises them.

S1, volume–shape grid (constant volume). Every phase seed is run at 11 affine cell shapes, each at three temperatures spanning the stability range of the phase and its neighbours (Table S3). The shapes are isotropic scalings by 0.970, 0.985, 1.000, 1.015 and 1.030, biaxial in-plane strains of ± 1 and $\pm 2\,\%$ (the substrate-clamping mode) and tetragonal strains of the c axis of $\pm 2\,\%$ (the Poisson response). Holding the cell fixed at known strains supplies the stress labels from which the force field learns the elastic response, on which the clamping study depends, and anchors the sampling at the experimental volume independently of the equation of state of the sampling model.

S2, heating and cooling ramps (constant pressure). Flexible constant-pressure MD at P^* is ramped linearly over 150 ps, from 100 to 600 K for $CsSnI_3$ ($3.3\,\mathrm{K\,ps^{-1}}$) and from 50 to 450 K for the hybrids ($2.7\,\mathrm{K\,ps^{-1}}$). The ramps start from each phase and run in both

Table S2: Materials Project[42] structures used as starting points for the sampling MD, with their space groups. Substitution sources were transmuted to the composition in the last column and rescaled to its pseudo-cubic lattice constant. The structures were retrieved through the Materials Project API on 28 September 2026.

ID	structure	space group	used as
used directly			
mp-568570	$CsSnI_3$, γ phase	$Pnma$	$CsSnI_3$
mp-27381	$CsSnI_3$, yellow (edge-sharing) phase	$Pnma$	$CsSnI_3$
mp-1094059	$MASnI_3$, antipolar MA order	$Pnma$	$MASnI_3$
mp-995238	$MASnI_3$, polar MA order	$P1$	$MASnI_3$
mp-1102654	$FASnI_3$	$Amm2$	$FASnI_3$
mp-27636	Cs_2SnI_6	$Fm\bar{3}m$	reference
mp-27194	SnI_2	$C2/m$	reference
mp-614603	CsI	$Fm\bar{3}m$	reference
substitution sources			
mp-1014168	$CsPbBr_3$	$I4/mcm$	$CsSnI_3$
mp-1120725	$CsPbBr_3$	$Amm2$	$CsSnI_3$
mp-27394	$CsSnCl_3$	$P2_1/c$	$CsSnI_3$
mp-995203	$MAPbI_3$	$Pnma$	$MASnI_3$
mp-995214	$MAPbI_3$	$Pnma$	$MASnI_3$
mp-1194604	$MAPbI_3$	$Pnma$	$MASnI_3$
mp-995227	$MAPbI_3$	Cc	$MASnI_3$
mp-2646983	$FASnBr_3$	Cm	$FASnI_3$
mp-977014	$FAPbI_3$	$Pmm2$	$FASnI_3$

directions, the hybrids are cooled from two orientation draws, and heating ramps at $P^* \pm 0.2\,\mathrm{GPa}$ are added. These ramps generate transition-crossing configurations, both hysteresis branches and the spontaneous transition strain, which constant-volume runs suppress. They are several hundred times faster than the ramps used to locate the transition temperatures (main text, *Phase identification*) and serve for sampling only.

S3, near-T_c boundary conditions (constant pressure). The β and α seeds are run for 100 ps at three temperatures bracketing each transition, 340 K, 380 K and 430 K for $CsSnI_3$, 250 K, 275 K and 300 K for $MASnI_3$ and 200 K, 240 K and 280 K for $FASnI_3$, under the two mechanical conditions of the production study: an in-plane tension of 50 MPa, with pressures of $(P^* - 0.05, P^* - 0.05, P^*)$ GPa, and a rigid in-plane clamp, realised by an effectively infinite in-plane barostat modulus with the normal free at P^*. The response of tilts and cation orientations to clamping is largest near T_c, where a force field trained only on free cells could not be trusted.

S4, robustness. Constant-volume runs of 100 ps beyond the production range, at 700 K for $CsSnI_3$ and 500 K for the hybrids, supply configurations that keep the force field from failing when production MD fluctuates beyond the training range. The polar-aligned $MASnI_3$ seed, run at 150 K and 300 K, tests the stability of aligned MA order.

S5 and S5′, non-diagonal and larger cells. The non-diagonal supercells are run at constant volume at the upper two temperatures of their phase for 100 ps each, which samples tilt and orientation correlations free of period-two constraints. Stage S5′ adds $3\times3\times2$ $CsSnI_3$ cells (360 atoms), a 54-formula-unit non-diagonal supercell of a pseudo-cubic $3 \times 3 \times 3$ cell (270 atoms, α phase only) and $3\times3\times1$ hybrid cells (432 atoms), each run at one temperature per phase for 100 ps, at constant volume and, for the orthorhombic cells, also at constant pressure. These cells hold correlated tilt and orientation domains of about three pseudo-cubic units and are labelled directly.

Table S3: S1 temperatures (K) per phase seed. Each seed is run at 11 cell shapes at each temperature.

	γ	β	α
$CsSnI_3$	150, 300, 400	300, 400, 450	400, 500, 600
$MASnI_3$	50, 100, 150	150, 250, 300	250, 350, 450
$FASnI_3$	50, 100, 150	150, 200, 250	250, 350, 450

S6 and S7, Materials Project and substitution-derived seeds. These seeds are run at constant volume (non-orthogonal cells) or at constant pressure at P^* (orthorhombic cells), at temperatures appropriate to each structure, with heating ramps for the ordered γ-type seeds. They add phases, cation orderings and decomposition products that S1 to S5 cannot reach.

S8, dipole-targeted cation order (constant volume). Seeds with prescribed molecular-dipole arrangements were added after the orientation analysis described below showed that S1 to S7 contain no polar cells. For $MASnI_3$ and $FASnI_3$ in the α and γ frameworks (192 atoms), six arrangements are built: fully polar arrangements ($|\langle\mathbf{d}\rangle| \approx 0.97$) along a pseudo-cubic $\langle 100\rangle$, $\langle 110\rangle$ or $\langle 111\rangle$ direction, partially polar arrangements ($|\langle\mathbf{d}\rangle| = 0.60 \pm 0.03$) along $\langle 100\rangle$ or $\langle 111\rangle$ drawn from a von Mises–Fisher distribution, and an antipolar arrangement with dipoles alternating between layers. Only the rotation of each cation about its dipole axis is random. Because the sampling model relaxes polar order within picoseconds above 100 K, no equilibration is discarded: after a fixed-cell relaxation, 8 ps of constant-volume MD at 50 K and 150 K (γ) or 150 K and 300 K (α) are sampled every 0.1 ps. The runs thus cover the relaxation from the prescribed order to disorder and every net polarisation in between; at 50 K the polar order largely survives ($|\langle\mathbf{d}\rangle| \approx 0.7$ after 8 ps), while at 150 to 300 K it decays to 0.2 to 0.3.

Integration, thermostat, barostat and sampling frequency

Starting sequence. Every run starts with a fixed-cell FIRE[60] relaxation to a maximum force of 9.7×10^{-4} Ha Bohr^{-1} (at most 1000 steps), after which velocities are drawn at the starting temperature and 5 ps of constant-volume MD with stochastic velocity rescaling[50] ($\tau_T = 100$ fs) precede the production segment. Frames from these steps are discarded, except in the S2 ramps, which start from the equilibrated state.

Time step. The time step is 1.0 fs for $CsSnI_3$ and 0.5 fs for the hybrids, because of the N–H and C–H stretches. We checked both choices in 5 ps constant-energy runs from equilibrated states. For $CsSnI_3$ at 500 K the total energy is conserved to better than 0.04 ţHa per atom up to a time step of 2 fs, with no measurable drift. For $MASnI_3$ at 350 K the energy fluctuation grows from 0.22 to 0.92 ţHa per atom between time steps of 0.25 and 0.5 fs, a factor of 3.8 close to the Δt^2 scaling of a stable integration, and the drift at 0.5 fs is 0.55 ţHa per atom and picosecond. At 1 fs the fluctuation grows a further 6.5-fold, to 5.9 ţHa per atom, and the drift eighteen-fold, to 9.9 ţHa per atom and picosecond, so that time step is too long for the hybrids.

Barostat. Constant pressure uses stochastic cell rescaling[51] with $\tau_P = 1$ ps, acting on each axis, with barostat moduli of 16, 14 and 12 GPa for $CsSnI_3$, $MASnI_3$ and $FASnI_3$. The moduli of the three axes differ in their third decimal, because with equal pressures and equal moduli the engine performs shape-preserving isotropic scaling, which would suppress the spontaneous transition strain.

Run length. Production segments last 100 ps and the S2 ramps 150 ps. At 250 to 450 K the reorientation of MA^+ and FA^+ takes a few picoseconds, so a segment spans tens of reorientation times. At 50 to 150 K the orientations are essentially frozen on this time scale, and the diversity there comes from the independent orientation draws and the Materials Project orderings rather than from run length. Octahedral tilt fluctuations decorrelate on

the picosecond scale.

Sampling frequency and frame budget. Stages S1 to S7 are budgeted at 5×10^4 frames for the subsequent down-sampling, and the dipole-targeted stage S8 (3936 frames) is added on top. The budget is split between the compounds in proportion to their planned labelling budgets (70:87:87 for $CsSnI_3$:$MASnI_3$:$FASnI_3$), giving about 1.5×10^4, 1.75×10^4 and 1.75×10^4 frames. At a fixed segment length of 100 ps this sets the sampling interval to 1 ps for $CsSnI_3$ (every 1000 steps), 0.8 ps for the hybrids (every 1600 steps) and 3.2 ps for the large S5′ hybrid cells, which contain more independent local environments per frame. Consecutive frames remain correlated on these intervals by design: the pool is oversampled so that the down-sampling can select decorrelated, diverse frames. Each frame stores the positions, the cell and the energy, forces and stress of the sampling model, which are used only for descriptors and diagnostics.

Throughput and reproducibility. All jobs share one run seed (2026), so the engine evaluates jobs with identical protocols as a single batch of independent systems; batch members are always distinct structures or temperatures, so their trajectories differ. The remaining protocol groups are divided into shards that run concurrently on one GPU under the CUDA Multi-Process Service, which gives a 2.7-fold speed-up for 192-atom systems; the trajectories are bitwise identical to those of sequential execution. The S1 to S7 campaign of 449 jobs ran as 12 concurrent shards on a single NVIDIA A100 in 13 min: the 156 jobs of $CsSnI_3$ and its reference compounds in four shards, finished after 8 min, and the 293 hybrid jobs in eight shards, finished after 13 min. All hybrid jobs and 154 of the 156 $CsSnI_3$ jobs completed; the two exceptions are constant-volume runs of Cs_2SnI_6 at 300 K and 500 K, which the temperature guard of the engine stopped, because that compound lies outside the training domain of the sampling model. S8 was added afterwards as a separate campaign of 48 jobs in three shards (10 s), so the S1 to S7 inputs are unchanged by it. The whole pool is regenerated bit for bit from the deposited job files and seed structures.

Table S4: Sampling stages. Ensembles, run lengths and frame counts after quality control and the PAW-overlap cut. S6 also contributes 215 frames of the reference compounds Cs_2SnI_6, SnI_2 and CsI, included in the total.

Stage	Ensemble	Seeds	Length	$CsSnI_3$	$MASnI_3$	$FASnI_3$
S1	NVT, 11 shapes × 3 T	Glazer α, β, γ	100 ps	9817	12 298	12 277
S2	NPT ramps at P^* (±0.2 GPa)	Glazer	150 ps	1248	2140	2144
S3	NPT, −50 MPa / clamp, near T_c	β, α	100 ps	1200	1500	1500
S4	NVT hot; polar MA	α	100 ps	100	1051	245
S5	NVT, non-diagonal	Glazer	100 ps	1200	750	750
S5′	NVT + NPT, 270–432 atoms	Glazer	100 ps	200	186	186
S6	NVT/NPT	Materials Project	100 ps	310	875	250
S7	NVT/NPT	substitution	100 ps	300	1000	250
S8	NVT, from $t = 0$	dipole-targeted	8 ps	–	1968	1968
Total				14 375	21 768	19 570

Quality control and PAW-overlap filter

Integrity checks. Every frame is checked for framework and molecular integrity. A frame is rejected if its mean Sn–I coordination (within 3.8 Å) is below 5.5, if any Sn–Sn distance is 4.5 Å or less, if any pair of Cs, Sn or I atoms is closer than 2.5 Å, if a C–N bond is broken (one N per C for MA and two for FA within 1.7 Å), if an H atom is detached, or if an H$\cdots$I distance is below 2.0 Å. Frames failing these checks are not used as ordinary training data; the first three per run are retained separately as candidate labels at the onset of failure, where the sampling model is least reliable. In the pool, 1045 frames failed these checks.

PAW-overlap cut. Because the labels are computed with the projector augmented-wave method,[36] we also cut on the overlap of the augmentation spheres. For each atom pair we use the radii of the JTH datasets[37] and the overlap measure of our DFT code,

$$O_{ij} = \frac{r_i + r_j - d_{ij}}{r_i + r_j}, \tag{2}$$

where r_i is the PAW radius of atom i and d_{ij} the pair distance, and remove frames with $\max_{ij} O_{ij} \geq 30\,\%$. In the hybrid frames the largest overlap is set by the C–H bonds, with a median of 20.6 %, a 99.9th percentile of 25.5 % and a maximum of 27.96 %; $CsSnI_3$ frames

have no overlap. Because the integrity checks already remove the compressed configurations, the cut removes no frame from this pool and is kept as a safeguard for later generations. The released pool therefore contains 55 928 frames with or without the cut, and the 70 failure-onset frames are provided separately.

Down-sampling and labelled set

Down-sampling. Each atom of every pool structure was described by Gaussian multipole (GMP) descriptors[33,34] (orders 0 to 5, ten Gaussian widths from 0.2 to 2.0 Å, scaled to the unit interval). Environments closer than a cutoff distance in this space were treated as equivalent, which leaves 3816 distinct environments. A greedy set cover, weighted by cell volume as a proxy for the DFT cost, then selected the cheapest set of structures that together contain all of them; the cutoff was set so that the cover holds 400 to 500 structures, and it selected 403. One further structure was added from each sampling stratum (a combination of stage, compound, phase and cell deformation) that the cover had missed, the one farthest in descriptor space from the structures already chosen (116 structures). A second round added 207 structures of at most 200 atoms: 70 configurations from the onset of failures of the sampling model and 137 random structures, at most two per trajectory. Of these 726 structures, 718 were labelled (main text, section *Electronic structure*): 320 of $MASnI_3$, 247 of $FASnI_3$, 136 of $CsSnI_3$ and 15 of reference compounds, with 36 to 432 atoms. After the exclusion of one duplicate and two Cs_2SnI_6 structures with unphysical Cs–I contacts, the PBEsol set used for training contains 715 structures. The r^{2}SCAN set excludes the same three structures and ten $CsSnI_3$ cells whose r^{2}SCAN energies were affected by a numerical error in the meta-GGA implementation, since corrected; it contains 705 structures.

Active-learning labels

The PBEsol labels were extended in three rounds of active learning by 212 structures (main text, Table 2): cells of 192 atoms cut around the defect events of the hybrid MD, with their

surroundings re-equilibrated and the defect core fixed, images in which one N–H or C–H bond is stretched to between 1.03 and 1.6 Å, and 27 frames of earlier out-of-distribution MD. The r^{2}SCAN labels were computed for 185 of these structures and for 20 of the out-of-distribution frames. A further 120 r^{2}SCAN structures address the remaining failures of the r^{2}SCAN MD: scans of the Cs–Sn distance of a Cs ion entering a neighbouring cage (22 cells of 60 atoms), frames of $CsSnI_3$ (60 atoms), $MASnI_3$ and $FASnI_3$ (96 atoms) from MD at the highest temperatures (51), and scans of short C–I and H–I contacts in $FASnI_3$ (31) and of C–I and N–I contacts in $MASnI_3$ (16), in cells of 96 atoms. Three $FASnI_3$ scan points whose largest force exceeds 0.4 Ha Bohr^{-1} are not used, which leaves 117.

Molecular-cation orientation and dipole statistics

Because the response of the molecular cations to strain is central to this work, we analysed the orientations of all 668 848 MA^+ and FA^+ cations in the 41 338 hybrid frames of the pool. The dipole axis of MA^+ is the unit vector from C to N, and that of FA^+ the unit vector from the midpoint of the two N atoms to C, which lies along the C–H bond and the molecular dipole. Orientations are expressed in the pseudo-cubic frame of each frame, found as the rotation that best aligns the Sn–Sn nearest-neighbour vectors with the Cartesian axes, so that they are comparable across cell shapes, including the non-diagonal supercells.

Single-cation orientations. Figure S1 shows the orientation density relative to an isotropic distribution. Above 150 K both cations sample the sphere almost uniformly: 99.7 % of equal-area bins of the cubic fundamental zone are visited with at least 10 % of the isotropic density. At and below 150 K the distributions become anisotropic, as expected for orientationally ordered phases: MA^+ avoids the $\langle 111 \rangle$ directions (coverage 84 % in S1 to S7 and 86 % with S8), and FA^+ prefers directions near $\langle 110 \rangle$ (95.5 % and 97.5 %).

Collective order. In stages S1 to S7 the net polarisation $|\langle \mathbf{d} \rangle|$ of a frame, averaged over its cations, follows the distribution of randomly oriented dipoles at every temperature and

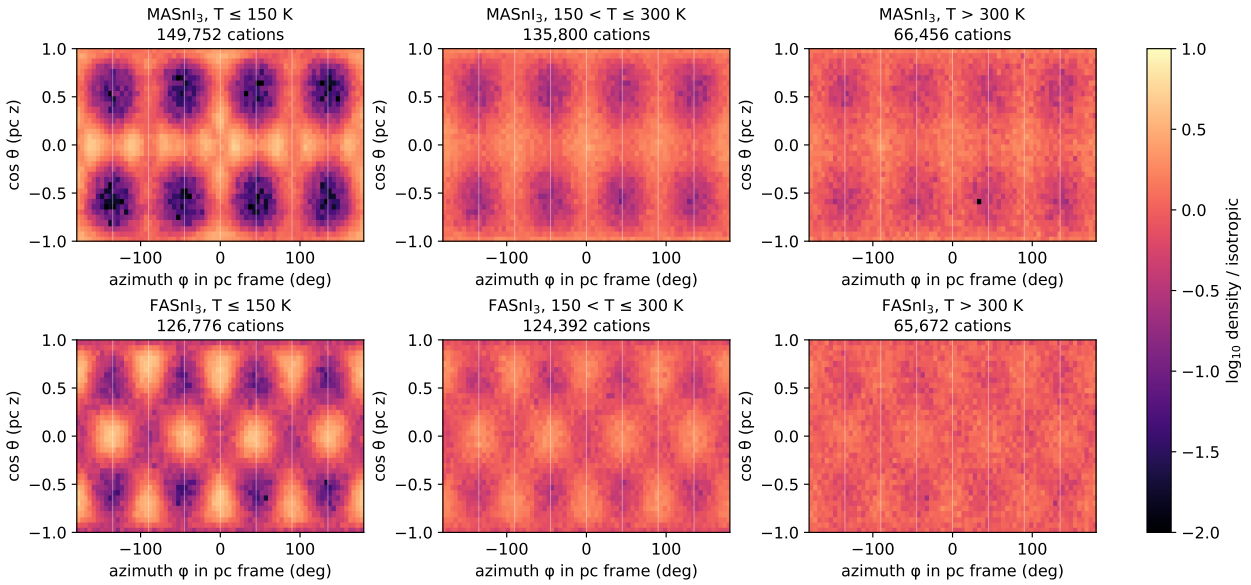


Figure S1: Orientation density of the molecular dipole axis (MA^+: C→N; FA^+: N–N midpoint →C) in the pseudo-cubic frame, relative to an isotropic distribution (logarithmic colour scale), for three temperature bands. Horizontal axis: azimuth; vertical axis: cosine of the polar angle, so equal areas are equal solid angles.

never exceeds 0.64. Polar configurations are absent even though polar seeds were included, because under the sampling model the fully polar $MASnI_3$ seed ($|\langle \mathbf{d} \rangle| = 0.97$) depolarises to 0.45 within 2 ps at 150 K, inside the discarded equilibration. This finding motivated stage S8. With S8 the pool contains 654 frames with $|\langle \mathbf{d} \rangle| > 0.6$ and 145 with $|\langle \mathbf{d} \rangle| > 0.8$, up to 0.98, spread over all intermediate values (Figure S2). Nearest-neighbour pairs of cations favour antiparallel alignment at low temperature and are nearly uncorrelated above 150 K (Figure S3).

DFT settings and convergence

All labels are computed with the PAW datasets of the JTH table,[37] using the 14-electron Sn dataset with the 4d states in the valence. The PAW radii are 0.527 Å (H), 0.798 Å (C), 0.638 Å (N), 1.329 Å (Sn), 1.217 Å (I) and 1.168 Å (Cs). The plane-wave cutoffs of 25 Ha for the wave functions and 250 Ha for the density were chosen in a convergence study on one cell of each compound (80 atoms of $CsSnI_3$, 192 atoms of $MASnI_3$ and of $FASnI_3$) against

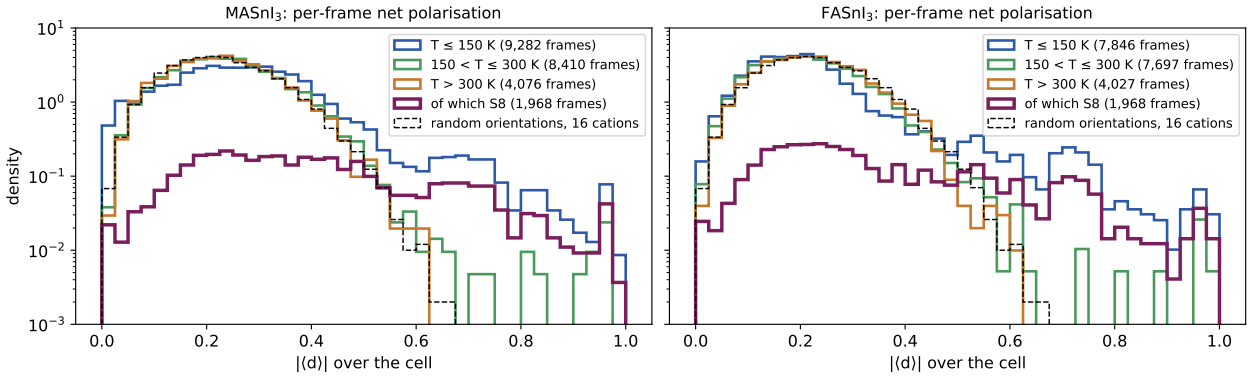


Figure S2: Per-frame net polarisation $|\langle \mathbf{d} \rangle|$ of the molecular cations, for three temperature bands (all stages), with the contribution of the dipole-targeted stage S8 (thick line) and randomly oriented dipoles in a 16-cation cell (dashed). Logarithmic density scale.

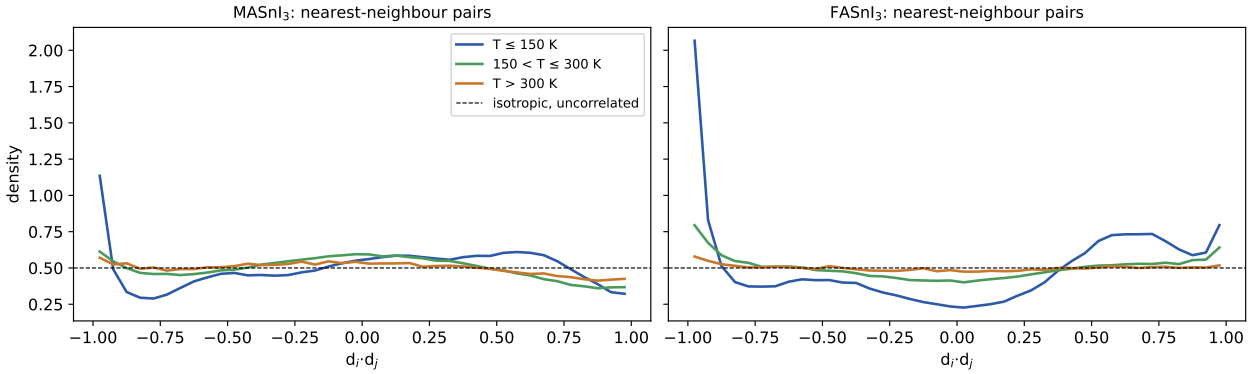


Figure S3: Distribution of the scalar product $\mathbf{d}_i \cdot \mathbf{d}_j$ of the dipole axes of nearest-neighbour cation pairs in the pool, for $MASnI_3$ (left) and $FASnI_3$ (right) and three temperature bands. The dashed line is the uniform density of isotropic, uncorrelated pairs. At and below 150 K antiparallel pairs ($\mathbf{d}_i \cdot \mathbf{d}_j \approx -1$) are enriched; above it the distribution is close to uniform.

a reference at 60 Ha and 300 Ha (Table S5). The targets were 2×10^{-4} Ha per atom for the energy, 2×10^{-4} Ha Bohr^{-1} for the root-mean-square force error and 5×10^{-6} Ha Bohr^{-3} for the largest stress component. The total energy carries a nearly constant offset per Sn atom (2.2 mHa at 25 Ha), which cancels in all energy differences, so the offset-corrected energy error is the relevant one; with it, all three compounds meet every target. The convergence study was made with PBEsol, and the same cutoffs are used for r^{2}SCAN.

The calculations use UrsaDFT (version 0.1.0) with plane waves. The Brillouin zone is sampled on Γ-centred meshes with a spacing of 0.2 Å^{-1} ($2\times2\times2$ for the 80-atom and $2\times2\times3$ for the 192-atom cells), and the occupations follow a Fermi–Dirac distribution at 500 K. The

Table S5: Errors of the label settings (25 Ha / 250 Ha) against 60 Ha / 300 Ha.

compound	ΔE total (Ha/atom)	ΔE offset-corrected (Ha/atom)	ΔF RMS ($\mathrm{Ha\,Bohr^{-1}}$)	ΔF max ($\mathrm{Ha\,Bohr^{-1}}$)	max $\Delta\sigma$ ($\mathrm{Ha\,Bohr^{-3}}$)
$CsSnI_3$	5.7×10^{-4}	-2.7×10^{-6}	2.6×10^{-6}	1.2×10^{-5}	2.1×10^{-6}
$MASnI_3$	2.7×10^{-4}	2.3×10^{-5}	5.4×10^{-5}	2.0×10^{-4}	2.2×10^{-6}
$FASnI_3$	3.4×10^{-4}	3.7×10^{-5}	7.6×10^{-5}	3.6×10^{-4}	2.8×10^{-6}

self-consistent cycle is converged to a density residual (L2 norm) of 10^{-5}; tightening it to 10^{-7} changes energies by 10^{-11} Ha per atom, forces by $2 \times 10^{-6}\,\mathrm{Ha\,Bohr^{-1}}$ and stresses by less than $3 \times 10^{-9}\,\mathrm{Ha\,Bohr^{-3}}$. All calculations are in double precision. The initial PBEsol labels were computed for 718 structures of 36 to 432 atoms, most of them cells of 160 or 192 atoms, and the initial r^{2}SCAN labels for the same structures and three further 360-atom cells of $CsSnI_3$. These calculations ran on the AMD Instinct MI250X GPUs of the Frontier supercomputer at the Oak Ridge Leadership Computing Facility (OLCF); the first 512 PBEsol structures took on average 16.9 self-consistent iterations and about 114 node-hours in total.

Spin–orbit coupling. No calculation in this work includes spin–orbit coupling (SOC). In tin halide perovskites SOC lowers the conduction-band minimum and reduces the gap by several tenths of an eV,[61] so absolute gaps without SOC are not meaningful and none is reported as a result. We report only gap differences between matched states of the same compound, ΔE_g. This rests on the assumption that the SOC correction to the gap is the same in the two states being compared. The states compared here differ by strains below 1 %, by tilt angles of a few degrees and by cation orientations, and we expect the SOC shift to change much less than the gap itself across such states; this assumption is not tested here.

Force-field training and validation

Labels and data split. The energy labels are cohesive energies: the DFT free energy minus the free energies of the isolated atoms computed with the same settings. Forces and

the full stress tensor are fitted as well. The labelled structures are split by MD trajectory, so that all frames of one trajectory fall into the same subset, and stratified by compound, with a fixed seed: 555 structures from 228 trajectories for training, 60 from 30 trajectories for validation and 100 from 44 trajectories held out. By compound, $CsSnI_3$, $MASnI_3$, $FASnI_3$ and the reference compounds contribute 105, 247, 190 and 13 training structures, 12, 27, 21 and 0 validation structures, and 19, 46, 35 and 0 held-out structures. The r^{2}SCAN potentials use the same split without the excluded structures (546, 60 and 99). All six potentials are trained on the training set, the validation set monitors the training, and the held-out set is used for the assessment below.

MACE. Both MACE models have two message-passing layers with correlation order 3 and angular resolution $l = 3$, a radial cutoff of 6 Å (receptive field 12 Å), and 512 (MACE-MH-1) or 128 (MACE-OMAT) channels per irreducible representation. MACE-MH-1 is a multi-head model; its fine-tuning starts from the head trained on PBE data of the OMat set for the PBEsol labels and from the head trained on r^{2}SCAN data of the MatPES set for the r^{2}SCAN labels. Both models were fine-tuned for 200 epochs in double precision with the AMSGrad variant of Adam, a learning rate of 10^{-3}, batches of four structures, an exponential moving average of the weights (decay 0.99) and loss weights of 1, 100 and 10 for energies, forces and stresses. The reference atomic energies are set to zero because the labels are cohesive energies.

UrsaMLFF. UrsaMLFF describes each atom by radial and angular descriptors with cutoffs of 8 Å and 6 Å, built from Chebyshev radial basis sets of size 16 and 12 with 8 and 6 radial functions, and angular terms up to $l = 4$ for three-body and $l = 2$ and 1 for four- and five-body correlations; a Ziegler–Biersack–Littmark repulsion is added at short range. One hidden layer of 120 neurons per element maps the descriptors to atomic energies. The weights were initialised from our in-house general-purpose model of the same architecture, trained on 87 elements of a public 89-element data set[58] and reduced to H, C, N, Sn, I and

Cs, and fine-tuned with Adam in batches of 64 structures for 10 000 epochs. For the first 1500 epochs the learning rate is 5×10^{-3} and the loss weights of energy, forces and virial are 1, 100 and 1; thereafter the learning rate is 5×10^{-4} and the weights are 100, 50 and 30. The PBEsol potential was chosen among five candidates (this one, a fine-tuned public foundation model and three potentials trained from random initialisation) by the force error on the held-out set, which makes its held-out errors slightly optimistic; its validation errors (1.11×10^{-4} Ha per atom, 1.03×10^{-3} Ha Bohr^{-1} and 1.79×10^{-6} Ha Bohr^{-3}) are of the same size. The r^{2}SCAN potential was trained with the identical protocol. Training takes about 1.2 h on one A100 GPU.

Accuracy on held-out structures. Tables 3 (main text, r^{2}SCAN) and S6 (PBEsol) and Figures 2 and S4 compare the six potentials on the held-out structures; for UrsaMLFF these are the potentials retrained as described below. All reproduce the DFT forces with $R^2 \geq 0.99$ and the energies within a compound with $R^2 \geq 0.98$. The MACE models reproduce the forces 2.4 to 2.7 times more accurately than UrsaMLFF with PBEsol labels and 1.9 to 2.1 times more accurately with r^{2}SCAN labels, while UrsaMLFF reproduces the stresses better, by factors of 2.6 to 2.7 with PBEsol and 2.3 to 4.7 with r^{2}SCAN labels; the stress error of the MACE models is dominated by a constant pressure offset. Every model fits the r^{2}SCAN labels slightly less accurately than the PBEsol labels, except for the stresses of MACE-OMATand UrsaMLFF; the PBEsol UrsaMLFF is in addition trained on the refined data, which have no r^{2}SCAN counterpart (below).

Refinement of the training data. MD of the hybrids with the production protocol revealed molecular dissociation: protons detached from their cation, broken C–N bonds and two cations in one cage, in configurations whose energies UrsaMLFF placed 1 to 2×10^{-3} Ha per atom below DFT (Table S10). We therefore added DFT labels of such configurations in three rounds of active learning. Each round cut 192-atom cells around the defect events of the hybrid MD, re-equilibrated their surroundings with the defect core fixed, and added

Table S6: Errors of the potentials trained on the PBEsol labels on the 100 held-out structures: root-mean-square errors of the energy per atom, the force components and the stress components, and coefficients of determination R^2 (for the energy after subtracting the mean DFT energy of each compound). UrsaMLFF: potential retrained on the refined training data; MACE: fine-tuned on the 555 initial structures.

model	E (Ha/atom)	F (Ha Bohr^{-1})	σ (Ha Bohr^{-3})	R^2_E	R^2_F	R^2_σ
UrsaMLFF	6.23×10^{-5}	5.15×10^{-4}	9.11×10^{-7}	0.9940	0.9986	0.9959
MACE-MH-1	3.29×10^{-5}	1.94×10^{-4}	2.46×10^{-6}	0.9983	0.9998	0.9705
MACE-OMAT	7.99×10^{-5}	2.15×10^{-4}	2.37×10^{-6}	0.9901	0.9998	0.9727

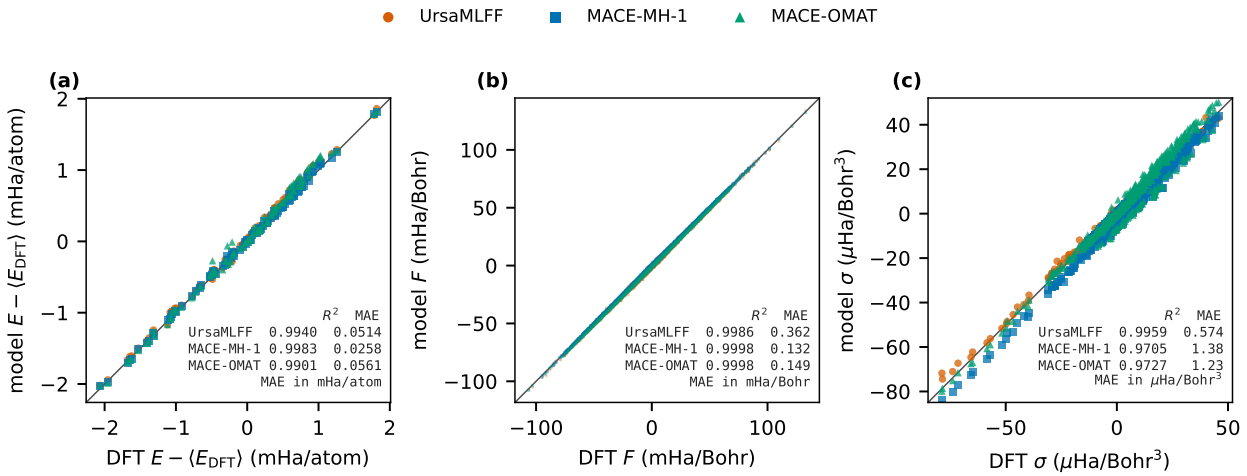


Figure S4: Predicted against DFT values on the 100 held-out structures for the potentials trained on the PBEsol labels; panels and insets as in Figure 2. UrsaMLFF is the potential retrained on the refined training data.

images in which one N–H or C–H bond is stretched to between 1.03 and 1.6 Å. Together with 27 frames of earlier out-of-distribution MD, 212 new structures were labelled with the settings above; 210 join the training set and two the held-out set, which then hold 765 and 102 structures (60 for validation). UrsaMLFF was retrained on the refined data, with one equivariant message-passing step added to its descriptors (radial and angular cutoffs of 6 and 5 Å, one hidden layer of 64 neurons, 49 391 parameters), from random initialisation for 1000 epochs, and the retrained candidate kept is the one whose MD of the hybrids remained free of molecular defects. Its errors on the training, validation and held-out sets are 4.7×10^{-5}, 6.4×10^{-5} and 6.2×10^{-5} Ha per atom for energies and 5.0×10^{-4}, 4.6×10^{-4} and 5.3×10^{-4} Ha Bohr^{-1} for forces, with stress errors of 7.7×10^{-7} and 1.2×10^{-6} Ha Bohr^{-3} on the validation and held-out sets; on the 100 held-out structures shared with the other potentials (Table 3, Figure 2) they are 6.2×10^{-5} Ha per atom, 5.2×10^{-4} Ha Bohr^{-1} and 9.1×10^{-7} Ha Bohr^{-3}. The final r^{2}SCAN potential was trained in the same way, for 3000 epochs, on the r^{2}SCAN training set of 866 structures, which holds the 546 initial structures, 203 active-learning structures and the 117 further structures described in section ; of three seeds, the one with the lower validation force error was kept. Its errors on the validation set and on the 99 held-out structures shared with the MACE models are 8.5×10^{-5} and 9.2×10^{-5} Ha per atom for energies, 4.6×10^{-4} and 5.2×10^{-4} Ha Bohr^{-1} for forces and 8.0×10^{-7} and 8.5×10^{-7} Ha Bohr^{-3} for stresses. The transition temperatures of $MASnI_3$ and $FASnI_3$, the clamping study of the hybrids and the comparison of exchange–correlation functionals use the retrained potentials; the production runs of $CsSnI_3$ and the cell-size study with r^{2}SCAN labels in the Supporting Information use the potentials trained before the refinement.

Table S7 lists the architectures and training settings of the two classes of potentials described in the main text. Table S8 breaks the held-out errors down by compound. For UrsaMLFF (the retrained potentials) the energy and stress errors are largest for $FASnI_3$; the MACE models have their largest stress errors mostly for $CsSnI_3$, in line with the constant

pressure offset noted in the main text. The force errors of each potential vary by less than half between the compounds.

The production UrsaMLFF was chosen among five candidates of the same architecture that differ in their initial weights (Table S9): fine-tuning from our in-house general-purpose model, fine-tuning from a public foundation model, and training from random initialisation with three seeds. The candidate fine-tuned from the in-house model has the lowest force error on the held-out set and was selected on that basis, which makes its held-out errors slightly optimistic. The candidates trained from random initialisation have force errors of 1.30 to 1.33×10^{-3} Ha Bohr^{-1}, against 1.17×10^{-3} Ha Bohr^{-1} for the selected potential, and energy and stress errors of similar size. Both MACE models reproduce the forces at least five times more accurately than any UrsaMLFF candidate.

Table S10 gives the test of whether MD with each potential stays within the training data. On frames from MACE runs, and on all $CsSnI_3$ frames, the force disagreement between UrsaMLFF and MACE-MH-1 is close to its value on the held-out set. On frames from the UrsaMLFF runs of the hybrids it is 1.5 to 1.6 times larger for $MASnI_3$ and 2.9 to 3.3 times larger for $FASnI_3$, and the energy of UrsaMLFF lies 0.5 to 1.6×10^{-3} Ha per atom below that of MACE-MH-1. The DFT calculations on such frames reported in the main text confirm that these configurations lie outside the training data of UrsaMLFF.

Throughput. Accuracy is only one requirement; the production MD of this work, about 45 ns in cells of 2000 to 4900 atoms plus the cell-size tests, also needs speed. Figure S5 shows the MD throughput of the three potentials on one A100 GPU, with D3, for cells below 500 atoms, between 500 and 1000 atoms, of about 2000 atoms and of about 10^4 atoms, each measured alone on an otherwise idle GPU; Table S11 lists the values. Up to 2300 atoms, UrsaMLFF runs 960 to 2540 MD steps per second and the MACE models 7 to 25; at 10^4 atoms, UrsaMLFF runs 550 to 1140 steps per second and the MACE models 1.7 to 4.7. UrsaMLFF is thus 70 to 135 times faster than the faster MACE model up to 2300 atoms

Table S7: Architectures and training settings of the fine-tuned potentials. The same settings are used for the PBEsol and the r^{2}SCAN labels.

	UrsaMLFF	MACE-MH-1 / MACE-OMAT
descriptors	radial and angular, cutoffs 8 Å and 6 Å; Chebyshev basis of size 16 and 12 with 8 and 6 radial functions; angular order 4 (three-body), 2 (four-body) and 1 (five-body)	two equivariant message-passing layers, correlation order 3, $l = 3$; radial cutoff 6 Å, receptive field 12 Å
network	one hidden layer of 120 neurons per element; Ziegler–Biersack–Littmark repulsion at short range	512 (MH-1) or 128 (OMAT) channels per irreducible representation
initial weights	in-house general-purpose model (87 elements), reduced to H, C, N, Sn, I, Cs	MACE-MH-1 (head of OMat PBE data for PBEsol, of MatPES r^{2}SCAN data for r^{2}SCAN) or MACE-OMAT-0-medium
optimiser	Adam, batches of 64 structures, 10 000 epochs	AMSGrad, batches of 4 structures, 200 epochs, weight average with decay 0.99
learning rate	5×10^{-3} for epochs 0–1500, then 5×10^{-4}	10^{-3}
loss weights $E/F/\sigma$	1/100/1, then 100/50/30	1/100/10
precision	double	double (training), single (MD)

Table S8: Root-mean-square errors on the held-out structures, by compound, for the six fine-tuned potentials. UrsaMLFF is the retrained potential of Table 3.

labels	model	compound (structures)	E (Ha/atom)	F (Ha Bohr^{-1})	σ (Ha Bohr^{-3})
PBEsol	UrsaMLFF	$CsSnI_3$ (19)	4.35×10^{-5}	5.36×10^{-4}	6.69×10^{-7}
		$MASnI_3$ (46)	4.41×10^{-5}	5.03×10^{-4}	7.36×10^{-7}
		$FASnI_3$ (35)	8.66×10^{-5}	5.21×10^{-4}	1.19×10^{-6}
	MACE-MH-1	$CsSnI_3$ (19)	3.07×10^{-5}	2.60×10^{-4}	3.58×10^{-6}
		$MASnI_3$ (46)	1.83×10^{-5}	1.79×10^{-4}	2.33×10^{-6}
		$FASnI_3$ (35)	4.62×10^{-5}	1.79×10^{-4}	1.78×10^{-6}
	MACE-OMAT	$CsSnI_3$ (19)	8.65×10^{-5}	2.66×10^{-4}	4.00×10^{-6}
		$MASnI_3$ (46)	4.15×10^{-5}	2.02×10^{-4}	1.31×10^{-6}
		$FASnI_3$ (35)	1.09×10^{-4}	2.08×10^{-4}	2.25×10^{-6}
r^{2}SCAN	UrsaMLFF	$CsSnI_3$ (18)	6.25×10^{-5}	6.13×10^{-4}	6.83×10^{-7}
		$MASnI_3$ (46)	8.20×10^{-5}	5.09×10^{-4}	6.94×10^{-7}
		$FASnI_3$ (35)	1.21×10^{-4}	5.15×10^{-4}	7.44×10^{-7}
	MACE-MH-1	$CsSnI_3$ (18)	3.86×10^{-5}	3.00×10^{-4}	3.99×10^{-6}
		$MASnI_3$ (46)	1.68×10^{-5}	2.23×10^{-4}	3.92×10^{-6}
		$FASnI_3$ (35)	6.52×10^{-5}	2.55×10^{-4}	4.14×10^{-6}
	MACE-OMAT	$CsSnI_3$ (18)	5.71×10^{-5}	3.06×10^{-4}	4.14×10^{-6}
		$MASnI_3$ (46)	5.92×10^{-5}	2.44×10^{-4}	1.07×10^{-6}
		$FASnI_3$ (35)	1.44×10^{-4}	2.79×10^{-4}	9.44×10^{-7}

Table S9: Held-out errors (100 structures, PBEsol labels) of the five UrsaMLFF candidates, which share the architecture and differ in their initial weights, and of the two MACE models. $F_{\max}$: largest force-component error.

potential	E (Ha/atom)	F (Ha Bohr^{-1})	σ (Ha Bohr^{-3})	$F_{\max}$ (Ha Bohr^{-1})
UrsaMLFF, fine-tuned from the in-house general-purpose model (production)	7.99×10^{-5}	1.17×10^{-3}	1.74×10^{-6}	2.86×10^{-2}
fine-tuned from the public foundation model (stopped after 5984 epochs)	9.04×10^{-5}	1.32×10^{-3}	2.35×10^{-6}	2.09×10^{-2}
random initialisation, seed 1	8.75×10^{-5}	1.33×10^{-3}	1.93×10^{-6}	3.64×10^{-2}
random initialisation, seed 2	6.77×10^{-5}	1.32×10^{-3}	1.79×10^{-6}	1.94×10^{-2}
random initialisation, seed 3	5.91×10^{-5}	1.30×10^{-3}	1.49×10^{-6}	2.43×10^{-2}
MACE-MH-1, fine-tuned	3.29×10^{-5}	1.94×10^{-4}	2.46×10^{-6}	2.43×10^{-3}
MACE-OMAT, fine-tuned	7.99×10^{-5}	2.15×10^{-4}	2.37×10^{-6}	2.22×10^{-3}

Table S10: Force disagreement between UrsaMLFF and MACE-MH-1 (PBEsol labels, without D3) on frames of the 960- and 864-atom D3 ramps, relative to their disagreement on the held-out set of the same compound. Absolute: ratio of the root-mean-square force differences; relative: the same for the difference divided by the force magnitude, which removes the temperature scaling. ΔE: energy of UrsaMLFF minus that of MACE-MH-1 per atom, relative to its held-out mean.

compound	frames from	T (K)	frames	absolute	relative	ΔE (Ha/atom)
$MASnI_3$	UrsaMLFF runs	50–150	147	1.48	3.18	-6.30×10^{-4}
	UrsaMLFF runs	150–250	150	1.57	2.31	-5.90×10^{-4}
	UrsaMLFF runs	250–400	225	1.59	1.75	-4.80×10^{-4}
	MACE runs	50–150	98	0.52	1.14	-1.90×10^{-5}
	MACE runs	150–250	100	0.65	0.99	8.00×10^{-6}
	MACE runs	250–400	150	0.80	0.91	2.10×10^{-5}
$FASnI_3$	UrsaMLFF runs	50–150	147	2.94	5.18	-1.50×10^{-3}
	UrsaMLFF runs	150–250	150	3.13	3.85	-1.60×10^{-3}
	UrsaMLFF runs	250–400	225	3.30	3.15	-1.60×10^{-3}
	MACE runs	50–150	98	0.66	1.33	-2.70×10^{-6}
	MACE runs	150–250	100	0.78	1.07	-1.90×10^{-6}
	MACE runs	250–400	150	0.92	0.95	-1.10×10^{-6}
$CsSnI_3$	UrsaMLFF runs	200–350	224	0.76	0.90	-1.20×10^{-4}
	UrsaMLFF runs	350–450	150	0.89	0.88	-1.00×10^{-4}
	UrsaMLFF runs	450–600	222	0.99	0.86	-8.80×10^{-5}
	MACE runs	200–350	150	0.73	0.90	-3.40×10^{-5}
	MACE runs	350–450	100	0.86	0.88	-4.60×10^{-6}
	MACE runs	450–600	148	0.97	0.87	2.20×10^{-5}

and about 240 times faster at 10^4 atoms. In the production cells this means 386 ($CsSnI_3$) and 83 ns ($MASnI_3$) per day for UrsaMLFF against 2.9 and 0.84 ns per day for MACE-OMAT. At the measured rates, the production MD of this work takes about 10 h of GPU time with UrsaMLFF, against about 40 days with MACE-OMAT and 59 days with MACE-MH-1. This is why UrsaMLFF carries the production simulations although its force error is two to three times larger; the MACE models serve to validate it (Table 3 and section *Transition temperatures compared with experiment*).

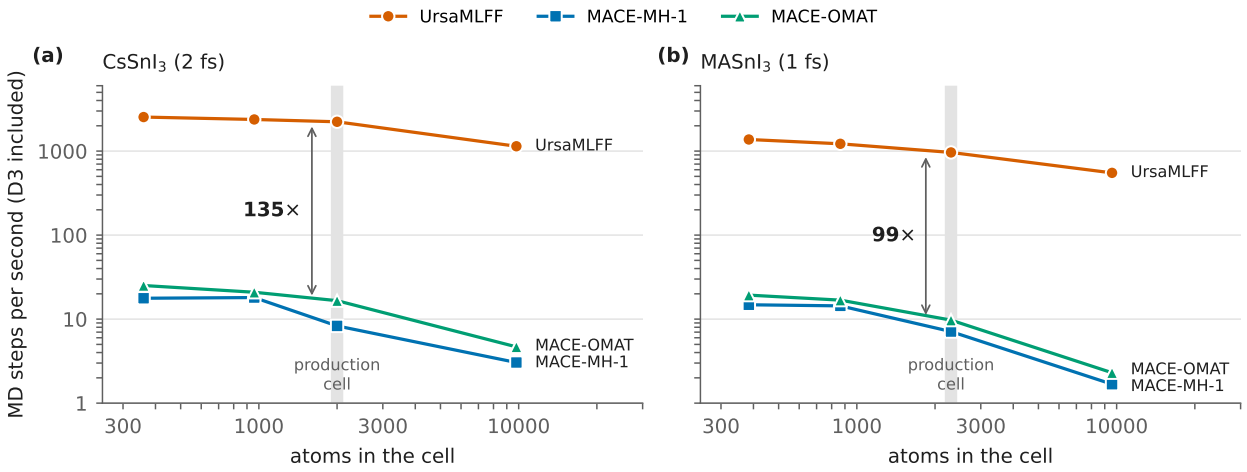


Figure S5: MD throughput of the fine-tuned potentials on one NVIDIA A100 GPU, including D3, against the number of atoms in the cell: (a) $CsSnI_3$, (b) $MASnI_3$. UrsaMLFF runs in the production MD engine in double precision, the MACE models in the MD driver of the model comparison in single precision. Each value was measured alone on an otherwise idle GPU. Grey band: production cell; arrow: speed-up of UrsaMLFF over the faster MACE model in that cell. The values, with the corresponding nanoseconds per day, are listed in Table S11.

Table S11 lists the MD throughput values shown in Figure S5, with the corresponding simulated time per day at the production time steps.

Phase-transition protocol

All heating and cooling ramps of this work, that is, the comparison of the six fine-tuned potentials, the cell-size study and the production cells, follow one protocol, and only the cell

Table S11: MD throughput of the fine-tuned potentials on one NVIDIA A100 GPU, in MD steps per second, including D3. UrsaMLFF: the production MD engine, one system per GPU, double precision. MACE: the MD driver used for the model comparison, single precision. In brackets: nanoseconds per day at the production time steps (2 fs for $CsSnI_3$, 1 fs for the hybrids). Each value was measured alone on an otherwise idle GPU.

$CsSnI_3$ model	360 atoms	960 atoms	2000 atoms	9800 atoms
UrsaMLFF	2541 (4.4e+02)	2379 (4.1e+02)	2232 (3.9e+02)	1141 (2e+02)
MACE-MH-1	18 (3.1)	18 (3.1)	8.3 (1.4)	3 (0.53)
MACE-OMAT	25 (4.3)	21 (3.6)	17 (2.9)	4.7 (0.81)
$MASnI_3$ model	384 atoms	864 atoms	2304 atoms	9600 atoms
UrsaMLFF	1374 (1.2e+02)	1221 (1.1e+02)	964 (83)	550 (48)
MACE-MH-1	15 (1.3)	14 (1.2)	7.1 (0.61)	1.7 (0.15)
MACE-OMAT	19 (1.7)	17 (1.5)	9.7 (0.84)	2.3 (0.2)

size changes. The cells are supercells of the $\sqrt{2} \times \sqrt{2} \times 2$ setting: 960 atoms for $CsSnI_3$ ($4 \times 4 \times 3$) and 864 atoms for $MASnI_3$ and $FASnI_3$ ($3 \times 3 \times 2$) in the model comparison, and 360 to 4900 atoms for $CsSnI_3$ and 384 to 4800 atoms for the hybrids in the cell-size study, which includes the production cells of 2000 ($5 \times 5 \times 4$) and 2304 atoms ($4 \times 4 \times 3$). The time step is 2 fs for $CsSnI_3$ and 1 fs for the hybrids. Each run starts with a FIRE minimisation at fixed cell, followed by 5 ps at constant volume and 20 ps at constant pressure at the start temperature, and continues with a linear temperature ramp at 2 K/ps between 200 K and 600 K for $CsSnI_3$ or between 50 K and 400 K for the hybrids. Heating runs start from the orthorhombic structure and cooling runs from the cubic structure. For the hybrids with the retrained UrsaMLFF, the cell of the orthorhombic starting structure is first set, by an affine scaling, to the mean cell of the model at 50 K in the same cell size (main text, *Starting structure of the hybrid heating runs*). The 18 432-atom runs ($8 \times 8 \times 6$) follow the same protocol at 1 K/ps. The holds that bracket the equilibrium tilt loss of $MASnI_3$ (2304 atoms) last 150 ps at 250, 265 and 280 K; they start either from the crystal prepared in the 50 K cell and heated to the hold temperature at 5 K/ps, or from the heating frame of a production run

closest to the hold temperature, in the cubic phase. Temperature is controlled by stochastic velocity rescaling with a time constant of 100 fs, and pressure by stochastic cell rescaling with independent axes and a time constant of 1 ps.

Dispersion is included with D3 and Becke–Johnson damping, using the parameters of the training functional (PBEsol: $s_6 = 1$, $s_8 = 2.9491$, $a_1 = 0.4466$, $a_2 = 6.1742$ Å; r^{2}SCAN: $s_6 = 1$, $s_8 = 0.78981$, $a_1 = 0.49484$, $a_2 = 5.730\,84$ Å), a pair cutoff of 15 Å and a coordination-number cutoff of 6 Å. UrsaMLFF is run five times per compound, direction and cell size at the external pressure P_{ext} of its functional: -0.065, -0.043 and -0.058 GPa for $CsSnI_3$, $MASnI_3$ and $FASnI_3$ with PBEsol and -0.039, -0.150 and -0.183 GPa with r^{2}SCAN. Each MACE model is run once, at zero pressure. The transition temperatures are determined with the criterion of the main text (*Phase identification and transition temperatures*).

The MACE runs use an MD driver built on ASE[62] that implements the thermostat and barostat of the engine, with the MACE models evaluated in single precision and D3 evaluated on the GPU. We validated the driver against the engine with the PBEsol UrsaMLFF in three runs of 100 ps at 400 K: the temperature and volume fluctuations and the cell shape agree, and the mean volume is 0.16 % lower, equivalent to a pressure difference of less than 20 MPa. The D3 implementation of the driver agrees with that of the engine to better than 4×10^{-9} Ha per atom.

Engine and cells. MD is run with the GPU engine of UrsaMLFF, which evaluates the force field and D3 in double precision (version 0.1.0, development build of commit 52159a3). The barostat compressibility is set from a bulk modulus of 15 GPa, and structures are minimised with FIRE[60] before the first MD stage. Because the cubic, tetragonal and orthorhombic phases are all orthogonal in the $\sqrt{2} \times \sqrt{2} \times 2$ setting, every transition can proceed by rescaling the three axes, at the price that the tetragonal axis is favoured along the cell's c axis. Pseudo-cubic lattice constants are $a/(\sqrt{2}n_1)$, $b/(\sqrt{2}n_2)$ and $c/(2n_3)$ for an $n_1 \times n_2 \times n_3$ supercell.

Heating and cooling ramps. Each ramp starts with a FIRE minimisation at fixed cell, 5 ps of constant-volume MD and 20 ps of constant-pressure MD at the start temperature, followed by a linear temperature ramp at 2 K/ps at constant pressure. $CsSnI_3$ is heated from the γ (*Pnma*) structure from 200 K to 600 K and cooled from the cubic structure from 600 K to 200 K. $MASnI_3$ and $FASnI_3$ are heated from orthorhombic structures and cooled from cubic structures with randomly oriented cations between 50 K and 400 K. The heating runs start from the Materials Project *Pnma* structure of $CsSnI_3$ (mp-568570) at the experimental lattice,[7] the *Pnma* structure of $MASnI_3$ with antipolar MA order (mp-1094059), and an $a^-a^-c^+$ framework of $FASnI_3$ at the experimental lattice[18] with random FA orientations. The cooling runs start from cubic cells with lattice constants of 6.206 Å, 6.2397 Å and 6.316 Å.[7,18,55] The upper ends of the ramps lie 160 K ($CsSnI_3$) and 125 to 150 K (hybrids) above the highest experimental transitions, within the temperature range of the training data (up to 600 K for $CsSnI_3$ and 450 K for the hybrids), and below the temperatures at which the compounds cease to be intact perovskites: $CsSnI_3$ melts at 451 °C (724 K),[7] and in thermogravimetry $FASnI_3$ and $MASnI_3$ begin to lose mass at about 420 K, the main decomposition of $MASnI_3$ setting in near 620 K.[63] Five independent runs with different initial velocities are made for each material and direction. Frames are written every 1 ps.

Cell-size convergence. The same protocol was repeated in cells of 360 to 4900 atoms for $CsSnI_3$ and 384 to 4800 atoms for the hybrids (Supporting Information). Cells with an odd number of repeats in the plane cannot hold an in-phase tilt about an in-plane axis; for $CsSnI_3$ even cells of 3600 and 4320 atoms were therefore added on cooling.

Model and functional comparison. To select the exchange–correlation functional of the training data, UrsaMLFF retrained on PBEsol and on r^{2}SCAN labels was heated five times per compound, with the D3 parameters and the external pressure of its functional, in two cells: 960 and 2000 atoms for $CsSnI_3$ ($4 \times 4 \times 3$, $5 \times 5 \times 4$) and 864 and 2304 atoms for the hybrids ($3 \times 3 \times 2$, $4 \times 4 \times 3$), the hybrids from the potential's own 50 K cell. To validate

UrsaMLFF, the two MACE models were run once with PBEsol and r^{2}SCAN labels in the smaller cells, with an MD driver that implements the same thermostat and barostat as the engine. The protocol is given in the Supporting Information.

Validation of the tilt analysis. We validated the PDynA analysis on structures with imposed tilts: an $a^-a^-c^+$ pattern of 6°/6°/8° returns correlations of −5.9°, −5.9° and +7.9°.

Phase transitions of the free-standing crystals

Table S15 gives the transition temperatures of the UrsaMLFF trained on r^{2}SCAN labels for all cell sizes; the PBEsol values are given in Table S19 (section). As for PBEsol, the heating transition temperatures of the production cells lie close to those of the largest cells, while the cooling values depend on the cell size. Table S17 shows, for $CsSnI_3$ in the 960-atom cell, the effect of D3, of the training functional, of the external pressure and of the model form. Without D3, all three PBEsol potentials place the transitions of $CsSnI_3$ 105 to 178 K below experiment while reproducing the lattice constant to within 0.7 %. D3 contracts the lattice by about 2 % and raises the transitions by 155 to 215 K. Run at the experimental volume, the PBEsol potentials with D3 place the transitions 65 to 100 K below experiment, so most of the effect of D3 on the transition temperatures acts through the volume. Table S18 lists all values of the comparison of the two retrained UrsaMLFF potentials shown in the main text. Table S16 compares the heating transitions of the hybrids, with the retrained UrsaMLFF, for the two starting cells discussed in the main text. Table S12 gives the tilt-ordering temperatures for ramps of 2 to 0.1 K/ps, with the final r^{2}SCAN potential and with PBEsol. Slower ramps raise the cooling onsets of the hybrids: with r^{2}SCAN, $MASnI_3$ orders at 208, 248, 270, 277 and 280 K and $FASnI_3$ at 237, 240, 242, 247 and 250 K at 2, 1, 0.5, 0.25 and 0.1 K/ps. Below 0.5 K/ps the onsets change by at most 7 K, which sets the production rate of 0.25 K/ps. With r^{2}SCAN, every $FASnI_3$ run orders as $a^0a^0c^+$ at every rate, and every $MASnI_3$ run as $a^0a^0c^-$ between 2 and 0.25 K/ps. At 0.1 K/ps the two $MASnI_3$ runs

do not reach a single tilt system: at 200 K one carries an antiphase tilt of 4° about c with in-phase tilts of about 1° about the other axes, the other an in-phase tilt of 3° about a and an antiphase tilt of 2.5° about c. With PBEsol $FASnI_3$ is in-phase at every rate, while $MASnI_3$ orders in only some of the runs. For $CsSnI_3$ with r^{2}SCAN the transition temperatures below 1 K/ps agree within the scatter between runs; at 0.1 K/ps the $\beta \leftrightarrow \alpha$ transition lies at (355 ± 18) K on cooling and (377 ± 3) K on heating and the $\gamma \leftrightarrow \beta$ transition at (310 ± 8) K and (322 ± 4) K, against 370, 385, 317 and 323 K at 0.25 K/ps.

Table S12: Tilt-ordering temperatures (K) against ramp rate, $CsSnI_3$ 2000 atoms and hybrids 2304 atoms, criteria of Table 4: mean ± standard error over the runs with a transition; [a]: not all runs show the transition. Cooling starts in the cubic phase ($CsSnI_3$ at 450 or 500 K, hybrids at 400 K); heating starts from the end state of the cooling run at the same rate, except at 2 K/ps, where the hybrids start from the 50 K template and $CsSnI_3$ from the γ structure. –: not run. Three runs per entry, except $MASnI_3$ r^{2}SCAN cooling at 0.1 K/ps 2; $FASnI_3$ r^{2}SCAN cooling at 0.1 K/ps 2.

compound	transition	potential	ramp	2 K/ps	1 K/ps	0.5 K/ps	0.25 K/ps	0.1 K/ps
$CsSnI_3$	$\beta \leftrightarrow \alpha$	r^{2}SCAN	cooling	335 ± 19	322 ± 6	360 ± 18	370 ± 18	355 ± 18
	$\beta \leftrightarrow \alpha$	r^{2}SCAN	heating	402 ± 8	370 ± 25	392 ± 17	385 ± 3	377 ± 3
	$\beta \leftrightarrow \alpha$	PBEsol	cooling	–	–	413 ± 17	435 ± 13	–
	$\beta \leftrightarrow \alpha$	PBEsol	heating	465 ± 9	–	442 ± 21	465 ± 8	–
	$\gamma \leftrightarrow \beta$	r^{2}SCAN	cooling	287 ± 26	277 ± 7	315 ± 6	317 ± 2	310 ± 8
	$\gamma \leftrightarrow \beta$	r^{2}SCAN	heating	339 ± 14	307 ± 24	318 ± 2	323 ± 2	322 ± 4
	$\gamma \leftrightarrow \beta$	PBEsol	cooling	–	–	372 ± 16	380 ± 26	–
	$\gamma \leftrightarrow \beta$	PBEsol	heating	409 ± 4	–	385 ± 23	387 ± 21	–
$MASnI_3$	tilt order	r^{2}SCAN	cooling	208 ± 26	248 ± 14	270 ± 8	277 ± 7	280[a]
	tilt order	r^{2}SCAN	heating	289 ± 8	–	290 ± 13	290 ± 6	–
	tilt order	PBEsol	cooling	–	190[a]	228 ± 8[a]	230[a]	–
	tilt order	PBEsol	heating	310 ± 13	–	230 ± 5[a]	240[a]	–
	second tilted axis	r^{2}SCAN	heating	207 ± 3	–	none	none	–
	second tilted axis	PBEsol	cooling	–	185[a]	188 ± 28[a]	none	–
	second tilted axis	PBEsol	heating	214 ± 2	–	188 ± 42[a]	none	–
$FASnI_3$	tilt order	r^{2}SCAN	cooling	237 ± 9	240 ± 6	242 ± 2	247 ± 2	250 ± 0
	tilt order	r^{2}SCAN	heating	184 ± 3	–	255 ± 3	255 ± 3	–
	tilt order	PBEsol	cooling	–	255 ± 10	252 ± 7	265 ± 6	–
	tilt order	PBEsol	heating	245 ± 9	–	265 ± 3	272 ± 2	–
	second tilted axis	r^{2}SCAN	heating	none	–	90[a]	none	–
	second tilted axis	PBEsol	cooling	–	83 ± 4	100[a]	80 ± 5[a]	–
	second tilted axis	PBEsol	heating	107[a]	–	145[a]	137 ± 6	–

Cells of about 20 000 atoms. The coolings at 0.25 K/ps were repeated three times in cells of 23 040 atoms for $CsSnI_3$ and 18 432 atoms for the hybrids, 7 to 10 nm across. Table S13

compares them with the production cells. They form the same tilt systems, with tilt angles within about 1° and transition temperatures within 15 K. In one $MASnI_3$ run the antiphase tilt forms domains whose phases are shifted by one layer, so that the net tilt of the cell cancels while each domain keeps the antiphase pattern; the angles of Table S13 are insensitive to such domains. The layered low-temperature tilt of $FASnI_3$ is weak in the larger cells (layer alternation of 0.4 to 1.2°, against about 3° in the production cells), presumably because it also forms domains whose alternation is out of phase and cancels in the layer averages. One $MASnI_3$ run in which two cations briefly shared a cage near 390 K is left out of the averages.

Table S13: Cell size: tilt systems, tilt angles (degrees; model temperature in K in parentheses) and cooling transition temperatures (K) with the r^{2}SCAN potential at 0.25 K/ps in cells of about 2000 (2k: $CsSnI_3$ 2000, hybrids 2304 atoms) and 20 000 atoms (20k: $CsSnI_3$ 23 040, hybrids 18 432 atoms), analysed as in Table 6. T_c: transition into the phase on cooling, mean ± standard error, criteria of Table 4; runs flagged by the integrity scan are left out. [a]: not every run shows the transition; [b]: loss of the layer alternation of the $P4bm$ structure.

compound	phase	tilt system (2k / 20k)	angle 2k	angle 20k	T_c 2k (K)	T_c 20k (K)
$CsSnI_3$	β ($P4/mbm$)	$a^0a^0c^+$ / $a^0b^-c^+$	+6.2 (343)	+7.2 (346)	370 ± 18	372 ± 4
	γ ($Pnma$)	$a^-a^-c^+$	9.9 / 6.7 (266)	9.7 / 6.7 (269)	317 ± 2	320 ± 10
$MASnI_3$	tetragonal ($I4cm$)	$a^0a^0c^-$	−8.5 (202)	−8.1 (187)	277 ± 7	262 ± 2
	tetragonal ($I4/mcm$)	$a^0a^0c^-$	−9.6 (142)	−8.8 (127)		
$FASnI_3$	tetragonal ($P4/mbm$)	$a^0a^0c^+$	+6.4 (209)	+6.4 (204)	247 ± 2	242 ± 2
	low T ($P4bm$)	c^+, layers	9.7 / 6.2 (109)	8.7 / 7.1 (104)	127 ± 13[b]	100[ab]

Low-temperature phase of $MASnI_3$. $MASnI_3$ does not form its low-temperature phase near 110 K on cooling at any rate, while a crystal prepared in the ordered $a^-a^-c^+$ structure keeps its second tilt on heating up to 190 K (0.25 K/ps). To bracket the transition, 2304-atom cells were held for 1.5 ns at 100, 130, 160 and 190 K, starting from the frame at that temperature of a cooling run (tetragonal, $a^0a^0c^-$) and of a heating run from the ordered structure (Table S14). Between 100 and 160 K neither start changes its tilt system: the tetragonal cells gain no second tilt and the ordered cells keep theirs. At 190 K the ordered start had already turned antiphase. Both structures are therefore stable for nanoseconds in this range; the transition requires a cooperative reorientation of the MA cations that does not occur on the simulated time scales, and its temperature in the model lies between 50

and 190 K.

Table S14: Holds of 2304-atom $MASnI_3$ cells for 1.5 ns with the r^{2}SCAN potential: domain-insensitive tilt amplitudes (degrees) about the three pseudo-cubic axes, largest first (+ in-phase, − antiphase), in 50 ps time averages at the start and at the end of the hold.

	from the cooled crystal		from the ordered crystal	
T (K)	start	end	start	end
100	−9.9, −0.3, −0.3	−10.0, −0.5, +0.2	+10.1, −4.0, −4.0	+10.2, −4.0, −4.0
130	−9.8, −0.4, −0.4	−8.2, −0.7, −0.5	+9.8, −3.9, −3.8	+9.9, −4.0, −3.9
160	−9.4, −0.4, −0.2	−9.6, −0.3, −0.3	+9.4, −3.6, −3.6	+9.5, −3.8, −3.7
190	−9.1, −0.2, −0.2	−9.2, −0.3, +0.2	−8.2, −0.5, −0.1	−9.0, −0.3, −0.2

Static energies of the tilt senses

Before the free-energy calculations of the main text, we compared the static energies of the two tilt senses for relaxed cation arrangements. These calculations used the r^{2}SCAN potential before its final retraining and the cubic metric of the PBEsol potential ($FASnI_3$ 6.127 Å, $MASnI_3$ 6.058 Å), 2.7 to 2.9 % below that of the final potential; they give the same signs as the free energies.

The static energy of a single cell does not show which sense a cation prefers, because it depends on how the cations are arranged. In the four-formula-unit cells used for DFT, the antiphase–in-phase difference at 8° scatters by 1 to 3 mHa per formula unit (standard deviation over random arrangements), as much as its mean or more. We therefore computed it with the r^{2}SCAN potential for 20 to 160 random arrangements in cells of 4 to 128 formula units, tilted in-phase or antiphase by 4, 8 or 12° about c at a fixed cubic metric. Each arrangement is placed in both cells and the cations are relaxed with Sn and I fixed; D3, evaluated at the relaxed structures, changes the differences at 8° by less than 0.07 mHa. The scatter falls as the inverse square root of the cell size, and the sign of the mean does not depend on it (Figure S6b). In the cells of 16 to 128 formula units at 8°, $FASnI_3$ is lower in-phase by 0.8 to 1.3 mHa per formula unit and $MASnI_3$ lower antiphase by 1.2 to 2.0 mHa;

Table S15: Transition temperatures (K) against cell size for UrsaMLFF trained on r^{2}SCAN labels, with D3 at the constant pressure P_{ext}, at 2 K/ps: mean ± standard error over the runs with a transition (runs with a transition / runs). Cells are $n_1 \times n_2 \times n_3$ supercells of the $\sqrt{2} \times \sqrt{2} \times 2$ setting; the 3600- and 4320-atom cells (even in-plane repeats) were run on cooling only. Noise: the largest tilt correlation (deg) over the hottest 40 K of the heating runs, the level the transition criterion has to exceed. The PBEsol values are given in the main text.

$CsSnI_3$ atoms (cell)	$\gamma \leftrightarrow \beta$ heating	$\gamma \leftrightarrow \beta$ cooling	$\beta \leftrightarrow \alpha$ heating	$\beta \leftrightarrow \alpha$ cooling	noise
360 (3×3×2)	322 ± 10 (5/5)	269 ± 19 (4/5)	374 ± 11 (5/5)	288 ± 32 (4/5)	1.5
960 (4×4×3)	311 ± 10 (5/5)	308 ± 8 (5/5)	373 ± 12 (5/5)	372 ± 12 (5/5)	1.0
2000 (5×5×4)	308 ± 5 (5/5)	249 ± 5 (4/5)	380 ± 8 (5/5)	302 ± 13 (5/5)	0.7
2880 (6×6×4)	309 ± 3 (5/5)	275 ± 14 (5/5)	396 ± 4 (5/5)	330 ± 10 (5/5)	0.7
3600 (6×6×5)	–	255 ± 9 (5/5)	–	320 ± 9 (5/5)	–
4320 (6×6×6)	–	275 ± 11 (5/5)	–	329 ± 7 (5/5)	–
4900 (7×7×5)	301 ± 3 (5/5)	272 ± 7 (5/5)	397 ± 4 (5/5)	307 ± 7 (5/5)	0.5
$MASnI_3$ atoms (cell)	$\gamma \leftrightarrow \beta$ heating	$\gamma \leftrightarrow \beta$ cooling	$\beta \leftrightarrow \alpha$ heating	$\beta \leftrightarrow \alpha$ cooling	noise
384 (2×2×2)	167 ± 9 (5/5)	none (0/5)	374 ± 4 (5/5)	344 ± 22 (5/5)	2.9
864 (3×3×2)	172 ± 7 (5/5)	none (0/5)	361 ± 14 (5/5)	340 ± 12 (5/5)	3.0
2304 (4×4×3)	172 ± 7 (5/5)	none (0/5)	364 ± 13 (5/5)	323 ± 5 (5/5)	2.5
4800 (5×5×4)	158 ± 2 (5/5)	none (0/5)	367 ± 8 (5/5)	326 ± 4 (5/5)	2.7
$FASnI_3$: loss of any tilt order atoms (cell)	heating	cooling			noise
384 (2×2×2)	104 ± 15 (4/5)	none (0/5)			0.9
864 (3×3×2)	150 ± 15 (4/5)	none (0/5)			0.5
2304 (4×4×3)	140 ± 10 (5/5)	none (0/5)			0.3
4800 (5×5×4)	156 ± 1 (5/5)	none (0/5)			0.2

Table S16: Heating transition temperatures (K) of the hybrids with the retrained UrsaMLFF for the two starting cells: the model's 50 K cell (used in the main text) and the cell of the size study's starting structures, 4–7 % ($MASnI_3$) and 2–3 % ($FASnI_3$) longer, which the barostat compresses during the 50 K equilibration. Mean ± standard error (runs with a transition / runs); one 4800-atom $MASnI_3$ run from the oversized cell is left out for a cage defect formed during its 50 K stage, two were stopped. The 384-atom $MASnI_3$ cell has a residual tilt correlation close to the 3° criterion.

	$MASnI_3$ $\gamma \leftrightarrow \beta$		$MASnI_3$ $\beta \leftrightarrow \alpha$		$FASnI_3$ tilt loss	
atoms	50 K cell	oversized	50 K cell	oversized	50 K cell	oversized
384	243 ± 7 (5/5)	212 ± 5 (5/5)	350 ± 16 (5/5)	357 ± 14 (5/5)	259 ± 9 (5/5)	254 ± 4 (5/5)
864	222 ± 4 (5/5)	189 ± 3 (5/5)	290 ± 8 (5/5)	245 ± 8 (5/5)	245 ± 9 (5/5)	257 ± 7 (5/5)
2304	217 ± 3 (5/5)	152 ± 2 (5/5)	302 ± 9 (5/5)	210 ± 6 (5/5)	248 ± 6 (5/5)	242 ± 4 (5/5)
4800	215 ± 1 (5/5)	154 ± 2 (2/2)	281 ± 7 (5/5)	204 ± 2 (2/2)	251 ± 4 (5/5)	233 ± 4 (5/5)

Table S17: $CsSnI_3$ in a 960-atom cell (2 K/ps): effect of D3, of the training functional, of the external pressure P and of the model form on the transition temperatures (K) and on a_{pc} (Å; relative error against experiment). MACE: one run. $P = -0.845$ GPa restores the experimental volume at 500 K for UrsaMLFF with D3; $P = 0.93$ GPa compresses UrsaMLFF without D3 to the volume with D3.

model	labels	D3	P (GPa)	$\gamma \leftrightarrow \beta$ heat	$\gamma \leftrightarrow \beta$ cool	$\beta \leftrightarrow \alpha$ heat	$\beta \leftrightarrow \alpha$ cool	a_{pc} 300 K	a_{pc} 500 K
UrsaMLFF	PBEsol	no	0	257 ± 20 (3)	225 (1)	297 ± 8 (3)	245 ± 15 (3)	6.162 (+0.2%)	6.206 (+0.0%)
UrsaMLFF	PBEsol	yes	0	414 ± 2 (3)	398 ± 9 (3)	464 ± 14 (3)	452 ± 20 (3)	6.018 (-2.1%)	6.065 (-2.3%)
UrsaMLFF	r^{2}SCAN	no	0	267 ± 15 (3)	238 ± 22 (2)	287 ± 5 (3)	278 ± 8 (2)	6.271 (+2.0%)	6.325 (+1.9%)
UrsaMLFF	r^{2}SCAN	yes	0	319 ± 7 (3)	300 ± 25 (2)	370 ± 9 (3)	315 ± 33 (3)	6.212 (+1.0%)	6.260 (+0.9%)
UrsaMLFF	PBEsol	yes	−0.065	415 ± 9 (3)	387 ± 14 (3)	440 ± 3 (3)	405 ± 22 (3)	6.029 (-1.9%)	6.074 (-2.1%)
UrsaMLFF	r^{2}SCAN	no	−0.039	250 ± 4 (3)	245 (1)	295 ± 9 (3)	250 (1)	6.277 (+2.1%)	6.333 (+2.0%)
UrsaMLFF	r^{2}SCAN	yes	−0.039	310 ± 10 (3)	305 ± 10 (3)	357 ± 13 (3)	370 ± 21 (3)	6.218 (+1.1%)	6.267 (+1.0%)
UrsaMLFF	PBEsol	yes	−0.845	290 ± 4 (3)	295 ± 0 (2)	340 ± 7 (3)	305 ± 5 (2)	6.156 (+0.1%)	6.212 (+0.1%)
UrsaMLFF	PBEsol	no	0.93	380 ± 15 (3)	315 ± 18 (3)	444 ± 26 (3)	372 ± 21 (3)	6.031 (-1.9%)	6.069 (-2.2%)
MACE-MH-1	PBEsol	no	0	252 (1)	none (1)	292 (1)	235 (1)	6.187 (+0.6%)	6.234 (+0.5%)
MACE-MH-1	PBEsol	yes	0	407 (1)	430 (1)	467 (1)	455 (1)	6.050 (-1.6%)	6.100 (-1.7%)
MACE-MH-1	r^{2}SCAN	yes	0	312 (1)	275 (1)	347 (1)	305 (1)	6.235 (+1.4%)	6.293 (+1.4%)
MACE-OMAT	PBEsol	no	0	217 (1)	225 (1)	262 (1)	245 (1)	6.193 (+0.7%)	6.240 (+0.5%)
MACE-OMAT	PBEsol	yes	0	417 (1)	445 (1)	477 (1)	445 (1)	6.051 (-1.6%)	6.103 (-1.6%)
MACE-OMAT	r^{2}SCAN	yes	0	317 (1)	305 (1)	352 (1)	305 (1)	6.237 (+1.4%)	6.295 (+1.4%)
experiment				362	351	440	426	6.148	6.2057

Table S18: Choice of the label functional with UrsaMLFF retrained on PBEsol and on r^{2}SCAN labels, each with the D3 parameters and the external pressure P_{ext} of its functional, in two cells per compound (2 K/ps): transition temperatures on heating (K; mean ± standard error, runs with a transition / runs) and pseudo-cubic lattice constants (Å; $CsSnI_3$ at 300 K, hybrids at 298 K). The hybrid heating runs start from the potential's own 50 K cell.

labels	atoms	$CsSnI_3$			$MASnI_3$			$FASnI_3$	
		$\gamma \leftrightarrow \beta$	$\beta \leftrightarrow \alpha$	a_{pc}	$\gamma \leftrightarrow \beta$	$\beta \leftrightarrow \alpha$	a_{pc}	tilt loss	a_{pc}
PBEsol	960 / 864	460 ± 15 (5/5)	471 ± 14 (5/5)	6.027	222 ± 4 (5/5)	290 ± 8 (5/5)	6.098	245 ± 9 (5/5)	6.160
PBEsol	2000 / 2304	414 ± 5 (5/5)	477 ± 10 (5/5)	6.028	217 ± 3 (5/5)	302 ± 9 (5/5)	6.096	248 ± 6 (5/5)	6.161
r^{2}SCAN	960 / 864	333 ± 10 (5/5)	389 ± 11 (5/5)	6.210	198 ± 8 (5/5)	300 ± 14 (5/5)	6.267	194 ± 6 (3/3)	6.357
r^{2}SCAN	2000 / 2304	329 ± 17 (3/3)	395 ± 7 (3/3)	6.209	199 ± 3 (5/5)	306 ± 7 (5/5)	6.268	186 ± 4 (4/4)	6.360
experiment		362	440	6.148	∼110	∼275	6.240	225–250	6.316

in the 128-formula-unit cells all but one FA arrangement and every MA arrangement fall on these sides. Both preferences peak near 8° and fall to a few tenths of a millihartree at 12°, where $FASnI_3$ turns slightly antiphase (Figure S6c). The sense is therefore chosen where the tilt forms, near the transition; at large static tilts $FAPbI_3$ is antiphase as well.[64]

The cause is the shape of the cage (Figure S6a). A tilt about c turns each face of the A-site cage into a rectangle, long along one in-plane axis L. In-phase tilting gives the upper and lower faces the same long axis, a box of uniform cross-section; antiphase tilting crosses them, so the long axis turns by 90° from the lower to the upper face. Fitting the energy difference of the small cells to the cation orientations gives the contribution of a cation pointing along each axis (Figure S6d). Both cations favour in-phase tilting when they lie along L (FA 3.7 and MA 1.6 mHa per formula unit) and antiphase tilting when they lie along the short in-plane axis $L_\perp$. They differ along c: an FA cation standing along c favours in-phase by 8.2 mHa, an MA cation antiphase by 4.3 mHa. Planar FA carries an NH_2 group at each end; lying along L or standing along c, it meets two equivalent faces of the box, while in the crossed cage one end is always squeezed. Averaged over the three axes, FA favours in-phase and MA antiphase, by about the amounts found in the large cells. Relaxation turns the FA cations towards L in the in-phase cell ($\langle(\hat{u}\cdot L)^2\rangle$ rises from 0.33 to 0.6–0.7 for the N–N axis $\hat{u}$).

The deposited structures and the MD agree. In both structures of $FASnI_3$ (Table 1) the FA nitrogens lie along the long axis of their cage, with the N–N axes of neighbouring cages perpendicular in the plane and repeated along c;[8] the refinement of tetragonal $MASnI_3$ places the C–N axis along c,[4] the orientation that favours antiphase stacking most, although with large displacement parameters. In the in-phase phase of the MD the FA cations keep their N–N axis in the plane (mean $|\cos|$ with the tilt axis 0.16 to 0.29, against 0.5 for random orientations), while in the antiphase phase the MA cations turn almost freely (0.46 to 0.48), so for MA the preference is an average over orientations.

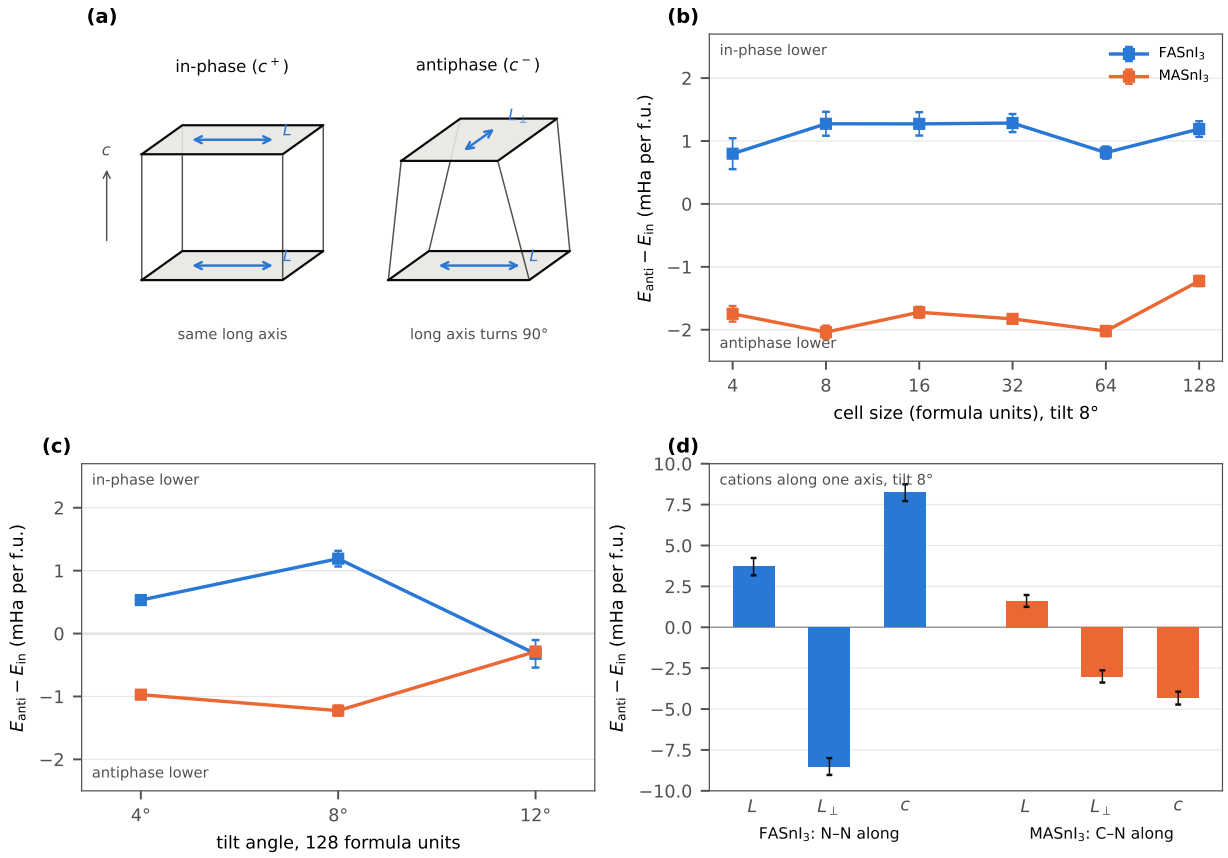


Figure S6: Tilt sense and cation orientation. (a) A tilt about c makes each face of the A-site cage a rectangle with long axis L; in-phase stacking keeps L at the lower and upper faces, antiphase stacking turns it by 90°. (b) Energy of the antiphase relative to the in-phase cell at a tilt of 8° against cell size: mean and standard error over 20 to 160 random cation arrangements, cations relaxed with Sn and I fixed. (c) The same against the tilt angle in the 128-formula-unit cells. (d) Contribution of cations lying along L, $L_\perp$ or c, from a fit of the energy difference of the 4- and 8-formula-unit cells at 8° to the cation orientations (FA: N–N axis; MA: C–N axis). Energies of the r^{2}SCAN potential before its final retraining, at the cubic metric of the PBEsol potential ($FASnI_3$ 6.127 Å, $MASnI_3$ 6.058 Å).

Results with PBEsol labels

This section collects the results on the free crystals obtained with the potentials trained on the PBEsol labels.

Comparison of the two label sets

We chose between the two label sets by comparing the phase transitions of UrsaMLFF trained on each, with the D3 parameters and the external pressure of its functional, in two cells per compound, 960 and 2000 atoms for $CsSnI_3$ and 864 and 2304 atoms for the hybrids, each heated five times at 2 K/ps (protocol in the Supporting Information). Figure S7 shows the transition temperatures on heating and the lattice constants; the two cell sizes agree within the scatter, except for the $\gamma \leftrightarrow \beta$ transition of $CsSnI_3$ with PBEsol labels, which lies 46 K higher in the smaller cell. For $CsSnI_3$ the two functionals err in opposite directions: with PBEsol labels the transitions lie 31 to 98 K above experiment, with r^{2}SCAN labels 29 to 51 K below. Both place the orthorhombic-to-tetragonal transition of $MASnI_3$ far too high (by 88 to 112 K) and its transition to the cubic phase 15 to 31 K too high. The loss of tilt order in $FASnI_3$ lies within the experimental range with PBEsol labels and 31 to 39 K below it with r^{2}SCAN labels. PBEsol is thus closer to experiment for the three transitions into the untilted phase, and r^{2}SCAN for the two $\gamma \leftrightarrow \beta$ transitions; averaged over the five transitions, the deviations are similar (45 to 51 K with PBEsol, 45 to 47 K with r^{2}SCAN). The lattice constants separate the functionals clearly: PBEsol with D3 underestimates them by 2.0 to 2.5 %, r^{2}SCAN with D3 overestimates them by 0.4 to 1.0 %. Three of the ten r^{2}SCAN runs of $FASnI_3$ developed molecular defects above 260 K, after the loss of tilt order, and two r^{2}SCAN runs of $CsSnI_3$ in the larger cell two Cs ions in one cage above 520 K; these runs are left out of the averages. The r^{2}SCAN training data contain no active-learning structures, which the PBEsol data needed to keep the hybrid MD free of such defects.

The clamping study of the main text uses the PBEsol labels, because it rests on the transitions into the untilted phase, which the PBEsol potential places closer to experiment for $CsSnI_3$; the price is a lattice compressed by about 2 %, and part of the PBEsol agreement comes from a cancellation of errors (section). The results of the main text on the free crystals use the r^{2}SCAN labels.

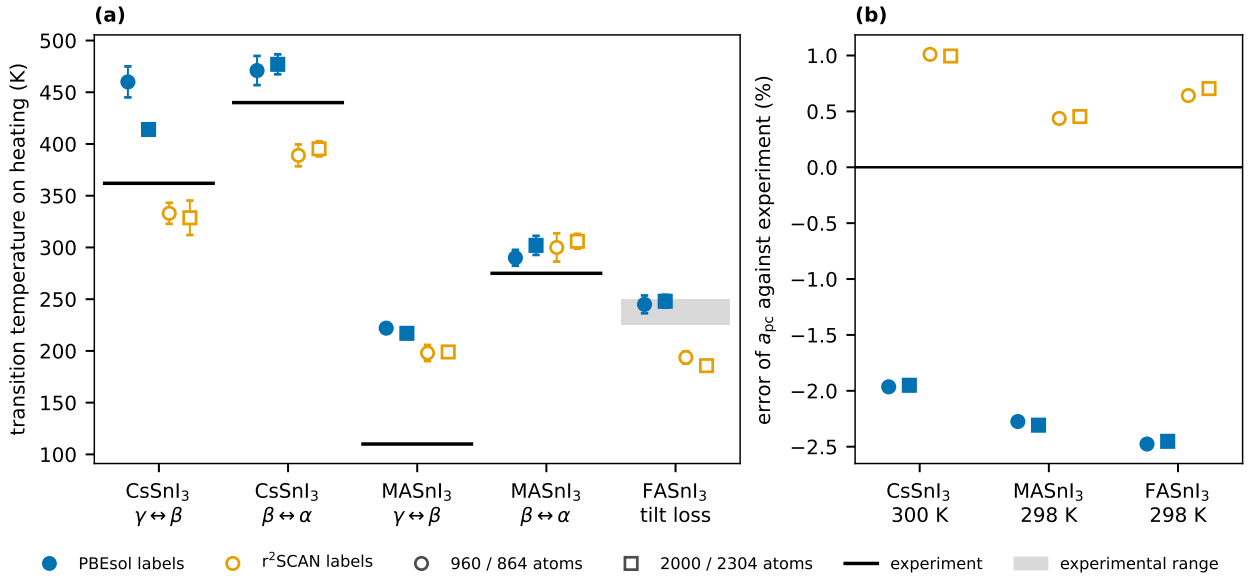


Figure S7: Comparison of PBEsol and r^{2}SCAN training data. (a) Transition temperatures on heating of UrsaMLFF retrained on PBEsol (filled) and r^{2}SCAN labels (open), each with the D3 parameters and the external pressure of its functional, in cells of 960 and 2000 atoms ($CsSnI_3$; circles and squares) and 864 and 2304 atoms (hybrids) at 2 K/ps, against experiment (black lines; grey band: experimental range for $FASnI_3$).[7,18–20] Mean and standard error of five runs, without the runs with defects (Supporting Information). (b) Error of the pseudo-cubic lattice constant against experiment.[7,18,55]

Convergence of the phase transitions with system size

The transition temperatures of the PBEsol UrsaMLFF were determined in cells of 360 to 4900 atoms ($CsSnI_3$) and 384 to 4800 atoms (hybrids), five runs per cell size and direction, with the heating and cooling protocol given in the Supporting Information. Table S19 lists the transition temperatures, Figure S8 the Glazer tilt order along the ramps, and Figures S10 and 4 (section *Octahedral tilt patterns*) the tilt patterns of the phases.

Phase sequences. On heating, $CsSnI_3$ passes from $a^-a^-c^+$ to $a^0a^0c^+$ to an untilted cubic phase, the tilt systems of the experimental $Pnma$, $P4/mbm$ and $Pm\bar{3}m$ phases.[7] $MASnI_3$ starts in $a^-a^-c^+$, with an in-phase tilt of 12° about the long axis and antiphase tilts of about 3.5° about the other two. The antiphase tilts are lost near 217 K, leaving $a^0a^0c^+$, and the in-step stacking of the remaining tilt is lost near 300 K. Close to this transition the tilt about the long axis also forms a layered pattern, with in-phase and antiphase parts of similar size, in which successive layers tilt by about 11° and 4° (Figure S9a); the nearest-neighbour tilt correlation of our transition criterion averages this state to nearly zero although its tilt persists. The intermediate in-phase state has no experimental counterpart, where the tetragonal phase has the antiphase tilt $a^0a^0c^-$,[4] and marks a limitation of the model for $MASnI_3$. $FASnI_3$ orders only as $a^0a^0c^+$, the tilt system of its experimental tetragonal phase,[18] and loses it at 248 K; the orthorhombic phase of $FASnI_3$ below 125–150 K does not form in our runs.

Convergence on heating. For $CsSnI_3$ the tilt-order curves of all cells from 960 atoms upward coincide (Figure S8), and the heating transition temperatures stay within 407–411 K ($\gamma \leftrightarrow \beta$) and 449–462 K ($\beta \leftrightarrow \alpha$) up to 4900 atoms; the 360-atom cell places both transitions higher and has the largest residual tilt correlation in the cubic phase. For the hybrids the 384-atom cell is not converged: its residual tilt correlation in the disordered phase, 2.5° for $MASnI_3$, is close to the 3° criterion and places the $\beta \leftrightarrow \alpha$ transition of $MASnI_3$ at 350 K. From 864 atoms on, every run shows every transition, the residual correlation falls to 0.5° to 1.5°, and the heating transitions lie at 215–222 K ($\gamma \leftrightarrow \beta$) and 281–302 K ($\beta \leftrightarrow \alpha$) for $MASnI_3$ and at 245–251 K for $FASnI_3$. Cells of about 2000 atoms (2000 for $CsSnI_3$, 2304 for the hybrids) are therefore the smallest in which the transition statistics are converged, and we use them for production.

Starting structure of the hybrid heating runs. The starting structures of the hybrid heating runs were first built with cells 4–7 % ($MASnI_3$) and 2–3 % ($FASnI_3$) longer than the

Table S19: Transition temperatures (K) against cell size for UrsaMLFF trained on PBEsol labels, with D3 at the constant pressure P_{ext}, at 2 K/ps: mean ± standard error over the runs with a transition (runs with a transition / runs). Cells are $n_1 \times n_2 \times n_3$ supercells of the $\sqrt{2} \times \sqrt{2} \times 2$ setting; the 3600- and 4320-atom cells (even in-plane repeats) were run on cooling only. Noise: the largest tilt correlation (deg) over the hottest 40 K of the heating runs, the level the transition criterion has to exceed. The r^{2}SCAN values are given in the Supporting Information. Hybrids: UrsaMLFF retrained on the refined training data; their heating ramps start from structures with the model's 50 K cell, their cooling ramps in the cubic phase (starting-cell dependence: Supporting Information).

$CsSnI_3$ atoms (cell)	$\gamma \leftrightarrow \beta$ heating	$\gamma \leftrightarrow \beta$ cooling	$\beta \leftrightarrow \alpha$ heating	$\beta \leftrightarrow \alpha$ cooling	noise
360 (3×3×2)	429 ± 6 (5/5)	395 ± 11 (4/5)	508 ± 25 (5/5)	476 ± 28 (5/5)	2.2
960 (4×4×3)	407 ± 8 (5/5)	360 ± 34 (5/5)	456 ± 10 (5/5)	416 ± 20 (5/5)	1.3
2000 (5×5×4)	411 ± 7 (5/5)	327 ± 13 (5/5)	462 ± 17 (5/5)	377 ± 13 (5/5)	1.0
2880 (6×6×4)	411 ± 5 (5/5)	352 ± 16 (5/5)	449 ± 7 (5/5)	375 ± 18 (5/5)	0.9
3600 (6×6×5)	–	314 ± 16 (5/5)	–	332 ± 16 (5/5)	–
4320 (6×6×6)	–	302 ± 22 (5/5)	–	334 ± 13 (5/5)	–
4900 (7×7×5)	407 ± 3 (5/5)	314 ± 8 (5/5)	461 ± 9 (5/5)	341 ± 4 (5/5)	0.8
$MASnI_3$ atoms (cell)	$\gamma \leftrightarrow \beta$ heating	$\gamma \leftrightarrow \beta$ cooling	$\beta \leftrightarrow \alpha$ heating	$\beta \leftrightarrow \alpha$ cooling	noise
384 (2×2×2)	243 ± 7 (5/5)	188 ± 28 (4/5)	350 ± 16 (5/5)	336 ± 14 (4/5)	2.5
864 (3×3×2)	222 ± 4 (5/5)	170 ± 0 (2/5)	290 ± 8 (5/5)	237 ± 12 (5/5)	1.5
2304 (4×4×3)	217 ± 3 (5/5)	130 (1/5)	302 ± 9 (5/5)	211 ± 13 (4/5)	0.7
4800 (5×5×4)	215 ± 1 (5/5)	180 ± 13 (3/5)	281 ± 7 (5/5)	150 ± 25 (4/5)	0.5
$FASnI_3$: loss of any tilt order atoms (cell)	heating	cooling			noise
384 (2×2×2)	259 ± 9 (5/5)	257 ± 14 (5/5)			1.4
864 (3×3×2)	245 ± 9 (5/5)	243 ± 8 (5/5)			0.7
2304 (4×4×3)	248 ± 6 (5/5)	232 ± 11 (5/5)			0.6
4800 (5×5×4)	251 ± 4 (5/5)	220 ± 9 (5/5)			0.5

model's 50 K cell, which the barostat compresses during the equilibration at 50 K. Crystals prepared this way have the tilt pattern, the cation arrangement and the energy (within 4×10^{-5} Ha per formula unit) of crystals started in the 50 K cell, but carry about 20 % more octahedral distortion, with a mean Sn off-centring of 0.113 against 0.093 Å, and keep it on heating. $MASnI_3$ prepared in the oversized cell loses the in-step stacking of its tilt at (210 ± 6) K instead of (302 ± 9) K and its second tilted axis at (152 ± 2) K instead of (217 ± 3) K (2304 atoms), whereas $FASnI_3$ changes within the error ((242 ± 4) K against (248 ± 6) K; Table S16). Every hybrid heating run of this work therefore starts from the model's 50 K cell. Because the nearest-neighbour criterion does not detect a layered tilt, we also bracketed the equilibrium tilt loss of $MASnI_3$ with holds of 150 ps at 250, 265 and 280 K, each started once from a tilted and once from a cubic crystal (two runs each, 2304 atoms). At 265 K both tilted starts keep a layered tilt of about 8° about one axis and one cubic start forms it; at 280 K both cubic starts stay untilted and both tilted starts lose their coherent tilt within the hold. The equilibrium tilt loss of the model thus lies between 265 and 280 K, and the heating transition of the production cell lies about 30 K above it, a superheating of the clean crystal at 2 K/ps.

Cooling. On cooling, the transitions occur far below their heating temperatures, and the cooling values depend on the cell size. The $\beta \leftrightarrow \alpha$ transition of $CsSnI_3$ falls from 476 K in the 360-atom cell to 332–341 K in the cells of 3600 to 4900 atoms, and $\gamma \leftrightarrow \beta$ from 395 to 302–314 K, a hysteresis of 60 to 120 K in the cells of 2000 atoms and more against 11 to 14 K in experiment.[7] The state reached on cooling is also less ordered: at 210 K the largest tilt correlation is 5.6 to 11.1° after cooling against 13.0 to 13.2° after heating, consistent with cooling into several tilt domains that the larger cells can hold. Cells with odd in-plane repeats cannot hold an in-phase tilt about an in-plane axis and end less ordered than the even 2880- to 4320-atom cells, but the even cells give the same cooling transition temperatures as the odd 4900-atom cell, so commensurability is not the cause. $FASnI_3$ orders on cooling

in every run, 10 to 25 K below its heating transition. $MASnI_3$ cools into tilt domains: its local tilt about one axis reaches 7° to 8° at 150 K, but domains of opposite stacking keep its average tilt correlation close to the 3° criterion, so that its cooling transition temperatures scatter between 150 and 237 K; the 18 432-atom runs show these domains directly (section *Cage and cation angles in 18 432-atom cells*). A second tilted axis appears on cooling in one to four of five runs, so that $MASnI_3$ rarely reaches its orthorhombic phase, as found for $MAPbI_3$.[22,24] We attribute these effects to the kinetics of nucleating an ordered tilt pattern at 2 K/ps, quote transition temperatures on heating, and measure shifts under clamping on heating as well.

Transition temperatures compared with experiment

Table S20 compares the heating transition temperatures and lattice constants of the production cells with experiment and with MACE-MH-1 + D3, trained on the same labels. For $CsSnI_3$ and $FASnI_3$ the two potentials agree within 6 K and 0.5 %. For $MASnI_3$, MACE-MH-1 places both transitions lower; in the same 864-atom cell UrsaMLFF gives (222 ± 4) K and (290 ± 8) K, against 202 and 237 K, so the difference between the models is largest for the transition to the cubic phase.

The agreement of the absolute transition temperatures with experiment should not be over-read. Without D3, the same UrsaMLFF reproduces the lattice constants to within 1 % but places the transitions of $CsSnI_3$ 105 and 143 K too low ((257 ± 20) K and (297 ± 8) K in a 960-atom cell). D3 raises them by about 160 K, mostly by compressing the lattice by about 2 %: run at the experimental volume, UrsaMLFF + D3 gives (290 ± 4) K and (340 ± 7) K. The agreement in Table S20 therefore rests partly on a cancellation between the over-binding of D3 and the under-stabilisation of the tilted phases by UrsaMLFF. Trained on r^{2}SCAN labels, UrsaMLFF gives lattice constants within 1.0 % of experiment but transition temperatures of (329 ± 17) K and (395 ± 7) K for $CsSnI_3$ in the production cell (Figure S7). Absolute transition temperatures thus carry an uncertainty of order 100 K between equally defensible

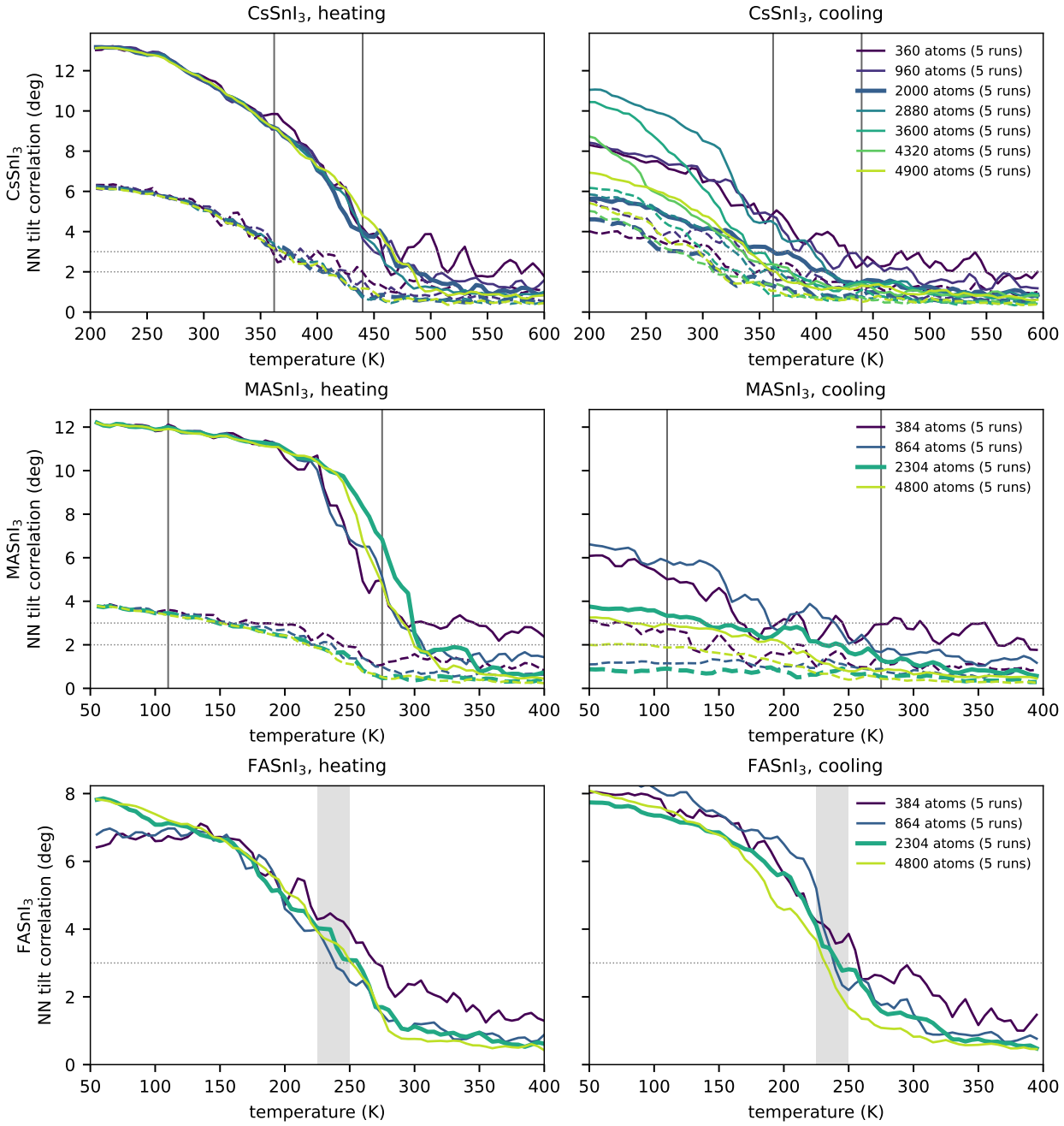


Figure S8: Glazer tilt order against temperature for every cell size (PBEsol UrsaMLFF + D3, 2 K/ps, mean of five runs in 10 K windows), on heating (left) and cooling (right). Solid lines: largest nearest-neighbour tilt correlation (lost at the transition to the cubic phase); dashed lines: second largest (lost when the second tilted axis disappears). Dotted lines: the thresholds of the transition criterion; vertical lines and grey band: experimental transition temperatures. The production cells (2000 and 2304 atoms) are drawn thicker. Hybrids: retrained UrsaMLFF; heating runs started from the model's 50 K cell.

Table S20: Transition temperatures on heating and pseudo-cubic lattice constants of the free-standing crystals: UrsaMLFF + D3 (PBEsol) in the production cells ($CsSnI_3$ 2000 atoms, $MASnI_3$ and $FASnI_3$ 2304 atoms), 2 K/ps, mean ± standard error of five runs; MACE-MH-1 + D3 (PBEsol labels) in the cells of the functional comparison ($CsSnI_3$ 960 atoms, $MASnI_3$ and $FASnI_3$ 864 atoms), 2 K/ps, one run; and experiment. a_{pc} at 300 K ($CsSnI_3$) and 298 K (hybrids). Hybrids: UrsaMLFF retrained on the refined training data, heating runs started from the model's 50 K cell.

compound	quantity	UrsaMLFF + D3	MACE-MH-1 + D3	experiment
$CsSnI_3$	$\gamma \leftrightarrow \beta$ (K)	411 ± 7	407	362[7]
	$\beta \leftrightarrow \alpha$ (K)	462 ± 17	467	440[7]
	a_{pc} (Å)	6.030	6.050	6.148[7]
$MASnI_3$	$\gamma \leftrightarrow \beta$ (K)	217 ± 3	202	∼110[19,21]
	$\beta \leftrightarrow \alpha$ (K)	302 ± 9	237	∼275[19,20]
	a_{pc} (Å)	6.103	6.133	6.2397[55]
$FASnI_3$	loss of tilt order (K)	248 ± 6	247	225–250[18]
	a_{pc} (Å)	6.160	6.189	6.316[18]

models, which is why every simulation temperature in the clamping study is anchored to the model's own T_c (section *Effect of process conditions on band-gap deviations*) and why we report only differences.

Cage and cation angles in 18 432-atom cells

To follow the tilt and cation patterns beyond the production cells, we heated and cooled cells of 18 432 atoms ($8 \times 8 \times 6$, about 7 nm across) of both hybrids at 1 K/ps, one run per direction, and analysed the structure averaged over 10 ps blocks, the static (time-averaged) and local (instantaneous) tilts, the Sn–I–Sn angles and the orientations of the cations (Figure S9).

$FASnI_3$ orders as a single in-phase domain that fills the whole cell, on heating and on cooling (Figure S9k,l). The tilt of the average structure exceeds 3° below 271 K on heating and below 269 K on cooling, so at this size and rate the transition shows no hysteresis. At 200 K the tilt is 6.4° and the in-plane Sn–I–Sn angle of the average structure 166.6°, against 6.3° and 167.4° measured for the $P4/mbm$ phase,[8] and the apical angle is 175°, against 180°. At 100 K the tilt, 7.2°, is the same in every layer to within 1.1°: the doubling of the c axis

with tilts alternating between 11.6° and 3.8° reported for the low-temperature $P4bm$ phase[8] does not form. On cooling the FA axes concentrate along ⟨100⟩ (a fraction of 0.8 at 50 to 90 K) and align with the tilt axis, without nearest-neighbour order of their dipoles.

$MASnI_3$, heated from the model's 50 K cell, starts in $a^-a^-c^+$ throughout the cell, with an in-phase tilt of 12° about the long axis that is the same in every layer, antiphase tilts of about 3.5° about the other two axes, and MA dipoles that lie perpendicular to the long axis and are ordered antiparallel between nearest neighbours (nearest-neighbour correlation 0.02 at 150 K). This cation order melts between 200 and 280 K, and with it the in-step stacking of the tilt: the antiphase tilts about the short axes are lost near 241 K, and from about 265 to 315 K the tilt about the long axis has in-phase and antiphase parts of similar size (4.9° and 5.7° at 270 K), the layered state of the holds, before the cell becomes cubic near 320 K (Figure S9a). The tilts of the model are larger than measured: at 200 K the in-phase tilt is 11.1° and the in-plane Sn–I–Sn angle of the average structure 156°, against 8.7° and 162.6° for the $I4cm$ phase,[4] and at 140 K 11.7° and 155° against 9.3° and 161.4°.[19] Cooled from the cubic phase, $MASnI_3$ splits into tilt domains several nanometres across with opposite in-phase stacking (Figure S9j): its local tilt reaches 7°, but the tilt of the average structure stays below 2°, and its cations do not order (nearest-neighbour dipole correlation 0.43 at 50 K).

The anneals of 18 432-atom films at the anchored temperatures ($MASnI_3$ 400 to 327 K, $FASnI_3$ 377 to 304 K; free, stress-controlled and rigidly clamped, one run each) reproduce the production cells: at the end of the hold no film has a tilt above 0.8° about any axis.

Phase sequences and tilt systems with PBEsol labels

Figure S10 and Tables S21 and S22 give the results of the protocol of the main text (section) with the PBEsol potential; for $MASnI_3$ two cooling runs were made in the 18 432-atom cell. The PBEsol potential reproduces the sequence of $CsSnI_3$, $a^0a^0c^+$ followed by $a^-a^-c^+$, and the in-phase tilt of $FASnI_3$, on cooling and on heating. For $MASnI_3$ two of the three 2304-

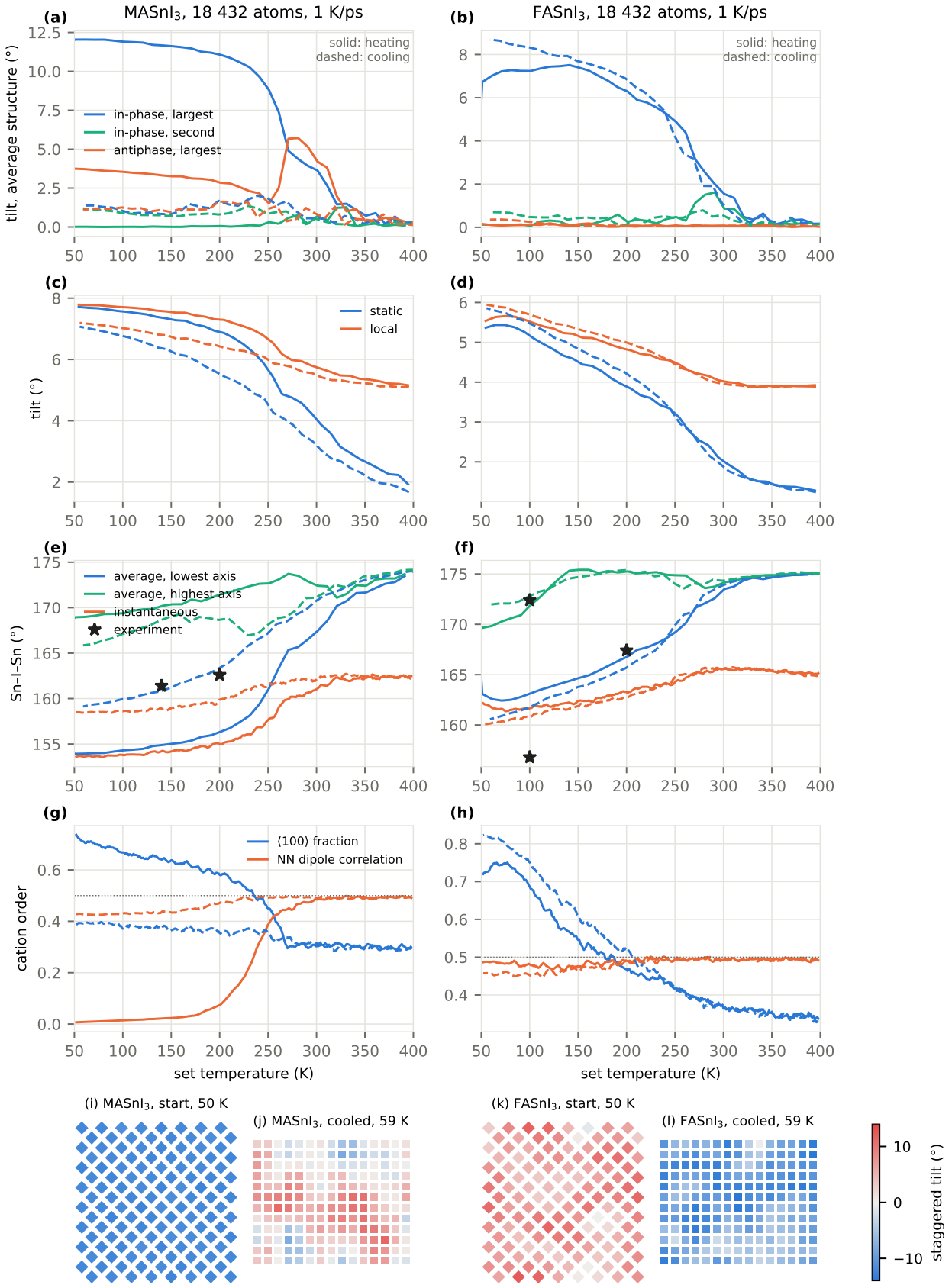


Figure S9: Cage and cation angles of $MASnI_3$ (left) and $FASnI_3$ (right) in 18 432-atom cells (retrained UrsaMLFF + D3, 1 K/ps, one run per direction, heating from the model's 50 K cell; solid heating, dashed cooling). (a, b) Tilt amplitudes of the structure averaged over 10 ps: largest and second in-phase, largest antiphase. (c, d) Static (time-averaged) and local (instantaneous rms) tilt. (e, f) Sn–I–Sn angles of the average structure about the most and

atom runs form weak in-phase tilts about one or two axes and the third does not order; both 18 432-atom runs order only below 185 K, one antiphase by about 4.5° and one in-phase by about 7°. PBEsol thus leaves the tilt sense of $MASnI_3$ undecided at this rate; at 0.25 K/ps all three 2304-atom runs end antiphase. On cooling it places the tilt order of $MASnI_3$ at (228 ± 8) K and that of $FASnI_3$ at (252 ± 7) K. For $CsSnI_3$ it is closer to experiment than the r^{2}SCAN potential: (413 ± 17) K and (372 ± 16) K against 426 and 351 K on cooling, and (442 ± 21) K against 440 K on heating. Tetragonal $FASnI_3$ tilts by 6.1° and γ-$CsSnI_3$ bends its equatorial and apical Sn–I–Sn bridges by 10.6° and 7.3°; β-$CsSnI_3$ tilts in-phase by 7.1°, and at low temperature the $FASnI_3$ layers differ by only 1°.

In the static tilt-sense scan (section) at 8°, the PBEsol potential places $FASnI_3$ lower in-phase by 0.8 to 1.0 mHa per formula unit and $MASnI_3$ lower antiphase by 0.6 to 1.2 mHa in cells of 16 to 128 formula units, about half the r^{2}SCAN preference for $MASnI_3$. The fitted contribution of a cation lying along L is 2.1 (FA) and 1.3 mHa (MA) per formula unit, and of a cation standing along c +6.0 (FA) and −2.6 mHa (MA). Where the preference is this weak, the cooling path decides the sense.

Table S21: Transition temperatures (K) at 0.5 K/ps with the PBEsol potential: mean ± standard error of three runs in cells of about 2000 atoms, on cooling from the cubic phase and on heating the cooled crystal, and one cooling run in cells of about 20 000 atoms. Experiment: heating / cooling. Where not all runs show a transition the mean is over those that do: $MASnI_3$ $\gamma \leftrightarrow \beta$ and $\beta \leftrightarrow \alpha$: two of the three runs; $FASnI_3$ second tilted axis: one of the three runs. . . . : no valid run.

			2k atoms		20k atoms
compound	transition	experiment	cooling	heating	cooling
$CsSnI_3$	$\gamma \leftrightarrow \beta$	362 / 351 [7]	372 ± 16	385 ± 23	385
	$\beta \leftrightarrow \alpha$	440 / 426 [7]	413 ± 17	442 ± 21	425
$MASnI_3$	$\gamma \leftrightarrow \beta$	114 / 108 [21]	188 ± 28	188 ± 42	none
	$\beta \leftrightarrow \alpha$	∼275 [19,20]	228 ± 8	230 ± 5	185
$FASnI_3$	second tilted axis	125–150 (powder) [18]	100	145	none
	tilt order	225–255 [8,18]	252 ± 7	265 ± 3	250

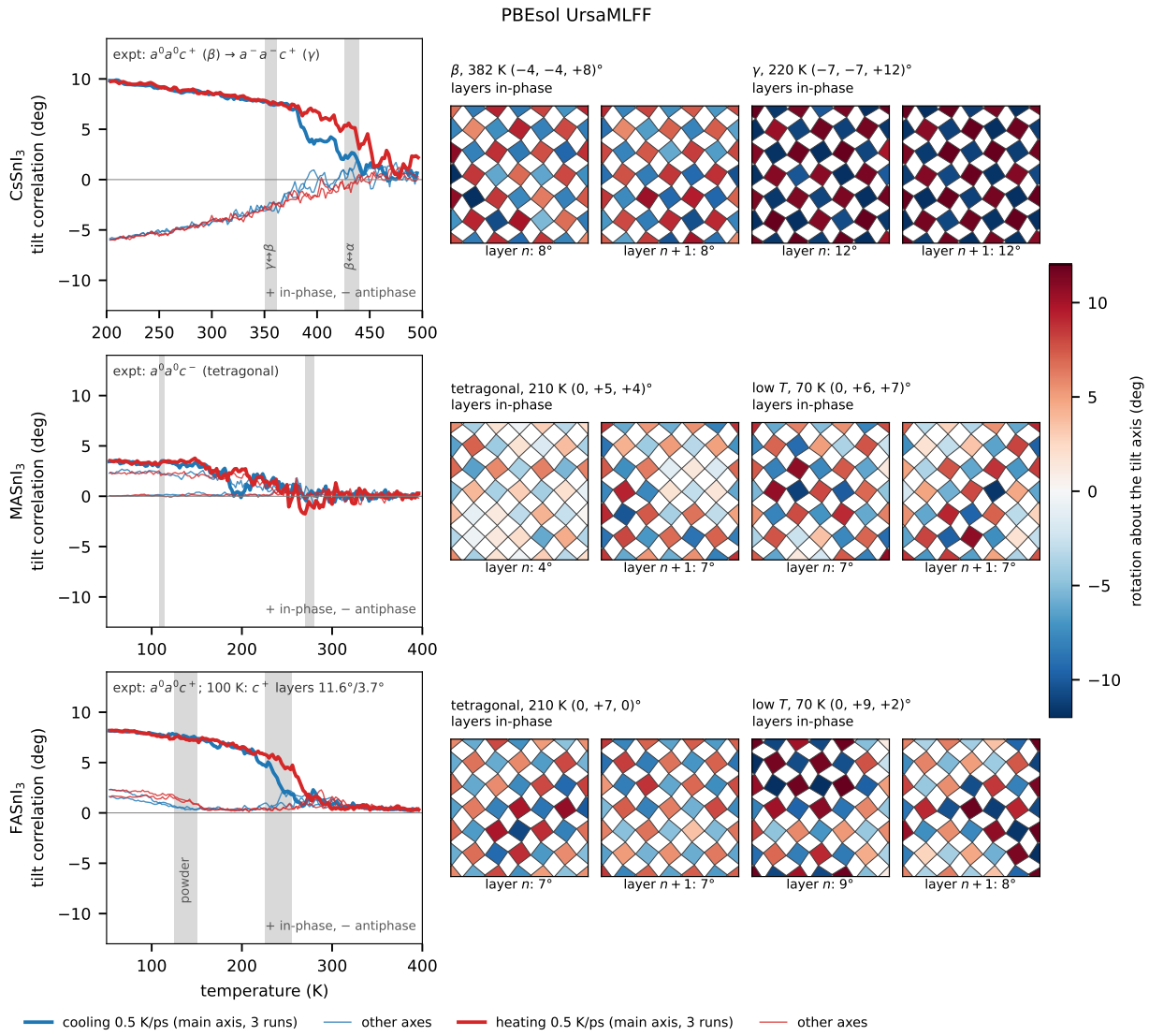


Figure S10: Tilt patterns with the PBEsol potential ($CsSnI_3$ 2000 atoms, hybrids 2304); panels as in Figure 4, except that every compound is heated back from its cooled crystal.

Table S22: Tilt systems and tilt angles (degrees) at 0.5 K/ps with the PBEsol potential against the deposited structures (Table 1), each compared at the same distance below its transition as in experiment (model temperature T). Time-averaged structures (7 ps), cooling runs; $CsSnI_3$ β: heating runs, middle of the β range (on cooling it spans only 5–50 K). Tilt system and angle from the amplitudes averaged over the runs. Single tilt: amplitude about the tilt axis (+ in-phase, − antiphase); *Pnma*: equatorial / apical Sn–I–Sn bending $(180° - \theta)/2$; *P4bm*: tilts of the alternating layers. Model amplitudes about each axis from the rotation correlation of neighbouring octahedra along it, insensitive to tilt domains. $MASnI_3$: the two runs that tilt.

	experiment			PBEsol		
compound	phase, T	tilt system	angle	T (K)	tilt system	angle
$CsSnI_3$	β (*P4/mbm*), 380 K[7]	$a^0a^0c^+$	+6.0	413	$a^0b^-c^+$	+7.1
	γ (*Pnma*), 300 K[7]	$a^-a^-c^+$	11.3 / 6.8	321	$a^-a^-c^+$	10.6 / 7.3
$MASnI_3$	tetragonal (*I4cm*), 200 K[4]	$a^0a^0c^-$	−8.6	152	$a^0b^+c^+$	+6.3
	tetragonal (*I4/mcm*), 140 K[19]	$a^0a^0c^-$	−9.3	92	$a^0b^+c^+$	+6.6
$FASnI_3$	tetragonal (*P4/mbm*), 200 K[8]	$a^0a^0c^+$	+6.3	214	$a^0a^0c^+$	+6.1
	low T (*P4bm*), 100 K[8]	c^+, layers	11.6 / 3.7	114	$a^0a^0c^+$	8.8 / 7.8

Application protocol: clamping and band gaps

Mechanical boundary conditions

Film geometry. The a and b axes of the cell are the in-plane directions of a (001)-oriented film and c is its normal. The stress is $\sigma = -W/V$, where W is the virial including the kinetic contribution; positive values are tensile.

Temperature anchoring. The absolute transition temperatures of a force field carry its systematic errors. To place the anneal and the operating point in the same phase and at the same distance from T_c as in a device, every device temperature T is mapped onto the model as $T_{sim} = T + [T_c(\text{model}) - T_c(\text{expt})]$, using the transition that the anneal-to-operation cooling path approaches: $\gamma \leftrightarrow \beta$ for $CsSnI_3$, $\beta \leftrightarrow \alpha$ for $MASnI_3$ and the loss of tilt order for $FASnI_3$, each on heating in the production cell. The device temperatures are the operating temperature (300 K), the damp-heat test (358 K), the anneal (373 K) and, for $CsSnI_3$ only, a cubic anneal at 440 K. The shifts are +49, +27 and +4 K for $CsSnI_3$, $MASnI_3$ and $FASnI_3$, giving the simulation temperatures of Table S23. A control set at the

unshifted device temperatures (one run per condition) tests the effect of the anchoring. The transition temperatures of the hybrids are those of heating runs started from the model's 50 K cell (main text, section *Starting structure of the hybrid heating runs*). The $MASnI_3$ shift of +27 K replaces a shift of −65 K obtained from runs started in an oversized cell; the $MASnI_3$ runs at 235, 293 and 308 K made with that shift are listed in Table S30 but not used in the main text. The $FASnI_3$ shift was computed from the earlier runs ((242 ± 4) K); the runs from the 50 K cell give (248 ± 6) K, a shift larger by 6 K, within the error, so the $FASnI_3$ campaigns were kept.

Table S23: Device temperatures and the anchored simulation temperatures (K). The shift is T_c(model) − T_c(expt) of the anchor transition.

	$CsSnI_3$	$MASnI_3$	$FASnI_3$
anchor transition	$\gamma \leftrightarrow \beta$	$\beta \leftrightarrow \alpha$	tilt loss
T_c model / expt	411 / 362	302 / 275	242[b] / 238[a]
shift	+49	+27	+4
operation (300 K)	349	327	304
damp heat (358 K)	407	385	362
anneal (373 K)	422	400	377
cubic anneal (440 K)	489	–	–

[a] Midpoint of the experimental range 225–250 K.[18] [b] From the runs started in the oversized cell; (248 ± 6) K from the 50 K cell.

Three brackets. A real film lies between two limits: a free-standing crystal and a crystal whose in-plane dimensions are fixed by a rigid substrate. We simulate both limits and one realistic intermediate state, each with three independent runs at the operating temperature and one at the damp-heat temperature. The free bracket (ref) applies a constant pressure P_{ext} on all three axes. The stress-controlled bracket (sctl) applies an in-plane tensile stress of 50 MPa ($P_{xx} = P_{yy} = P_{ext} - 0.05$ GPa) with the normal free ($P_{zz} = P_{ext}$), the residual stress level measured in perovskite films.[29] The rigid clamp with process history (proc) consists of 50 ps of free constant-pressure MD at the anneal temperature, after which a and b are frozen while c stays free; the cell is then cooled at 2 K/ps to the damp-heat temperature

(held 50 ps) and to the operating temperature (held 100 ps). The thermal contraction of the substrate itself is neglected; the residual stress of a film arises because it is much smaller than that of the perovskite.[29,31] For $CsSnI_3$ the same protocol is also started from the cubic anneal (proccub). Equilibrium runs start from a heating-ramp frame of the free crystal close to the target temperature, with the cell set to the five-run mean at that temperature, and consist of 5 ps of constant-volume MD followed by 150 ps at the target condition. The first 50 ps of the last stage are discarded and averages are taken over the remaining frames.

Elastic response. Two ladders at the operating temperature give the elastic constants of each model and the strain dependence of the structure: biaxial strains of ± 0.25, ± 0.5, ± 1 and $\pm 2\,\%$ applied to a and b (frozen, c free), and hydrostatic pressures of -0.3, -0.15, $+0.15$, $+0.3$ and $+0.6$ GPa relative to P_{ext}. The biaxial modulus $Y/(1-\nu)$, Poisson's ratio ν, Young's modulus Y and the bulk modulus B are fitted to the mean stresses and cells.

Transition temperatures under the clamp. Because the model supercools, the shift of a transition under the clamp is measured on heating. For $CsSnI_3$, cells are clamped in-plane at the mean free cell of the anneal state and heated from the operating temperature at 2 K/ps through both transitions (three runs). The transition temperatures are determined as for the free ramps, and $\Delta T_{\mathrm{c}} = T_{\mathrm{c}}(\mathrm{clamped}) - T_{\mathrm{c}}(\mathrm{free})$. A γ-phase cell held under the same clamp at the operating temperature (150 ps, three runs) provides the clamped γ state for the band-gap calculations. Clamped heating runs have so far been made for $CsSnI_3$ only.

Band gaps of the clamped and free ensembles

DFT on production frames. Band gaps are computed with the label settings of the main text (*Electronic structure*) on complete production frames of 2000 or 2304 atoms, so no subcell has to be cut out of the MD cell. The gap is the lowest unoccupied minus the highest occupied Kohn–Sham eigenvalue. The supercells sample only the Γ point. In the $\sqrt{2} \times \sqrt{2} \times 2$ setting the supercell vectors are $n_1(1,1,0)$, $n_2(-1,1,0)$ and $n_3(0,0,2)$ in units

of a_{pc}, so the pseudo-cubic R point, which carries the band extrema of the cubic phase and from which those of the tilted phases derive, folds onto Γ for every n_1, n_2, n_3; so does the in-phase tilt vector $M = (\frac{1}{2}, \frac{1}{2}, 0)$. In a pseudo-cubic supercell with an odd number of repeats along an axis, R would instead fold onto the zone boundary, and a Γ-only gap would be too large. Each DFT calculation also returns the energy of the frame, which tests UrsaMLFF in the clamped ensembles.

Γ-point sampling. The folding argument holds for the average structure. A thermal snapshot has no symmetry beyond the supercell lattice, and disorder could move a band edge to another point of the supercell zone. Table S24 collects every DFT calculation of this work that samples more than one k-point. In all 86 static structures and in all 62 MD snapshots of 192 and 240 atoms, both band edges are at Γ, by at least 213 meV for the valence band and 4 meV for the conduction band. The conduction band has the smaller margin because, without SOC, its minimum at R is threefold degenerate and includes heavy branches. In the 80-atom fcc-type cell at 600 K, one of two snapshots has its conduction-band minimum off Γ, and its Γ gap is 11 meV too large. Such an error is bounded by the width of the band-edge bands across the supercell zone, which falls roughly as the inverse square of the cell length: by a factor of about five from the 80-atom cell to the hybrid production cells and of about eight to the $CsSnI_3$ cells. The Γ gap can only overestimate the band-edge gap, and by similar amounts in clamped and free frames of the same compound, so the error largely cancels in ΔE_{g}.

Frame selection. Coordinates are never averaged: the gap of the mean structure differs from the mean gap because averaging removes the tilts that open the gap. Instead, we select individual frames. For every frame in the averaging window a vector of structural descriptors is computed: the root-mean-square tilt and the in-phase and antiphase tilt order about each pseudo-cubic axis, the mean, minimum and fifth percentile of the Sn–I–Sn angles, the mean, spread and 95th percentile of the Sn–I bond lengths, the off-centring of Sn in its octahedron,

the 95th percentile of the tilt angles, and for the hybrids the polarisation of the cation axes, the fractions of cations nearest to $\langle 100\rangle$, $\langle 110\rangle$ and $\langle 111\rangle$ and the nearest-neighbour orientational correlation. Within each run the autocorrelation time τ of the first principal component of these descriptors defines a spacing $\Delta = \max(2\tau, 5\,\mathrm{ps})$, and the frames of a bracket are thinned to at least Δ apart. The candidates of a bracket are divided into strata, run $\times$ first or second half of the window, or into clusters when k-means finds a well separated split of the descriptors (silhouette above 0.5 and the smaller cluster above 15 %), as expected for coexisting tilt states. Frames are drawn at random within strata with a fixed seed.

Two-level estimator. Level 1 isolates the elastic part of the gap change. Six free frames per material are each computed as they are and after an affine strain to the mean cell of the stress-controlled bracket, of the rigid clamp, and of the ± 0.3 GPa hydrostatic runs. Pairing the same configuration cancels the configurational noise, and the paired differences give the gap change at fixed internal structure and the deformation potential $\mathrm{d}E_\mathrm{g}/\mathrm{d}\ln V$. Level 2 compares the ensembles: the mean gap of each bracket is the stratum-weighted mean $\langle E_\mathrm{g}\rangle = \sum_k w_k m_k$ with the stratum weights w_k of the candidate pool, and its variance is $\sum_k w_k^2 s_k^2/n_k$; for a stratum with a single frame the spread of the bracket is used. A pilot of two frames per bracket was planned to set a Neyman allocation[65] of further frames in proportion to $w_k s_k$; its within-stratum spreads were too uncertain for that, so the three brackets at the operating temperature were completed to 16 frames each with the frames distributed over the strata in proportion to w_k. Level 1 uses the mean cells of the first three runs of each bracket, which differ from the eight-run means by at most 0.08 % in-plane for the hybrids and 0.16 % for the rigid clamp of $CsSnI_3$. A ridge regression of the gap on the descriptors was foreseen as a control variate, $\langle E_\mathrm{g}\rangle \approx \overline{f}_\mathrm{pool} + \overline{(E_\mathrm{g} - f)}_\mathrm{DFT}$, where f is evaluated on every candidate frame. On the present frames the frame-averaged descriptors do not predict the gap (Table S33), so the stratified means are reported without it. The difference between Level 2 and Level 1 is the part of the gap change carried by the relaxation

of tilts and cation orientations. For each bracket we report the mean gap shift relative to the free crystal at the same temperature and the smallest gap of the frames computed, because carriers collect in the low-gap tail.

For $CsSnI_3$ two further sets complete the estimators. Five Level-1 free frames are strained to the mean rigid-clamp cell of all eight runs and to the cell of the clamp set at the cubic anneal; the first gives an elastic part of (20 ± 11) meV, in agreement with the (19 ± 7) meV obtained at the three-run cell, and the two cells differ by (-4.3 ± 0.8) meV on the same frames. One MD frame per run of the free, stress-controlled and rigidly clamped brackets at the operating temperature (eight each), the three runs of the γ crystal held under the clamp, and two frames each of the cubic-anneal clamp and of the free crystal at the damp-heat temperature are relaxed with MACE-MH-1 + D3 (PBEsol labels; D3 as in the production MD) at the fixed mean cell of the frame's own run over its production window, to a largest force of 2×10^{-4} Ha Bohr^{-1}. Frames already computed as they are were preferred, so that 30 of the 31 relaxed frames have the gap of their MD frame as a partner. Only the atoms move: the relaxed frame keeps the thermal expansion and the mechanical state of its bracket, and for the rigid clamp the in-plane cell is the clamp itself. The cell is not relaxed, because the pressure offset of MACE at the UrsaMLFF cell (0.3–0.6 GPa of tension in every relaxed frame) would move it away from the ensemble it is compared with; since every bracket is relaxed in the same way, only differences between relaxed frames are used. The relaxation keeps the tilt pattern of every frame, so that the tilt variant of each rigid-clamp run is already present in its MD frame, and increases the tilt amplitudes by up to about 4° (Table S34).

Clamping study: all runs

Clamped reheating of $CsSnI_3$

Whether the substrate constraint moves the transitions of $CsSnI_3$ is measured on heating, because the model supercools on cooling. The end states of the free and the clamped films

Table S24: Location of the band edges in the DFT calculations of this work that sample more than one k-point (PBEsol and r^{2}SCAN, perovskite structures; Fermi–Dirac 500 K, no SOC; energies in meV). Frames: static, relaxed phases and frozen tilt patterns; MD, snapshots of molecular dynamics. N: calculations. At Γ: calculations with both band edges at Γ. $\Gamma - \min$: largest difference between the gap at Γ and the smallest gap on the mesh. Margins: smallest amount by which Γ is the valence-band maximum (VB) and the conduction-band minimum (CB) over the other mesh points; negative when the edge lies elsewhere. The 80-atom cell is the fcc-type supercell with vectors $(0,2,2)$, $(2,0,2)$ and $(2,2,0)$ in units of a_{pc}, onto whose Γ point R also folds; its snapshots are from 600 K.

compound	atoms	frames	mesh	functional	N	at Γ	$\Gamma - \min$	margin VB / CB
$CsSnI_3$	20	static	4×4×3	PBEsol, r^{2}SCAN	76	76	0	406 / 98
$CsSnI_3$	40	static	3×3×6	PBEsol	3	3	0	335 / 41
$CsSnI_3$	80	MD	2×2×2	r^{2}SCAN	2	1	11	54 / −11
$CsSnI_3$	80	static	3×3×3	PBEsol	3	3	0	336 / 41
$CsSnI_3$	120	static	3×3×2	PBEsol	4	4	0	336 / 41
$CsSnI_3$	240	MD	2×2×1	PBEsol	14	14	0	363 / 43
$MASnI_3$	192	MD	2×2×3	PBEsol, r^{2}SCAN	28	28	0	213 / 17
$FASnI_3$	192	MD	2×2×3	PBEsol, r^{2}SCAN	20	20	0	261 / 4

of the main-text protocol (r^{2}SCAN potential with D3, setup 2, three runs each) are heated at 0.25 K/ps from the operating temperature to 431 K (model temperatures, 39 K below the device scale), the clamped films with a and b still fixed at their anneal values. The $\gamma \leftrightarrow \beta$ transition does not move: 319 K clamped (321, 321, 316 K) against 321 K free (316, 321, 326 K), $\Delta T_{\mathrm{c}} = -2 \pm 3$ K. The free films lose their last tilt at 374 ± 9 K (381, 386, 356 K), whereas all three clamped films keep it up to the top of the ramp, so the $\beta \leftrightarrow \alpha$ transition rises by at least 57 K. Above the anneal temperature the clamped film cannot expand in the plane, and this in-plane compression relative to the free crystal favours the tilt about the normal. On the device scale this shift lies above about 410 K, outside the operating range, whereas the transition closest to operation, $\gamma \leftrightarrow \beta$, is unaffected.

Strain, tilt and cation orientation

Figure S11 and Table S25 give the response of the octahedral tilt and of the cation orientation to biaxial in-plane strain and to hydrostatic pressure at the operating temperature (r^{2}SCAN potential with D3). In $CsSnI_3$ the in-phase tilt about the film normal falls by about 2° per

percent of in-plane tension, from 11.8° at −2 % to 6.2° at +1 %, while the antiphase tilts about the in-plane axes grow; at +2 % the in-phase axis moves into the plane. Hydrostatic compression raises all tilts, by 2.7° per GPa about the normal, so that per unit change of volume the in-plane strain acts on the tilt about the normal four to five times more strongly than pressure. $MASnI_3$ and $FASnI_3$ stay untilted under tension. A compression of 0.5 % induces a weak antiphase tilt about the normal in $MASnI_3$ (2.6°), which lies closer to its transition, whereas $FASnI_3$ develops only a weak in-phase tilt about an in-plane axis at −2 % and in-phase tilts at +0.6 GPa; every induced tilt keeps the sense of the compound.

The cations respond differently. The N–N axes of FA rotate linearly with the strain, towards the film normal and ⟨100⟩ under compression and into the plane under tension (P_2 from +0.105 at −2 % to −0.065 at +2 %), without a tilt of the framework. The C–N axes of MA turn into the plane under compression, together with the induced tilt, and do not respond to tension. At the strains of the two substrate setups (0.2 to 0.4 %) the tilt of $CsSnI_3$ about the normal falls from 8.5 to 7.5–8.1°, the FA axes move slightly into the plane (P_2 lower by 0.005 to 0.02, f_{100} by about 0.008), and the MA orientations do not change.

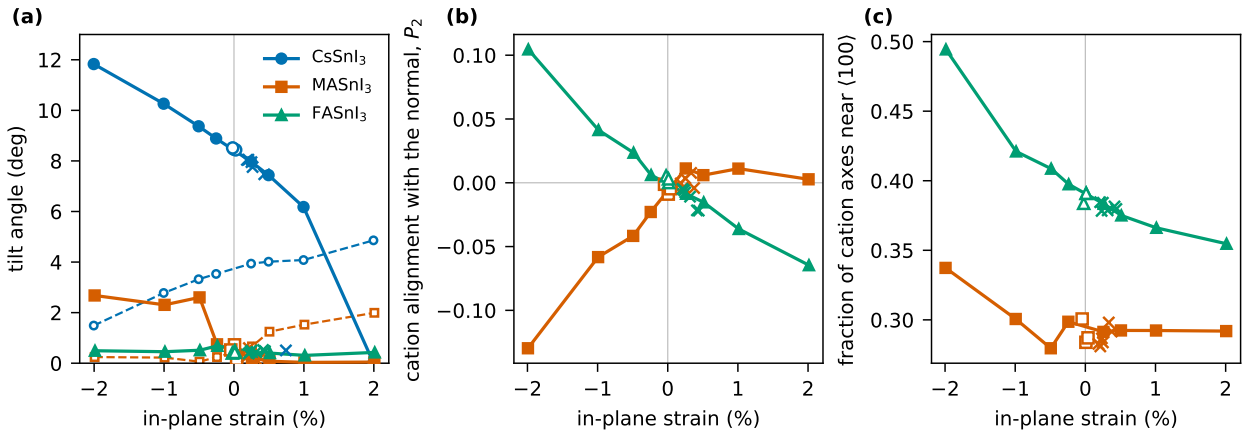


Figure S11: Response to biaxial in-plane strain at the anchored operating temperature (r^{2}SCAN potential with D3; ladder: one run per strain, a and b fixed, c free), plotted against the measured in-plane strain. Open symbols: the three free films; crosses: the films of setups 1 and 2. (a) Tilt about the film normal (solid; $CsSnI_3$ in-phase, $MASnI_3$ antiphase, $FASnI_3$ in-phase) and largest tilt about an in-plane axis (dashed). (b) Alignment of the cation axis with the film normal, $P_2 = \langle(3\cos^2\theta - 1)/2\rangle$ (MA C–N, FA N–N). (c) Fraction of cation axes nearest ⟨100⟩.

Table S25: Octahedral tilt and cation orientation against strain with the r^{2}SCAN potential and D3 at the anchored operating temperature: biaxial in-plane strain (a and b fixed, c free) and hydrostatic pressure added to P_{ext}, one run per point (last 100 ps); free films and the films of the two substrate setups, three runs each (150 ps operating hold). Tilt: rotation correlation of neighbouring octahedra (deg; + in-phase, − antiphase) about the film normal c and, for $CsSnI_3$, the mean about the in-plane axes. $P_2 = \langle(3\cos^2\theta - 1)/2\rangle$: alignment of the cation axis (MA C–N, FA N–N) with the film normal; f_{100}: fraction of cation axes nearest $\langle 100\rangle$. Run-to-run spread of the free films: tilt 0.1°, P_2 0.004, f_{100} 0.01.

	$CsSnI_3$ tilt		$MASnI_3$			$FASnI_3$		
state	normal	in-plane	tilt	P_2	f_{100}	tilt	P_2	f_{100}
biaxial −2 %	+11.8	−1.4	−2.7	−0.130	0.337	+0.5	+0.105	0.494
biaxial −1 %	+10.3	−2.7	−2.3	−0.058	0.301	+0.5	+0.041	0.421
biaxial −0.5 %	+9.4	−3.3	−2.6	−0.042	0.280	+0.5	+0.024	0.409
biaxial −0.25 %	+8.9	−3.5	−0.7	−0.023	0.299	+0.7	+0.006	0.397
free (3 runs)	+8.5	−3.6	−0.5	−0.005	0.291	+0.4	+0.003	0.389
biaxial +0.25 %	+8.0	−3.9	−0.2	+0.011	0.291	+0.4	−0.009	0.383
biaxial +0.5 %	+7.4	−4.0	−0.1	+0.006	0.292	+0.4	−0.015	0.375
biaxial +1 %	+6.2	−4.1	0.0	+0.011	0.292	+0.3	−0.036	0.366
biaxial +2 %	0.0[b]	+4.9 / −4.6[b]	0.0	+0.003	0.292	+0.4	−0.065	0.355
setup 1 (3 runs)	+8.0[a]	−3.8[a]	−0.4	−0.003	0.283	+0.5	−0.008	0.380
setup 2 (3 runs)	+7.8	−3.7	−0.2	+0.002	0.291	+0.5	−0.016	0.382
ΔP −0.3 GPa	+7.5	−2.5	−0.2	+0.017	0.294	+0.2	−0.001	0.337
ΔP −0.15 GPa	+8.1	−3.1	−0.8	−0.005	0.276	+0.4	−0.005	0.361
ΔP +0.15 GPa	+8.9	−4.2	−0.4	0.000	0.297	+0.7	+0.005	0.417
ΔP +0.3 GPa	+9.2	−4.5	−0.4	−0.001	0.290	+1.0	+0.004	0.465
ΔP +0.6 GPa	+9.9	−5.0	+0.5	+0.007	0.340	+3.3	−0.073	0.568

[a] Mean of the two films that keep the in-phase axis along the normal; in the third it moved into the plane. [b] The in-phase axis lies in the plane (tilts about the two in-plane axes).

Clamping study: detailed account

A perovskite absorber film is deposited and annealed at an elevated temperature, where it crystallises on and bonds to its substrate, and is then cooled to its operating temperature while its in-plane dimensions are restricted by the substrate. The residual stress left by this history is of order tens of MPa,[29,31] and the corresponding strain of about 0.5 %[30] is comparable to the spontaneous strain of the structural transitions. We use the PBEsol force field, whose free-crystal results are given in the Supporting Information, and PBEsol band gaps, in the converged cells, to ask how this clamping changes the structure of $CsSnI_3$, $MASnI_3$ and $FASnI_3$ at their operating temperature, and how much of the operating band gap it sets. The three compounds sample three regimes: $CsSnI_3$, annealed in its tetragonal phase, crosses the $\gamma \leftrightarrow \beta$ transition on cooling; $MASnI_3$ operates a few tens of kelvin above its transition; $FASnI_3$ crosses no transition between anneal and operation. The protocol, given in full in the Supporting Information, is summarised here.

All production MD uses the PBEsol UrsaMLFF with D3 at the external pressure P_{ext}, in the production cells of 2000 ($CsSnI_3$) and 2304 atoms (hybrids), with the a and b axes as the in-plane directions of a (001)-oriented film; none of the 88 hybrid runs of the clamping study contains a molecular defect. Each device temperature is shifted by the difference between the model's and the experimental transition temperature of the transition closest to the anneal-to-operation path ($CsSnI_3$ +49 K, $MASnI_3$ +27 K, $FASnI_3$ +4 K), so that anneal (373 K), damp-heat test (358 K) and operation (300 K) lie in the same phase and at the same distance from T_c as in a device. For $MASnI_3$ the operating point is then 327 K, 25 K above its heating transition and about 50 K above its equilibrium tilt loss (section *Starting structure of the hybrid heating runs*); the control set at the unshifted temperatures puts it at 300 K, about 25 K above the equilibrium tilt loss. For $CsSnI_3$ a cubic anneal (440 K) is added, and a control set uses the unshifted temperatures. Three mechanical brackets enclose a real film: a free-standing crystal; a stress-controlled film with an in-plane tension of 50 MPa; and a rigid in-plane clamp set at the anneal, after which the film is cooled at 2 K/ps with its in-plane

cell fixed. Each bracket is run three times at the operating temperature and once at the damp-heat temperature, and each run is held for 150 ps; five further runs of each operating-temperature bracket enlarge the frame pools of the band-gap selection. Biaxial strains of ± 0.25 to ± 2 % and hydrostatic pressures of -0.3 to $+0.6$ GPa give the elastic constants of the model and the strain response of the tilts and of the cation orientations, and clamped cells are heated through the transitions to obtain $\Delta T_c = T_c(\text{clamped}) - T_c(\text{free})$ on the heating branch. Band gaps are computed with PBEsol DFT at the Γ point, onto which the band edges fold for every cell of this construction (Supporting Information), on complete production frames selected by stratified sampling in a space of structural descriptors with a fixed seed. A paired estimator, which computes the same frames as they are and affinely strained to the clamped cells, isolates the elastic part of the gap change; an ensemble estimator, the stratified mean over the MD frames of each bracket, gives the total change. For $CsSnI_3$, one MD frame per run of each bracket is also relaxed at the fixed mean cell of its run; the gaps of these inherent structures carry no thermal noise, and their difference from the gaps of the MD frames they come from is the thermal part of the gap. The DFT energies of the same frames test UrsaMLFF on the clamped ensembles.

We are interested in four quantities: the shift of the transition temperatures under clamping; the structural response, that is, the tilt amplitudes and patterns and the orientation distribution of the molecular cations under each bracket, which tests the predicted ordering $MASnI_3 > FASnI_3 > CsSnI_3$ $(= 0)$ of the orientational response; the band-gap shift between the brackets, split into the elastic part and the part carried by tilts and cation orientations; and the effect of the anneal temperature as a manufacturing variable.

Elastic response. The elastic constants obtained from the strain and pressure ladders at the anchored operating temperature are listed in Table S26. $CsSnI_3$ is the softest in-plane and $FASnI_3$, about 60 K above its tilt transition, the stiffest. With these biaxial moduli an in-plane stress of 50 MPa corresponds to a biaxial strain of about 0.3 % in $CsSnI_3$, 0.19 %

in $MASnI_3$ and 0.16 % in $FASnI_3$, the range of residual strains measured in films.[30] The stress-controlled bracket of $MASnI_3$ indeed strains it in-plane by 0.19 %.

In-plane strain controls the tilt about the film normal. The biaxial ladders show the mechanism by which clamping acts on these films (Figure S12a). A tilt about the film normal contracts the in-plane Sn–Sn distances, so in-plane tension opposes it and in-plane compression favours it. In $CsSnI_3$ at 349 K the in-phase tilt about the normal falls steadily from 13.9° at −2 % strain to 9.9° in the free crystal and 8.8° at 0.5 %, and collapses between 0.5 % and 1 %; at 2 % a different pattern with the in-phase axis in the plane appears. The two hybrids have no tilt order at their operating points, and compression induces a tilt about the normal in both, $MASnI_3$ at half the strain of $FASnI_3$. In $MASnI_3$ at 327 K a compression of 0.5 % already induces a tilt of 4.3°, with in-phase and antiphase parts of similar size (the layered pattern), which becomes a pure in-phase tilt of 11.3° at −2 %; a compression of 0.25 % and every tension leave the normal untilted, and tension builds a weak antiphase tilt about an in-plane axis, 3.5° at 2 %. In $FASnI_3$ at 304 K compression induces an in-phase tilt about the normal only from −1 % on (4.6° at −1 %, 7.8° at −2 %), and tension leaves it untilted. $MASnI_3$ operates 25 K above its heating transition and $FASnI_3$ about 60 K above its own.

The orientation of the molecular cations responds much less (Figure S12b). The fraction of FA axes nearest to a pseudo-cubic ⟨100⟩ direction decreases steadily with tension, by 0.07 between −1 % and 1 % and by 0.13 across the ladder. The fraction of MA axes near ⟨100⟩ stays at 0.30 to 0.32 between −1 % and 2 % and rises by 0.18 at −2 %, where the in-phase tilt about the normal appears: the MA orientations follow the tilt pattern rather than the strain. At the level of these orientation statistics, the response to strain orders $FASnI_3$ > $MASnI_3$, contrary to our prediction; in $MASnI_3$ the strain acts through the octahedral tilt instead.

The three clamping brackets. Table S27 summarises the states reached under the three boundary conditions at the anchored operating temperature. The stress-controlled bracket adds the intended 50 MPa of in-plane tension in all three compounds. It reduces the tilt about the normal of $CsSnI_3$ from 9.9° to 9.1° in every run. Neither hybrid has a tilt about the normal at its operating point ($MASnI_3$ 1.5° and $FASnI_3$ 1.2°, the level of the untilted crystal), and the tension creates none: in all eight runs of $MASnI_3$ the tilt about the normal stays at 1.0 to 1.3°, and the tilt about the in-plane axes rises only from 1.5 to 1.7° on average. The cell of $MASnI_3$ expands by 0.19 % in-plane and contracts by 0.14 % along the normal, a nearly volumetric strain (volume 0.23 %), as that of $FASnI_3$.

The rigid clamp acts differently in each compound. $CsSnI_3$ is clamped in the β phase at the anneal and crosses the $\gamma \leftrightarrow \beta$ transition on cooling. The three clamped runs end in three different tilt states: one in the free γ pattern with its in-phase axis along the normal, one with the in-phase axis in the plane, and one with weak, mixed tilts. Their mean in-plane stress equals that of the free crystal, whereas a γ crystal prepared in the free orientation and then held under the same clamp carries 82 MPa of in-plane tension. The clamp thus selects or frustrates tilt variants rather than straining a single one: the film relieves the clamping stress by reorganising its tilt pattern. Which variant forms is decided kinetically at 2 K/ps, and the odd in-plane repeat of the production cell (5×5) cannot hold an in-phase tilt about an in-plane axis, which frustrates that variant. Clamped at the cubic anneal (440 K device, 489 K model), $CsSnI_3$ retains no tilt order at the operating temperature (all amplitudes below 3°). $MASnI_3$ is clamped at 400 K and cooled 73 K to its operating point. In all eight runs it stays untilted about the normal (at most 1.3°) and carries tilts of 1.2 to 2.8° about one in-plane axis and 97 MPa of in-plane tension on average; its cell expands by 0.35 % in-plane and contracts by 0.24 % along the normal. In $FASnI_3$ neither clamped bracket changes the octahedra (all tilt amplitudes below 1.6°) or the cation orientations ($f_{\langle 100 \rangle} = 0.37$ in every bracket). At the unshifted temperatures (300 K, about 25 K above its equilibrium tilt loss), $MASnI_3$ responds in the same way (one run per bracket: tilt about the normal at most 1.3°).

The anchoring of the temperatures is essential to these results. At the unshifted device temperatures (one run per bracket), $CsSnI_3$ is 111 K below its model $\gamma \leftrightarrow \beta$ transition at 300 K. All brackets, including both rigid-clamp histories, then end in the free γ pattern, and an unanchored simulation would conclude that clamping has no structural effect on $CsSnI_3$.

Transition temperatures under the clamp. Heated under the in-plane clamp of the anneal state, $CsSnI_3$ loses its second tilted axis at (404 ± 4) K and all tilt order at (448 ± 19) K (three runs; one run keeps some order to the top of its ramp at 482 K), against (411 ± 7) K and (462 ± 17) K for the free crystal. The shifts, $\Delta T_c = (-7 \pm 8)$ K and (-14 ± 25) K, are zero within their uncertainty. A clamp set by the anneal-state cell therefore shifts the $\gamma \leftrightarrow \beta$ transition of $CsSnI_3$ by less than about 15 K on heating, and the shift of the $\beta \leftrightarrow \alpha$ transition is not resolved. The structural effect of the clamp is on the tilt variants that form on cooling. Clamped heating runs were made for $CsSnI_3$ only.

Band-gap shifts. No DFT gaps were computed for $MASnI_3$ at its anchored operating temperature (327 K); its gaps at 235 K are given below.

We computed the band gaps of 222 production frames, all of which enter the analysis: 132 frames as they come from the MD of the brackets, 58 affinely strained frames of the paired estimator, 31 relaxed frames of $CsSnI_3$ and one frame computed twice (Tables S28 and S29, and Supporting Information). None of the 100 frames of the hybrids, 50 per compound(those of $MASnI_3$ from its runs at 235 K), contains a molecular defect (a proton off its cation, a broken C–N bond or a cation out of its cage). The same $CsSnI_3$ frame computed in two independent self-consistent runs gives gaps that agree to 0.02 meV, so the numerical noise of a gap is negligible against the shifts discussed here.

Table S28 gives the elastic part of the gap change, obtained by straining the same configurations to the mean cells of the clamped brackets. The gap increases with volume, with deformation potentials $a_V = \mathrm{d}E_g/\mathrm{d}\ln V$ of (3.51 ± 0.12) eV for $CsSnI_3$ and (3.49 ± 0.13) eV for $FASnI_3$. This is the sign expected for a valence-band maximum of antibonding Sn 5s–I

5p character,[6] which rises when the Sn–I bonds are compressed. Both clamped brackets expand the film in-plane and contract it along the normal, with a net increase of the volume, and therefore open the gap of $CsSnI_3$. At fixed atomic configuration, the stress-controlled film opens the gap by (18 ± 6) meV and the rigid clamp by (19 ± 7) meV in $CsSnI_3$, and by (-3 ± 4) meV and (1 ± 5) meV in $FASnI_3$ (three frames). The clamped cells of $FASnI_3$ differ little from the free one (in-plane 0.14 % and 0.26 %), and its shifts are zero within their uncertainty, slightly below their volume parts of 3 and 9 meV. In $CsSnI_3$ the shift exceeds the volume part by an amount comparable to its uncertainty.

The ensemble estimator compares the gaps of the MD frames of each bracket (Table S32). It contains the elastic part and the relaxation of the tilts and cation orientations that the clamp induces. With 16 frames per bracket for $CsSnI_3$ and 12 for $FASnI_3$, the ensemble shifts at the operating temperature are (20 ± 13) meV and (6 ± 12) meV for stress-controlled and rigidly clamped $CsSnI_3$, and (-6 ± 10) meV and (-11 ± 10) meV for $FASnI_3$. In $CsSnI_3$ neither ensemble shift differs from zero by more than 1.6 standard errors, and both agree with the elastic parts within about one combined standard error, so the ensembles alone cannot tell whether the relaxation of the structure adds to the elastic shift or removes it; for $CsSnI_3$ the relaxed frames below resolve this. The clamp leaves the gap of $FASnI_3$ unchanged within about 20 meV; its ensemble shifts agree with the elastic parts within 1.1 combined standard errors (differences of (-3 ± 11) meV and (-12 ± 11) meV), whereas the volume change alone would open the gap by 9 and 20 meV, and the ensembles lie (15 ± 10) meV and (31 ± 10) meV below it. The γ phase of $CsSnI_3$ held under the anneal clamp lies (44 ± 30) meV above the free crystal (two frames).

Distance from the tilt instability. How a film responds to a given in-plane strain depends on how far its operating point lies from the octahedral-tilt instability. $CsSnI_3$ operates in its tilted γ phase: tension reduces its tilt about the normal and opens its gap, and a clamp that reorganises the tilt variant lowers its inherent gap (below). The two hybrids

operate above their tilt transitions, $MASnI_3$ by 25 K and $FASnI_3$ by about 60 K. Neither the tension nor the rigid clamp creates a tilt in either, the strain is small and nearly volumetric, and the gap of $FASnI_3$ stays within the uncertainty of its volume part. Compression induces a tilt about the normal in both, in $MASnI_3$ at half the strain that $FASnI_3$ needs (0.5 % against 1 %), so the closer compound to its instability is the one whose tilt a residual strain changes first. How $MASnI_3$ responds in its tilted phase just below the transition, where a small tension moves its tilt from the normal into the plane, is shown in the Supporting Information.

The precision of the ensemble estimator is limited by the scatter of the gap from frame to frame, a standard deviation of 38 meV in $CsSnI_3$ and 25 meV in $FASnI_3$ over all MD frames of a compound (Table S33). Neither the volume of a frame, nor its mean Sn–I–Sn angle, nor its rms tilt angle predicts its gap: the leave-one-out errors of linear fits on these descriptors are at most a quarter below the scatter itself. The band edges of a frame of 2000 atoms thus depend on details of its instantaneous structure that frame averages do not capture, the descriptor-based control variate does not reduce the variance, and a standard error of 10 meV on the difference of two bracket means requires 29 and 12 frames per bracket for $CsSnI_3$ and $FASnI_3$.

Inherent structures of $CsSnI_3$. For $CsSnI_3$ we removed the frame-to-frame scatter by relaxing one MD frame per run of the free, stress-controlled and rigidly clamped brackets, eight per bracket, together with three γ-held, two cubic-anneal and two damp-heat frames (Table S29; every frame in Table S34). The atoms were relaxed with MACE-MH-1 + D3, the more accurate of the two potentials, at the fixed mean cell of the run, so that each relaxed frame keeps the thermal expansion and the mechanical state of its bracket and differs from its MD frame only by the thermal displacements. The gaps of the eight relaxed free frames agree to 4 meV, against 38 meV for MD frames, and over all relaxed frames the mean Sn–I–Sn angle and the volume predict the gap to 9 meV (leave-one-out), against a spread of 23 meV;

at fixed volume the gap closes by 35 meV per degree by which the bridges straighten. The scatter of the MD frames is therefore thermal, and the inherent structures resolve bracket differences of a few meV.

Residual stress in $CsSnI_3$. In the inherent structures the stress-controlled film lies (26 ± 2) meV above the free crystal, and the γ crystal held under the anneal clamp, which carries 82 MPa of in-plane tension in the MD, lies (20 ± 2) meV above it. Both keep the tilt pattern of the free crystal. Together with the elastic part at fixed configuration, (18 ± 6) meV, and the shift of the MD ensembles, (20 ± 13) meV, three estimators built on different frames give an opening of 18 to 26 meV for a residual in-plane tension of 50 MPa that leaves the tilt variant unchanged. This is the effect of residual stress on the band gap of $CsSnI_3$.

Processing history of $CsSnI_3$. The rigid clamp acts through the tilt variant instead. Its eight relaxed frames fall into two groups. The two runs that kept the in-phase tilt about the normal have gaps (27 ± 5) meV above the free crystal, like the stress-controlled film. The six runs in which that tilt weakened or moved into the plane have straighter bridges, with mean Sn–I–Sn angles of 155.4 to 156.2° against 154.3 to 154.7° in the free crystal and the kept variant, and gaps (29 ± 2) meV below the free crystal. The change of tilt variant thus lowers the inherent gap by (56 ± 5) meV, twice the size of the residual-stress effect and of opposite sign.

At the operating temperature thermal motion largely removes this difference. The MD frames lie on average (71 ± 11) meV above their relaxed structures in the free crystal, (61 ± 15) meV in the stress-controlled film and (53 ± 15) meV in the kept variant, but (122 ± 16) meV in the reorganised variant: the weakly tilted structures move furthest from their inherent structure, and their thermal tilts bend the bridges that the relaxation straightens. The MD frames of the two variants have mean gaps of 0.618 and 0.632 eV, within their standard errors of 15 to 20 meV of the free ensemble at 0.619 eV, and the rigid clamp as a whole shifts the ensemble gap by (6 ± 12) meV. This phonon-induced renormalisation of the gap[66]

depends on the state of the film, so a static structure, relaxed or averaged, does not give the operating gap of a clamped film, and its difference between states can differ from the operating one even in sign. For $CsSnI_3$ clamped at the regular anneal, the processing history therefore shifts the operating gap by (6 ± 12) meV, consistent with zero, through a change of tilt variant that lowers the inherent gap and a larger thermal opening of the reorganised structures that compensates it. The fractions of the two variants, six reorganised runs of eight, follow from cooling at 2 K/ps in a cell whose odd in-plane repeat frustrates one variant, and are not a prediction for a device; the MD frames of either variant lie within one standard error of the free ensemble.

Anneal temperature and damp heat. The paired estimator also gives the volume part of every bracket difference, $a_V \Delta \ln V$, from the mean volumes of the ensembles (Table S32). The anneal temperature enters through the in-plane cell that the clamp locks in. Clamped at the regular anneal, $CsSnI_3$ at its operating temperature has a mean volume 0.31 % larger than the free crystal, and clamped at the cubic anneal 0.57 % larger, with volume parts of 11 and 20 meV. Straining the same five free frames to the two clamp cells shows that the larger volume does not open the gap further: the cubic-anneal cell, which also contracts the film more along the normal (by 1.18 % against 0.70 %), opens it by (4.3 ± 0.8) meV less than the regular-anneal cell, which opens it by (20 ± 11) meV. The total effect of the cubic anneal, which leaves no tilt order at the operating temperature, is (-45 ± 34) meV in the ensemble and (4 ± 16) meV in the inherent structures, with two frames each, and is not resolved. Heating the free crystals from the operating to the damp-heat temperature opens the gap by 53 meV in $FASnI_3$ (one frame at the damp-heat temperature, against a frame-to-frame scatter of 25 meV), of which the thermal expansion accounts for 27 meV; in $CsSnI_3$ the two damp-heat frames give (-37 ± 20) meV in the ensemble and (18 ± 32) meV in the inherent structures, again unresolved.

The DFT energies of the same frames test UrsaMLFF on the production ensembles; forces and stresses were not computed in these runs. For $CsSnI_3$ the UrsaMLFF energy lies 1.1×10^{-4} Ha per atom below DFT, with a spread of 9×10^{-6} Ha per atom over 56 MD frames of seven brackets, so that energy differences between the ensembles are reproduced to about 1×10^{-5} Ha per atom. For $MASnI_3$ (frames of its runs at 235 K) and $FASnI_3$ the UrsaMLFF energy lies 3.7×10^{-5} Ha and 2.3×10^{-5} Ha per atom below DFT, with spreads of 4.0×10^{-6} Ha and 4.4×10^{-6} Ha per atom over 38 MD frames each, below its error on the held-out set (6.2×10^{-5} Ha per atom). The offsets of the free, stress-controlled and clamped ensembles differ by less than 5×10^{-6} Ha per atom ($MASnI_3$ -4.0×10^{-5}, -3.6×10^{-5} and -3.5×10^{-5} Ha; $FASnI_3$ -2.5×10^{-5}, -2.3×10^{-5} and -2.1×10^{-5} Ha per atom), so that energy differences between the hybrid ensembles are reproduced to about 5×10^{-6} Ha per atom. The clamped hybrid films thus stay within the training data of the force field.

Table S26: Elastic constants at the anchored operating temperature ($CsSnI_3$ 349 K, $MASnI_3$ 327 K, $FASnI_3$ 304 K), fitted to the biaxial-strain and hydrostatic-pressure ladders (one run per point).

compound	$Y/(1-\nu)$ (GPa)	ν	Y (GPa)	B (GPa)
$CsSnI_3$	15.9	0.35	10.3	14.0
$MASnI_3$	25.7	0.29	18.2	14.7
$FASnI_3$	31.7	0.17	26.3	13.6

Table S30 lists the production-window averages of every run of the clamping study with UrsaMLFF (PBEsol, D3). Each run is named by compound, protocol and seed. The protocols are: ref, the free crystal; sctl, the stress-controlled film with an in-plane tension of 50 MPa; proc, the rigid in-plane clamp set at the anneal and cooled to the operating temperature; proccub ($CsSnI_3$ only), the same clamp set at the cubic anneal; ref358 and sctl358, the free and stress-controlled states at the damp-heat temperature; clampgamma ($CsSnI_3$ only), the γ phase held under the anneal clamp; biax_mXXX and biax_pXXX, biaxial strains of $-$XXX and +XXX hundredths of a percent with a and b fixed and c free; and hyd_mXXX and hyd_pXXX, hydrostatic pressures of $-$XXX and +XXX hundredths of a GPa relative

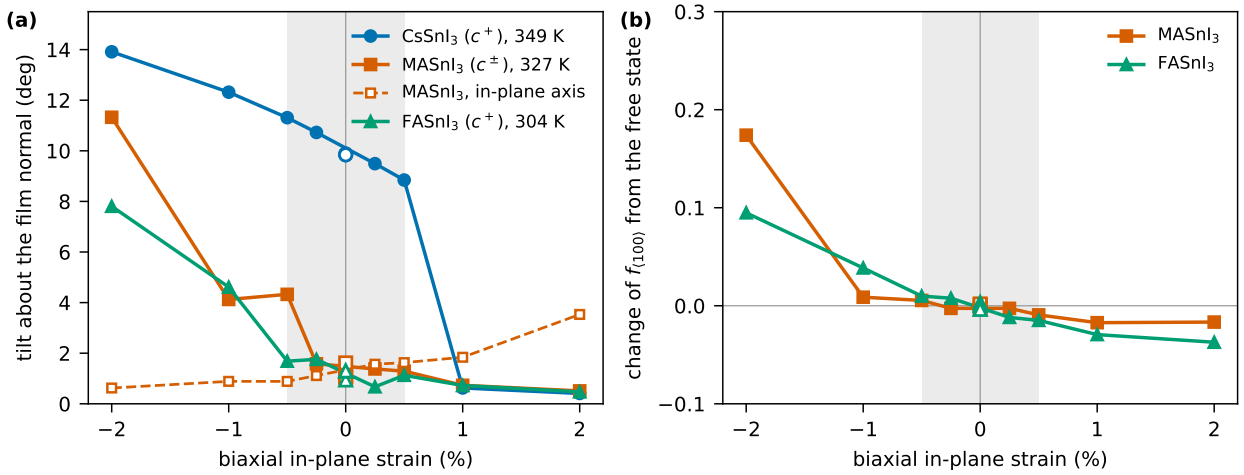


Figure S12: Response to biaxial in-plane strain at the anchored operating temperature (filled symbols, one run per strain; a and b fixed, c free). Open symbols: the three free runs. Grey band: $\pm 0.5\,\%$, the range of measured residual strains. (a) Amplitude of the octahedral tilt about the film normal: in-phase (c^+) for $CsSnI_3$ and $FASnI_3$; for $MASnI_3$ the larger of the in-phase and antiphase amplitudes, whose pattern changes along the ladder, and (dashed) its largest tilt about an in-plane axis. (b) Change of the fraction of cation axes nearest to $\langle 100 \rangle$, relative to the mean of the free runs ($MASnI_3$ 0.31, $FASnI_3$ 0.37).

Table S27: States at the anchored operating temperature under the three brackets. $\Delta\sigma_{\parallel}$: mean in-plane stress relative to the free runs (MPa, tensile positive). Tilt: amplitude about the film normal (deg; $CsSnI_3$ and $FASnI_3$ in-phase, $MASnI_3$ the larger of the in-phase and antiphase amplitudes), mean and individual runs. $f_{\langle 100 \rangle}$: fraction of cation axes nearest to $\langle 100 \rangle$. Three runs per entry (seeds 11 to 13; the hybrids have five more runs per bracket, described in the text).

compound	bracket	$\Delta\sigma_{\parallel}$	tilt (runs)	$f_{\langle 100 \rangle}$
$CsSnI_3$	free	0	9.9 (9.9, 9.8, 9.8)	–
	stress-controlled	+50	9.1 (9.2, 9.1, 9.0)	–
	rigid clamp	0	4.1 (1.6, 8.6, 2.2)	–
	γ held under clamp	+82	7.9 (7.8, 7.8, 8.1)	–
$MASnI_3$	free	0	1.2 (1.0, 1.6, 1.1)	0.31
	stress-controlled	+51	1.1 (1.0, 1.2, 1.1)	0.30
	rigid clamp	+70	1.1 (1.1, 0.8, 1.3)	0.30
$FASnI_3$	free	0	1.2 (0.9, 1.3, 1.3)	0.37
	stress-controlled	+49	1.3 (1.3, 1.6, 1.0)	0.37
	rigid clamp	+92	0.8 (0.5, 0.9, 1.0)	0.37

Table S28: Elastic part of the gap change under clamping (PBEsol, no SOC). Free frames at the operating temperature are computed as they are and after an affine strain to the mean cell of the stress-controlled bracket and of the rigid clamp. ΔE_g: mean paired change ± standard error over n frames (meV). $\Delta \ln V$: mean volume change (%). $a_V \Delta \ln V$: the part of ΔE_g accounted for by the volume change alone (meV), with the deformation potential $a_V = \mathrm{d}E_g/\mathrm{d}\ln V$ from the same frames strained to the mean cells of the ±0.3 GPa hydrostatic runs.

		stress-controlled			rigid clamp			
compound	n	ΔE_g	$\Delta \ln V$	$a_V \Delta \ln V$	ΔE_g	$\Delta \ln V$	$a_V \Delta \ln V$	a_V (eV)
$CsSnI_3$	6	18 ± 6	+0.32	+11.2	19 ± 7	+0.25	+8.8	3.51 ± 0.12
$FASnI_3$	3	−3 ± 4	+0.09	+3.2	1 ± 5	+0.27	+9.4	3.49 ± 0.13

$MASnI_3$: no DFT gaps at its anchored operating temperature (327 K); the frames computed at 235 K, below its tilt transition, are reported in the Supporting Information.

Table S29: Band-gap change of $CsSnI_3$ against the free crystal at the anchored operating temperature (PBEsol at Γ, no SOC; meV, mean ± standard error, n frames). Elastic: free MD frames computed as they are and affinely strained to the mean cell of the bracket (for the rigid clamps, the mean of all eight runs). Relaxed: MD frames relaxed at the fixed mean cell of their run, one per run (inherent structures); the free crystal has $\langle E_g \rangle = 0.538$ eV with a spread of 4 meV over eight runs. Thermal: E_g(MD frame) − E_g(the same frame relaxed), averaged over the pairs. MD ensemble: stratified mean of the frames as they come from the MD (Table S32). The rigid-clamp runs are split by the in-phase tilt about the film normal after relaxation: kept above 10°, as in every free, stress-controlled and γ-held frame, or weakened and reorganised.

	elastic	relaxed		thermal	MD ensemble	
state	ΔE_g (n)	n	ΔE_g		n	ΔE_g
free	–	8	0	71 ± 11	16	0
stress-controlled (50 MPa)	18 ± 6 (6)	8	26 ± 2	61 ± 15	16	20 ± 13
rigid clamp, all runs	20 ± 11 (5)	8	−15 ± 9	105 ± 16	16	6 ± 12
tilt about the normal kept	–	2	27 ± 5	53 ± 15	–	–
tilt reorganised	–	6	−29 ± 2	122 ± 16	–	–
γ held under clamp[a]	–	3	20 ± 2	106 ± 28	2	44 ± 30
rigid clamp, cubic anneal	16 ± 11 (5)	2	4 ± 16	33 ± 49	2	−45 ± 34
free, 358 K[b]	–	2	18 ± 32	26 ± 50	2	−37 ± 20

[a] Thermal part from two pairs (one relaxed frame has no MD partner). [b] Damp-heat device temperature; differences from the free crystal at the operating temperature.

to P_{ext}. Runs marked (nom) belong to the control set at the unshifted device temperatures. The free axes of every run sit at the target stress $-P_{\mathrm{ext}}$, so stresses relative to the free crystal are obtained by subtracting the free-crystal values. With the retrained UrsaMLFF every hybrid run completed without a molecular defect.

$MASnI_3$ at the anchor before the correction of the starting cell. The $MASnI_3$ runs at 235, 293 and 308 K (operation, damp heat and anneal) used the shift of −65 K derived from heating runs started in an oversized cell. At 235 K $MASnI_3$ lies about 35 K below the equilibrium tilt loss of the model, in a layered tilt about the normal (two of three free runs at 6.1° and 5.8°, one at 2.4°). There a biaxial strain between −0.25 % and 0.25 % removes the tilt about the normal, and tension moves it into the plane (an antiphase tilt of 5.9° at 1 %); the rigid clamp removes it in all eight runs and the tension of 45 MPa in five of eight, with a tetragonal cell (in-plane +0.6, normal −1.0 %, volume 0.2 %). DFT on these frames gives an elastic gap shift of (58 ± 18) meV for both clamped cells and ensemble shifts of (29 ± 13) meV (tension) and (40 ± 12) meV (clamp); at the same stress the frames whose tilt left the normal have a gap (46 ± 13) meV above those whose tilt stayed (Table S32). These runs show how $MASnI_3$ responds in its tilted phase near the transition; they are not the anchored operating state and are not used in the main text.

Table S30: Every run of the clamping study: production-window means. T: set temperature (K); a, b, c: cell (Å); σ: stress (GPa, tensile positive; the free axes sit at $-P_{\mathrm{ext}}$); tilt amplitudes (deg) in-phase (+) about the normal and antiphase (−) about x, y, z; f_{100}: fraction of cation axes nearest ⟨100⟩. Campaign suffix nom: unanchored device temperatures. $MASnI_3$ runs at 235 and 293 K: the anchor before the correction of the starting cell; at 327 and 385 K: the corrected anchor used in the main text.

run	T	a	b	c	σ_{xx}	σ_{yy}	σ_{zz}	$+z$	$-x$	$-y$	$-z$ / f_{100}
CsSnI3_ref_s11	349	42.763	42.337	48.665	0.064	0.055	0.049	9.9	5.1	5.1	0.3
CsSnI3_sctl_s11	349	42.816	42.487	48.559	0.114	0.105	0.049	9.2	5.3	5.3	0.3
CsSnI3_proc_s11	349	42.752	42.784	48.225	0.030	0.037	0.058	1.6	5.9	0.7	1.3
CsSnI3_ref_s12	349	42.652	42.449	48.695	0.053	0.068	0.060	9.8	4.9	4.8	0.3
CsSnI3_sctl_s12	349	42.712	42.590	48.592	0.102	0.117	0.061	9.1	5.0	5.1	0.3
CsSnI3_proc_s12	349	42.709	42.510	48.652	0.086	0.080	0.062	8.6	4.3	3.6	0.7
CsSnI3_ref_s13	349	42.828	42.305	48.661	0.085	0.061	0.055	9.8	5.1	5.1	0.3
CsSnI3_sctl_s13	349	42.873	42.479	48.550	0.134	0.110	0.055	9.0	5.3	5.0	0.3
CsSnI3_proc_s13	349	42.769	42.724	48.390	0.079	0.070	0.069	2.2	1.2	2.5	1.5
CsSnI3_ref358_s11	407	42.803	42.744	48.544	0.065	0.054	0.047	4.4	0.7	1.6	0.8
CsSnI3_sctl358_s11	407	42.979	42.925	48.243	0.114	0.103	0.047	0.8	1.5	3.1	0.9
CsSnI3_proccub_s11	349	42.897	42.960	48.109	0.130	0.145	0.068	0.9	0.9	0.8	1.0
CsSnI3_biax_m200_s11	349	41.992	41.361	49.369	-0.534	-0.483	0.052	13.9	3.9	4.0	0.1
CsSnI3_biax_m100_s11	349	42.420	41.783	49.057	-0.207	-0.199	0.050	12.3	4.3	4.4	0.2
CsSnI3_biax_m050_s11	349	42.635	41.994	48.879	-0.069	-0.077	0.050	11.3	4.7	4.8	0.2
CsSnI3_biax_m025_s11	349	42.742	42.099	48.782	0.006	-0.016	0.047	10.7	5.0	4.9	0.2
CsSnI3_biax_p025_s11	349	42.956	42.310	48.583	0.134	0.083	0.046	9.5	5.3	5.5	0.2
CsSnI3_biax_p050_s11	349	43.063	42.416	48.489	0.200	0.137	0.049	8.8	5.6	5.5	0.2
CsSnI3_biax_p100_s11	349	43.277	42.627	48.044	0.212	0.079	0.044	0.6	0.8	1.9	1.1
CsSnI3_biax_p200_s11	349	43.706	43.049	47.760	0.542	0.416	0.051	0.4	6.7	1.5	0.4
CsSnI3_hyd_m030_s11	349	42.980	42.946	48.889	0.364	0.355	0.349	6.4	2.7	3.0	0.9
CsSnI3_hyd_m015_s11	349	42.862	42.597	48.803	0.214	0.205	0.199	9.2	4.4	4.5	0.3
CsSnI3_hyd_p015_s11	349	42.680	42.093	48.525	-0.086	-0.096	-0.101	10.4	5.4	5.5	0.2

run	T	a	b	c	σ_{xx}	σ_{yy}	σ_{zz}	$+z$	$-x$	$-y$	$-z$ / f_{100}
CsSnI3_hyd_p030_s11	349	42.626	41.816	48.413	-0.236	-0.246	-0.251	11.1	5.9	5.8	0.2
CsSnI3_hyd_p060_s11	349	42.663	41.102	48.244	-0.535	-0.546	-0.552	12.5	6.4	6.5	0.2
CsSnI3_ref_s11 (nom)	300	42.942	41.882	48.676	0.065	0.056	0.050	11.3	5.9	5.9	0.2
CsSnI3_sctl_s11 (nom)	300	42.966	42.064	48.569	0.115	0.105	0.050	10.7	6.1	6.1	0.2
CsSnI3_proc_s11 (nom)	300	42.601	42.549	48.531	0.089	0.160	0.057	9.9	5.8	5.8	0.2
CsSnI3_ref358_s11 (nom)	358	42.737	42.429	48.657	0.065	0.055	0.049	9.5	4.8	4.7	0.3
CsSnI3_sctl358_s11 (nom)	358	42.803	42.547	48.564	0.114	0.105	0.048	8.9	5.1	5.1	0.3
CsSnI3_proccub_s11 (nom)	300	42.730	42.766	48.372	0.234	0.190	0.067	8.7	6.1	6.1	0.2
CsSnI3_clampheat_s11	465	42.760	42.747	48.718	-0.007	-0.009	0.076	1.5	0.7	0.8	2.9
CsSnI3_clampgamma_s11	349	42.760	42.747	48.446	0.147	0.136	0.048	7.8	4.9	5.2	0.3
CsSnI3_clampheat_s12	465	42.760	42.747	48.760	0.005	-0.001	0.082	5.5	1.2	0.9	1.0
CsSnI3_clampgamma_s12	349	42.760	42.747	48.465	0.138	0.153	0.060	7.8	5.4	4.6	0.3
CsSnI3_clampheat_s13	465	42.760	42.747	48.703	-0.008	-0.014	0.051	3.2	0.7	1.5	0.8
CsSnI3_clampgamma_s13	349	42.760	42.747	48.459	0.141	0.161	0.055	8.1	5.3	5.3	0.3
CsSnI3_ref_s14	349	42.764	42.309	48.725	0.068	0.058	0.071	10.0	4.9	5.0	0.2
CsSnI3_sctl_s14	349	42.813	42.463	48.623	0.119	0.107	0.072	9.4	5.1	5.1	0.3
CsSnI3_proc_s14	349	43.210	42.609	48.122	0.196	0.068	0.056	0.6	1.1	2.8	2.4
CsSnI3_ref_s15	349	42.633	42.480	48.702	0.055	0.077	0.065	9.8	4.8	4.7	0.3
CsSnI3_sctl_s15	349	42.701	42.621	48.587	0.106	0.128	0.066	9.0	5.1	5.1	0.2
CsSnI3_proc_s15	349	42.985	42.734	48.127	0.109	0.054	0.061	0.7	4.2	1.1	1.2
CsSnI3_ref_s16	349	42.714	42.360	48.737	0.065	0.060	0.072	10.0	4.7	4.7	0.2
CsSnI3_sctl_s16	349	42.782	42.508	48.618	0.115	0.110	0.072	9.2	5.2	4.9	0.3
CsSnI3_proc_s16	349	42.886	42.784	48.142	0.080	0.063	0.056	1.0	6.1	0.7	1.8
CsSnI3_ref_s17	349	42.638	42.470	48.709	0.058	0.071	0.061	9.8	4.2	4.6	0.3
CsSnI3_sctl_s17	349	42.530	42.792	48.582	0.111	0.119	0.060	9.0	4.4	5.2	0.3
CsSnI3_proc_s17	349	42.634	42.785	48.541	0.107	0.154	0.079	8.4	5.1	5.0	0.3
CsSnI3_ref_s18	349	42.729	42.373	48.706	0.068	0.064	0.063	10.0	4.8	4.9	0.2
CsSnI3_sctl_s18	349	42.729	42.596	48.582	0.120	0.112	0.063	9.0	4.9	5.1	0.3

run	T	a	b	c	σ_{xx}	σ_{yy}	σ_{zz}	$+z$	$-x$	$-y$	$-z$ / f_{100}
CsSnI3_proc_s18	349	42.825	42.669	48.516	0.163	0.166	0.070	8.6	5.2	5.3	0.2
MASnI3_ref_s11	235	34.406	34.286	36.784	0.052	0.028	0.032	5.1	0.8	0.6	6.1 / 0.30
MASnI3_sctl_s11	235	34.590	34.489	36.430	0.105	0.077	0.029	1.4	1.3	2.3	0.8 / 0.33
MASnI3_proc_s11	235	34.539	34.512	36.410	0.076	0.082	0.046	0.6	0.8	4.9	0.9 / 0.32
MASnI3_ref_s12	235	34.273	34.394	36.803	0.038	0.066	0.062	1.4	0.7	0.7	2.4 / 0.31
MASnI3_sctl_s12	235	34.483	34.622	36.427	0.075	0.096	0.047	1.0	0.7	1.1	0.7 / 0.32
MASnI3_proc_s12	235	34.386	34.832	36.367	0.117	0.204	0.044	1.1	0.8	2.1	0.7 / 0.33
MASnI3_ref_s13	235	34.369	34.319	36.769	0.052	0.044	0.034	4.8	0.6	0.6	5.8 / 0.31
MASnI3_sctl_s13	235	34.562	34.535	36.417	0.102	0.092	0.031	0.9	0.7	2.7	0.8 / 0.33
MASnI3_proc_s13	235	34.679	34.495	36.339	0.148	0.106	0.028	1.2	1.9	0.6	0.6 / 0.34
MASnI3_ref358_s11	293	34.601	34.483	36.651	0.064	0.039	0.039	1.5	1.2	1.1	1.2 / 0.31
MASnI3_sctl358_s11	293	34.672	34.552	36.590	0.102	0.082	0.032	1.1	1.6	1.5	1.3 / 0.31
MASnI3_biax_m200_s11	235	33.733	33.850	37.080	-0.290	-0.291	0.027	12.3	0.7	0.8	0.3 / 0.57
MASnI3_biax_m100_s11	235	34.077	34.196	36.930	-0.122	-0.099	0.026	5.9	0.5	0.5	5.8 / 0.34
MASnI3_biax_m050_s11	235	34.249	34.368	36.833	0.001	0.013	0.043	4.6	0.6	0.6	6.6 / 0.29
MASnI3_biax_m025_s11	235	34.336	34.455	36.724	0.051	0.087	0.034	4.2	0.6	0.6	5.5 / 0.31
MASnI3_biax_p025_s11	235	34.508	34.628	36.411	0.088	0.106	0.043	1.1	2.8	1.6	0.7 / 0.34
MASnI3_biax_p050_s11	235	34.594	34.714	36.355	0.176	0.200	0.051	0.8	0.9	1.1	0.7 / 0.32
MASnI3_biax_p100_s11	235	34.766	34.887	36.252	0.349	0.360	0.053	0.6	5.9	1.0	0.5 / 0.29
MASnI3_biax_p200_s11	235	35.110	35.232	36.036	0.641	0.658	0.055	0.3	0.7	8.1	0.3 / 0.27
MASnI3_hyd_m030_s11	235	34.661	34.541	36.986	0.352	0.330	0.331	2.3	0.7	0.7	6.3 / 0.29
MASnI3_hyd_m015_s11	235	34.555	34.449	36.821	0.199	0.179	0.182	1.4	1.0	1.1	2.0 / 0.31
MASnI3_hyd_p015_s11	235	34.352	34.240	36.499	-0.098	-0.123	-0.110	4.8	1.9	0.8	1.8 / 0.38
MASnI3_hyd_p030_s11	235	34.222	34.136	36.275	-0.246	-0.273	-0.266	6.0	1.0	0.9	1.0 / 0.53
MASnI3_hyd_p060_s11	235	33.797	33.593	36.552	-0.544	-0.570	-0.571	12.5	0.7	0.6	0.3 / 0.61
FASnI3_ref_s11	304	34.911	34.751	36.931	0.072	0.044	0.046	0.9	0.3	0.4	0.3 / 0.38
FASnI3_sctl_s11	304	34.982	34.805	36.918	0.121	0.094	0.049	1.3	0.3	0.4	0.3 / 0.37
FASnI3_proc_s11	304	34.970	34.849	36.916	0.146	0.122	0.066	0.5	0.3	0.3	0.3 / 0.37

run	T	a	b	c	σ_{xx}	σ_{yy}	σ_{zz}	$+z$	$-x$	$-y$	$-z$ / f_{100}
FASnI3_ref_s12	304	34.735	34.921	36.949	0.048	0.075	0.066	1.3	0.3	0.3	0.3 / 0.37
FASnI3_sctl_s12	304	34.788	34.960	36.927	0.092	0.122	0.069	1.6	0.3	0.3	0.4 / 0.37
FASnI3_proc_s12	304	34.671	35.229	36.895	0.132	0.216	0.059	0.9	0.3	0.3	0.3 / 0.36
FASnI3_ref_s13	304	34.876	34.803	36.928	0.064	0.053	0.046	1.3	0.4	0.3	0.4 / 0.37
FASnI3_sctl_s13	304	34.915	34.867	36.908	0.116	0.103	0.044	1.0	0.4	0.4	0.3 / 0.37
FASnI3_proc_s13	304	35.090	34.748	36.886	0.170	0.120	0.041	1.0	0.3	0.3	0.3 / 0.37
FASnI3_ref358_s11	362	35.012	34.832	37.025	0.075	0.045	0.046	0.8	0.4	0.3	0.3 / 0.35
FASnI3_sctl358_s11	362	35.066	34.887	36.993	0.122	0.095	0.047	0.6	0.3	0.3	0.4 / 0.35
FASnI3_biax_m200_s11	304	34.230	34.109	37.177	-0.513	-0.519	0.070	7.8	0.3	0.3	0.4 / 0.47
FASnI3_biax_m100_s11	304	34.579	34.457	37.065	-0.207	-0.231	0.056	4.6	0.3	0.3	0.4 / 0.41
FASnI3_biax_m050_s11	304	34.754	34.631	36.994	-0.076	-0.081	0.062	1.7	0.3	0.4	0.4 / 0.38
FASnI3_biax_m025_s11	304	34.841	34.718	36.947	0.024	0.000	0.048	1.8	0.3	0.3	0.3 / 0.38
FASnI3_biax_p025_s11	304	35.016	34.892	36.888	0.179	0.160	0.063	0.7	0.3	0.4	0.3 / 0.36
FASnI3_biax_p050_s11	304	35.103	34.980	36.831	0.273	0.236	0.054	1.1	0.4	0.3	0.3 / 0.36
FASnI3_biax_p100_s11	304	35.278	35.154	36.762	0.412	0.413	0.059	0.7	0.4	0.4	0.3 / 0.34
FASnI3_biax_p200_s11	304	35.627	35.502	36.625	0.673	0.665	0.052	0.5	0.3	0.3	0.3 / 0.34
FASnI3_hyd_m030_s11	304	35.169	35.007	37.204	0.376	0.349	0.352	0.8	0.3	0.3	0.4 / 0.34
FASnI3_hyd_m015_s11	304	35.055	34.864	37.063	0.222	0.197	0.198	0.8	0.3	0.3	0.3 / 0.36
FASnI3_hyd_p015_s11	304	34.771	34.612	36.763	-0.077	-0.104	-0.102	0.9	0.3	0.3	0.3 / 0.41
FASnI3_hyd_p030_s11	304	34.656	34.501	36.630	-0.229	-0.257	-0.249	0.9	0.3	0.4	0.4 / 0.43
FASnI3_hyd_p060_s11	304	34.350	34.192	36.501	-0.528	-0.555	-0.553	7.6	0.4	0.3	0.4 / 0.49
MASnI3_ref_s11 (nom)	300	34.612	34.486	36.672	0.056	0.024	0.035	1.6	1.0	1.3	1.4 / 0.31
MASnI3_sctl_s11 (nom)	300	34.686	34.569	36.594	0.113	0.083	0.039	0.9	1.5	2.1	1.3 / 0.30
MASnI3_proc_s11 (nom)	300	34.665	34.573	36.583	0.113	0.095	0.048	1.3	2.5	1.5	1.0 / 0.30
MASnI3_ref358_s11 (nom)	358	34.691	34.560	36.724	0.058	0.028	0.031	1.0	1.1	1.1	1.2 / 0.29
MASnI3_sctl358_s11 (nom)	358	34.749	34.606	36.680	0.106	0.077	0.031	1.0	1.3	1.2	1.1 / 0.30
FASnI3_ref_s11 (nom)	300	34.911	34.749	36.930	0.072	0.044	0.048	1.6	0.3	0.3	0.3 / 0.37
FASnI3_sctl_s11 (nom)	300	34.970	34.798	36.905	0.125	0.097	0.055	0.9	0.4	0.3	0.3 / 0.37

run	T	a	b	c	σ_{xx}	σ_{yy}	σ_{zz}	$+z$	$-x$	$-y$	$-z$ / f_{100}
FASnI3_proc_s11 (nom)	300	34.960	34.834	36.922	0.131	0.124	0.068	1.6	0.3	0.3	0.4 / 0.36
FASnI3_ref358_s11 (nom)	358	34.993	34.839	37.019	0.073	0.044	0.049	0.9	0.4	0.4	0.3 / 0.35
FASnI3_sctl358_s11 (nom)	358	35.078	34.879	36.998	0.129	0.095	0.055	0.7	0.4	0.3	0.3 / 0.35
MASnI3_ref_s14	235	34.350	34.336	36.771	0.039	0.037	0.050	1.1	0.7	0.8	2.5 / 0.34
MASnI3_sctl_s14	235	34.491	34.479	36.559	0.087	0.081	0.048	2.9	2.7	1.5	2.3 / 0.33
MASnI3_proc_s14	235	34.592	34.594	36.349	0.131	0.137	0.049	0.6	3.6	0.8	0.7 / 0.32
MASnI3_ref_s15	235	34.411	34.490	36.537	0.028	0.041	0.028	1.7	1.8	1.6	1.2 / 0.36
MASnI3_sctl_s15	235	34.493	34.583	36.431	0.086	0.101	0.029	1.1	0.9	2.5	1.0 / 0.33
MASnI3_proc_s15	235	34.391	34.760	36.369	0.105	0.186	0.064	0.7	2.7	0.9	0.7 / 0.34
MASnI3_ref_s16	235	34.326	34.368	36.788	0.039	0.048	0.033	5.0	0.7	0.6	6.2 / 0.30
MASnI3_sctl_s16	235	34.377	34.421	36.756	0.086	0.098	0.035	4.5	0.6	0.7	6.2 / 0.30
MASnI3_proc_s16	235	34.854	34.361	36.344	0.216	0.110	0.048	0.7	1.3	4.0	0.6 / 0.32
MASnI3_ref_s17	235	34.313	34.365	36.782	0.045	0.051	0.047	1.4	0.7	0.7	2.3 / 0.31
MASnI3_sctl_s17	235	34.397	34.438	36.699	0.092	0.102	0.050	4.8	0.7	1.1	5.0 / 0.33
MASnI3_proc_s17	235	34.639	34.581	36.335	0.164	0.149	0.026	1.1	1.2	0.7	0.8 / 0.32
MASnI3_ref_s18	235	34.405	34.389	36.683	0.050	0.048	0.045	2.4	0.9	1.1	3.1 / 0.34
MASnI3_sctl_s18	235	34.413	34.407	36.757	0.098	0.093	0.044	4.6	0.6	0.7	5.9 / 0.30
MASnI3_proc_s18	235	34.611	34.626	36.325	0.147	0.154	0.030	0.6	5.5	1.4	0.6 / 0.31
FASnI3_ref_s14	304	34.852	34.808	36.968	0.057	0.052	0.071	1.2	0.3	0.3	0.3 / 0.37
FASnI3_sctl_s14	304	34.878	34.857	36.920	0.104	0.099	0.065	1.0	0.4	0.4	0.3 / 0.38
FASnI3_proc_s14	304	34.996	34.959	36.867	0.208	0.200	0.055	1.9	0.3	0.3	0.3 / 0.37
FASnI3_ref_s15	304	34.770	34.891	36.922	0.044	0.063	0.040	1.0	0.3	0.4	0.3 / 0.37
FASnI3_sctl_s15	304	34.827	34.942	36.907	0.098	0.113	0.044	1.1	0.3	0.3	0.3 / 0.36
FASnI3_proc_s15	304	34.674	35.238	36.915	0.130	0.216	0.075	1.1	0.3	0.4	0.4 / 0.37
FASnI3_ref_s16	304	34.800	34.855	36.928	0.053	0.063	0.050	1.4	0.3	0.3	0.4 / 0.37
FASnI3_sctl_s16	304	34.855	34.923	36.910	0.107	0.123	0.052	0.9	0.3	0.4	0.4 / 0.36
FASnI3_proc_s16	304	35.257	34.639	36.896	0.210	0.124	0.060	0.9	0.3	0.4	0.4 / 0.36
FASnI3_ref_s17	304	34.781	34.855	36.934	0.052	0.065	0.059	2.0	0.3	0.3	0.3 / 0.37

run	T	a	b	c	σ_{xx}	σ_{yy}	σ_{zz}	$+z$	$-x$	$-y$	$-z$ / f_{100}
FASnI3_sctl_s17	304	34.857	34.914	36.925	0.102	0.114	0.057	0.9	0.3	0.4	0.3 / 0.37
FASnI3_proc_s17	304	35.032	34.910	36.863	0.204	0.177	0.037	0.9	0.3	0.3	0.3 / 0.36
FASnI3_ref_s18	304	34.841	34.837	36.953	0.060	0.059	0.064	0.7	0.3	0.3	0.3 / 0.37
FASnI3_sctl_s18	304	34.906	34.881	36.937	0.110	0.111	0.065	0.8	0.4	0.4	0.3 / 0.36
FASnI3_proc_s18	304	34.982	35.025	36.856	0.211	0.220	0.043	0.9	0.4	0.4	0.3 / 0.36
MASnI3_ref_s11	327	34.633	34.506	36.652	0.059	0.033	0.039	0.9	1.3	1.2	1.0 / 0.31
MASnI3_sctl_s11	327	34.691	34.567	36.621	0.110	0.083	0.038	0.9	1.3	1.1	1.0 / 0.31
MASnI3_proc_s11	327	34.694	34.573	36.628	0.118	0.082	0.052	1.1	1.2	1.2	1.1 / 0.30
MASnI3_ref_s12	327	34.486	34.631	36.658	0.030	0.059	0.057	1.6	1.1	1.4	1.1 / 0.31
MASnI3_sctl_s12	327	34.555	34.692	36.642	0.077	0.113	0.054	1.2	1.4	2.1	1.0 / 0.30
MASnI3_proc_s12	327	34.414	34.944	36.583	0.090	0.204	0.048	0.8	1.8	1.4	0.8 / 0.31
MASnI3_ref_s13	327	34.577	34.516	36.662	0.049	0.035	0.031	1.0	1.2	1.0	1.1 / 0.32
MASnI3_sctl_s13	327	34.641	34.602	36.606	0.097	0.089	0.031	1.1	1.4	1.2	1.1 / 0.31
MASnI3_proc_s13	327	34.764	34.489	36.596	0.129	0.065	0.036	1.3	1.3	1.2	0.9 / 0.30
MASnI3_ref358_s11	385	34.718	34.586	36.740	0.056	0.029	0.031	0.9	1.2	1.0	0.8 / 0.30
MASnI3_sctl358_s11	385	34.758	34.641	36.704	0.112	0.076	0.038	0.9	1.2	1.2	1.1 / 0.30
MASnI3_biax_m200_s11	327	33.858	33.721	37.220	-0.426	-0.461	0.037	11.3	0.6	0.5	0.5 / 0.49
MASnI3_biax_m100_s11	327	34.204	34.065	37.043	-0.242	-0.278	0.034	4.1	0.6	0.6	4.1 / 0.32
MASnI3_biax_m050_s11	327	34.376	34.237	36.920	-0.103	-0.133	0.037	4.1	0.5	0.7	4.3 / 0.32
MASnI3_biax_m025_s11	327	34.463	34.323	36.836	-0.046	-0.075	0.034	1.6	0.9	0.8	1.5 / 0.31
MASnI3_biax_p025_s11	327	34.636	34.495	36.654	0.059	0.027	0.039	0.9	1.0	1.6	1.4 / 0.31
MASnI3_biax_p050_s11	327	34.722	34.581	36.598	0.134	0.104	0.032	1.3	1.6	1.4	1.1 / 0.30
MASnI3_biax_p100_s11	327	34.895	34.753	36.477	0.285	0.259	0.031	0.7	1.7	1.8	0.7 / 0.30
MASnI3_biax_p200_s11	327	35.240	35.097	36.272	0.584	0.555	0.039	0.4	3.5	1.9	0.5 / 0.30
MASnI3_hyd_m030_s11	327	34.871	34.729	36.919	0.350	0.324	0.333	0.9	1.0	1.2	1.4 / 0.30
MASnI3_hyd_m015_s11	327	34.743	34.608	36.788	0.209	0.175	0.182	1.1	1.7	1.0	1.7 / 0.30
MASnI3_hyd_p015_s11	327	34.512	34.363	36.545	-0.097	-0.126	-0.118	2.5	1.2	1.1	1.2 / 0.33
MASnI3_hyd_p030_s11	327	34.376	34.251	36.417	-0.236	-0.265	-0.263	3.9	0.9	1.1	1.2 / 0.38

run	T	a	b	c	σ_{xx}	σ_{yy}	σ_{zz}	$+z$	$-x$	$-y$	$-z$ / f_{100}
MASnI3_hyd_p060_s11	327	34.200	34.086	36.090	-0.548	-0.576	-0.563	3.3	1.0	0.8	1.0 / 0.45
MASnI3_ref_s14	327	34.548	34.537	36.702	0.041	0.039	0.054	1.7	1.2	1.3	1.6 / 0.31
MASnI3_sctl_s14	327	34.618	34.598	36.640	0.089	0.082	0.046	1.0	1.1	1.0	1.0 / 0.31
MASnI3_proc_s14	327	34.690	34.691	36.577	0.162	0.161	0.048	0.9	1.3	1.5	0.8 / 0.30
MASnI3_ref_s15	327	34.518	34.591	36.666	0.030	0.047	0.027	1.5	1.1	1.4	1.3 / 0.31
MASnI3_sctl_s15	327	34.580	34.666	36.577	0.081	0.103	0.022	1.1	2.1	1.4	0.9 / 0.31
MASnI3_proc_s15	327	34.439	34.937	36.618	0.098	0.216	0.057	0.7	1.9	1.1	0.8 / 0.31
MASnI3_ref_s16	327	34.533	34.586	36.649	0.033	0.051	0.037	1.5	1.2	1.5	1.3 / 0.31
MASnI3_sctl_s16	327	34.605	34.645	36.621	0.085	0.098	0.033	1.1	1.3	1.2	1.2 / 0.30
MASnI3_proc_s16	327	34.962	34.419	36.591	0.208	0.089	0.057	1.0	1.7	1.3	0.9 / 0.30
MASnI3_ref_s17	327	34.527	34.576	36.683	0.035	0.047	0.044	2.1	1.3	1.5	1.7 / 0.31
MASnI3_sctl_s17	327	34.590	34.640	36.614	0.093	0.103	0.045	1.3	1.4	1.4	1.0 / 0.31
MASnI3_proc_s17	327	34.712	34.617	36.550	0.134	0.112	0.037	0.7	1.4	2.8	0.8 / 0.31
MASnI3_ref_s18	327	34.557	34.569	36.705	0.046	0.041	0.046	1.5	1.2	1.2	1.6 / 0.30
MASnI3_sctl_s18	327	34.627	34.612	36.641	0.099	0.098	0.055	0.9	1.3	1.3	1.3 / 0.30
MASnI3_proc_s18	327	34.721	34.734	36.535	0.180	0.178	0.022	0.7	1.0	2.0	0.9 / 0.31

Band-gap frame selection

Table S31 lists the candidate pools of all brackets, constructed as described in the application protocol. The DFT calculations of this work were made on complete production frames. For each compound, six free frames at the operating temperature, two from each of the first three runs and at least Δ apart, are each computed as they are and after an affine strain to the mean cells of the stress-controlled runs, of the rigid-clamp runs and of the hydrostatic runs at -0.3 and $+0.3$ GPa. Two pilot frames from different runs or strata are computed for every other bracket. The free, stress-controlled and rigid-clamp brackets at the operating temperature are then completed to 16 frames each from all eight runs, filling the run $\times$ time-half strata in proportion to their weights and keeping every frame at least Δ from any other frame of the same run. Of the 234 calculations, 224 enter the analysis: 149 on frames as they come from the MD, 72 on affinely strained frames and three repeats of a free frame. One rigid-clamp frame of $CsSnI_3$ did not return. Ten $FASnI_3$ frames are excluded because the MD had moved a proton off its cation (three frames) or a cation through a window of its cage into a neighbouring cage (seven frames); both are artefacts of the force field outside its training data, detected by a test of every computed frame. All calculations reached the density residual of the labels, and the gap is read from the final eigenvalues at Γ. The paired gap changes are given in Table S28 of the main text, the bracket means in Table S32 and the frame-to-frame scatter in Table S33. The lists of frames, strata and pairings are available with the data.

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Table S31: Candidate pools of the band-gap frame selection: production-window frames of all runs of a bracket, the autocorrelation time τ of the first principal component of the descriptors, the frame spacing $\Delta = \max(2\tau, 5\,\mathrm{ps})$, the number of candidates and the strata.

compound	bracket	frames	τ (ps)	Δ (ps)	candidates	strata
$CsSnI_3$	ref	808	3.1	6.2	114	seed × time half
$CsSnI_3$	sctl	808	3.2	6.5	113	seed × time half
$CsSnI_3$	proc	400	3.2	6.5	56	seed × time half
$CsSnI_3$	ref358	101	13.0	26.0	4	seed × time half
$CsSnI_3$	sctl358	101	10.0	20.0	5	seed × time half
$CsSnI_3$	proccub	51	2.0	5.0	10	seed × time half
$CsSnI_3$	clampgamma	303	4.7	9.3	30	seed × time half
$MASnI_3$	ref	808	6.6	13.2	56	seed × time half
$MASnI_3$	sctl	808	6.6	13.2	56	seed × time half
$MASnI_3$	proc	400	3.0	6.0	65	seed × time half
$MASnI_3$	ref358	101	1.0	5.0	20	seed × time half
$MASnI_3$	sctl358	101	1.0	5.0	20	seed × time half
$FASnI_3$	ref	808	1.6	5.0	160	seed × time half
$FASnI_3$	sctl	808	1.0	5.0	160	seed × time half
$FASnI_3$	proc	400	1.1	5.0	80	seed × time half
$FASnI_3$	ref358	101	1.0	5.0	20	seed × time half
$FASnI_3$	sctl358	101	1.0	5.0	20	seed × time half

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Table S32: Band gaps of the MD frames of each bracket (PBEsol at Γ, no SOC; absolute values are not predictions). n: frames. $\langle E_g \rangle$: stratified mean (eV) with its standard error (meV). min: smallest gap (eV). ΔE_g: difference from the free crystal at the same temperature (meV), with the combined standard error. $a_V \Delta \ln V$: the part of ΔE_g that the change of the mean volume of the ensembles accounts for, with a_V of Table S28 (meV). $MASnI_3$: frames at 235 and 293 K, the temperatures before the correction of its anchor.

compound	bracket	n	$\langle E_g \rangle$	s.e.	min	ΔE_g	$a_V \Delta \ln V$
$CsSnI_3$	free	16	0.619	9	0.552	–	–
$CsSnI_3$	stress-controlled	16	0.639	9	0.554	20 ± 13	+9.2
$CsSnI_3$	rigid clamp	16	0.625	9	0.570	6 ± 12	+10.7
$CsSnI_3$	free, 358 K[a]	2	0.582	18	0.564	−37 ± 20	+25.7
$CsSnI_3$	stress-controlled, 358 K[a]	2	0.654	6	0.649	72 ± 18	+7.5
$CsSnI_3$	rigid clamp, cubic anneal	2	0.575	33	0.542	−45 ± 34	+19.9
$CsSnI_3$	γ held under clamp	2	0.664	28	0.635	44 ± 30	+16.0
$MASnI_3$	free	12	0.465	10	0.391	–	–
$MASnI_3$	stress-controlled	12	0.494	9	0.437	29 ± 13	+6.7
$MASnI_3$	rigid clamp	12	0.505	7	0.456	40 ± 12	+8.0
$FASnI_3$	free	12	0.521	9	0.475	–	–
$FASnI_3$	stress-controlled	12	0.515	6	0.472	−6 ± 10	+9.0
$FASnI_3$	rigid clamp	12	0.510	5	0.484	−11 ± 10	+19.9

[a] Damp-heat device temperature; the simulation temperature is anchored as in Table S23. For the free crystal at 358 K, ΔE_g is the change from the free crystal at the operating temperature; for the stress-controlled film at 358 K, from the free crystal at 358 K.

Table S33: Frame-to-frame scatter of the band gap. SD: standard deviation of the gaps of all MD frames of a compound (meV). Leave-one-out error (meV) of a linear fit of the gap of a frame on its $\ln V$, its mean Sn–I–Sn angle, its rms tilt angle, or $\ln V$ and the angle together. n_{10}: frames per bracket needed for a standard error of 10 meV on a difference of two bracket means, $2\,\mathrm{SD}^2/(10\,\mathrm{meV})^2$.

compound	frames	SD	$\ln V$	angle	tilt	$\ln V$ + angle	n_{10}
$CsSnI_3$	56	38	39	37	38	35	29
$MASnI_3$	38	34	34	33	36	28	23
$FASnI_3$	38	25	25	21	24	19	12

Table S34: Relaxed $CsSnI_3$ frames (MACE-MH-1 + D3, atoms relaxed at the fixed mean cell of the run to a largest force of 2×10^{-4} Ha Bohr^{-1}). E_g: PBEsol gap at Γ of the relaxed frame and of the MD frame it was relaxed from (eV). Tilts after relaxation (deg): in-phase about the film normal (c^+), antiphase about the in-plane axes (a^-, mean of the two), and the larger in-phase tilt about an in-plane axis. $\bar{\theta}$: mean Sn–I–Sn angle (deg).

bracket	seed	E_g relaxed	E_g MD	c^+ normal	a^-	c^+ in-plane	$\bar{\theta}$
free	11	0.542	0.609	11.6	7.7	0.3	154.3
free	12	0.537	0.601	11.6	7.7	0.0	154.4
free	13	0.547	0.617	11.6	7.7	0.2	154.3
free	14	0.537	0.654	11.7	7.6	0.2	154.3
free	15	0.536	0.603	11.5	7.7	0.2	154.5
free	16	0.535	0.552	11.7	7.6	0.2	154.4
free	17	0.536	0.588	11.6	7.7	0.2	154.4
free	18	0.537	0.648	11.6	7.7	0.3	154.4
stress-controlled	11	0.568	0.660	11.2	7.9	0.4	154.4
stress-controlled	12	0.563	0.585	11.2	7.9	0.2	154.5
stress-controlled	13	0.562	0.666	11.2	7.9	0.2	154.5
stress-controlled	14	0.562	0.554	11.3	7.8	0.1	154.5
stress-controlled	15	0.565	0.632	11.2	7.9	0.2	154.6
stress-controlled	16	0.562	0.641	11.3	7.8	0.1	154.5
stress-controlled	17	0.566	0.595	11.2	7.9	0.1	154.5
stress-controlled	18	0.567	0.669	11.2	7.9	0.2	154.5
rigid clamp	11	0.512	0.597	0.6	4.6	5.4	155.8
rigid clamp	12	0.512	0.613	8.5	6.7	1.1	155.4
rigid clamp	13	0.515	0.638	2.3	4.2	5.2	155.9
rigid clamp	14	0.508	0.700	0.9	4.2	2.5	156.2
rigid clamp	15	0.509	0.609	0.7	4.7	0.4	155.8
rigid clamp	16	0.502	0.631	0.1	4.8	0.1	155.7
rigid clamp	17	0.570	0.638	10.9	8.0	0.2	154.7
rigid clamp	18	0.561	0.598	10.9	8.0	0.6	154.7
γ held under clamp	11	0.558	0.635	10.7	8.1	0.2	154.7
γ held under clamp	12	0.558	0.692	10.7	8.0	0.2	154.7
γ held under clamp	13	0.560	–	10.8	8.1	0.2	154.6
rigid clamp, cubic anneal	11	0.526	0.608	0.1	2.5	4.2	156.1
rigid clamp, cubic anneal	11	0.558	0.542	0.4	1.3	6.4	155.6
free, 358 K	11	0.588	0.564	11.4	4.0	0.7	155.2
free, 358 K	11	0.524	0.600	5.2	4.5	0.5	156.3

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